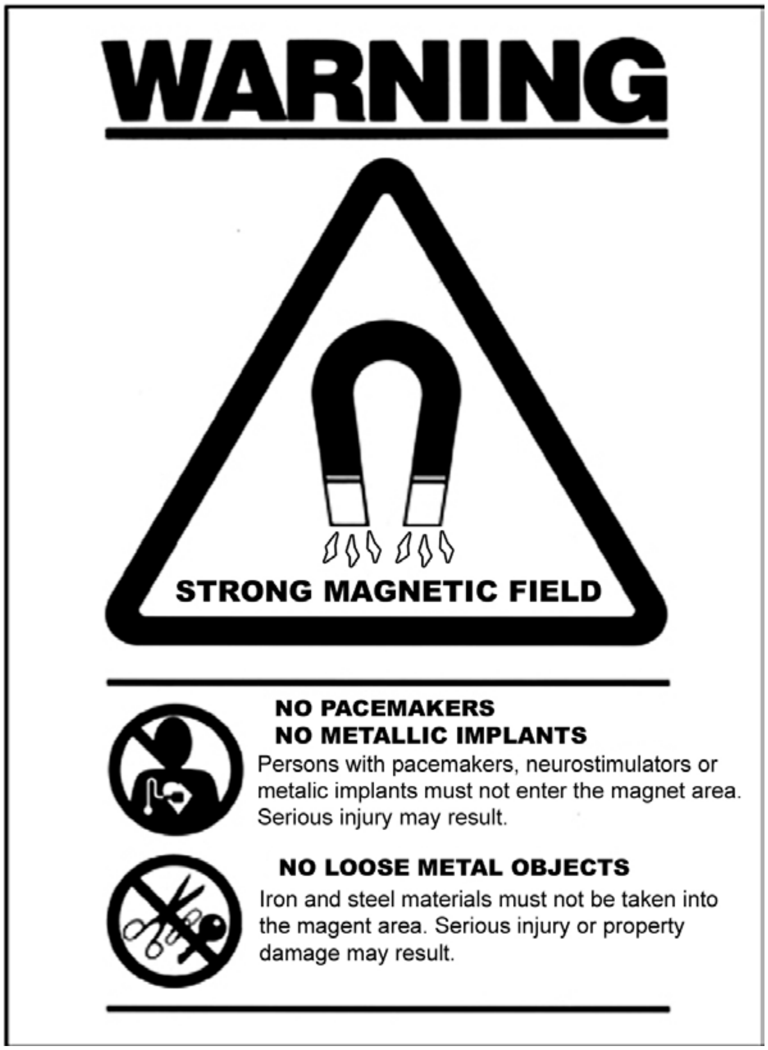
GE Healthcare

Signa Oncology Patient Transport Table Option

Service Methods

OPERATING DOCUMENTATION

5308106
Revision 1


WARNING
STRONG MAGNETIC FIELD
NO PACEMAKERS
NO METALLIC IMPLANTS
Persons with pacemakers, neurostimulators or metalic implants must not enter the magnet area. Serious injury may result.
NO LOOSE METAL OBJECTS
Iron and steel materials must not be taken into the magent area. Serious injury or property damage may result.

Important Information

LANGUAGE

ПРЕДУПРЕЖДЕНИЕ (BG)

- ТОВА УПЪТВАНЕ ЗА РАБОТА Е НАЛИЧНО САМО НА АНГЛИЙСКИ ЕЗИК.
- АКО ДОСТАВЧИКЪТ НА УСЛУГАТА НА КЛИЕНТА ИЗИСКА ЕЗИК, РАЗЛИЧЕН ОТ АНГЛИЙСКИ, ЗАДЪЛЖЕНИЕ НА КЛИЕНТА Е ДА ОСИГУРИ ПРЕВОД.
- НЕ ИЗПОЛЗВАЙТЕ ОБОРУДВАНЕТО ПРЕДИ ДА СТЕ СЕ КОНСУЛТИРАЛИ И РАЗБРАЛИ УПЪТВАНЕТО ЗА РАБОТА.
- НЕСПАЗВАНЕТО НА ТОВА ПРЕДУПРЕЖДЕНИЕ МОЖЕ ДА ДОВЕДЕ ДО НАРАНЯВАНЕ НА ДОСТАВЧИКА НА УСЛУГАТА, ОПЕРАТОРА ИЛИ ПАЦИЕНТ В РЕЗУЛТАТ НА ТОКОВ УДАР ИЛИ МЕХАНИЧНА ИЛИ ДРУГА ОПАСНОСТ.

警告 (ZH-CN)

- 本维修手册仅提供英文版本。
- 如果维修服务提供商需要非英文版本，客户需自行提供翻译服务。
- 未详细阅读和完全理解本维修手册之前，不得进行维修。
- 忽略本警告可能对维修人员，操作员或患者造成触电、机械伤害或其他形式的伤害。

VÝSTRAHA (CS)

- TENTO PROVOZNÍ NÁVOD EXISTUJE POUZE V ANGLICKÉM JAZYCE.
- V PŘÍPADĚ, ŽE EXTERNÍ SLUŽBA ZÁKAZNÍKŮM POTŘEBUJE NÁVOD V JINÉM JAZYCE, JE ZAJIŠTĚNÍ PŘEKLADU DO ODPOVÍDAJÍCÍHO JAZYKA ÚKOLEM ZÁKAZNÍKA.
- NESNAŽTE SE O ÚDRŽBU TOHOTO ZAŘÍZENÍ, ANIŽ BYSTE SI PŘEČETLI TENTO PROVOZNÍ NÁVOD A POCHOPILI JEHO OBSAH.
- V PŘÍPADĚ NEDODRŽOVÁNÍ TÉTO VÝSTRAHY MŮŽE DOJÍT K PORANĚNÍ PRACOVNÍKA PRODEJNÍHO SERVISU, OBSLUŽNÉHO PERSONÁLU NEBO PACIENTŮ VLIVEM ELEKTRICKÉHOP PROUDU, RESPEKTIVE VLIVEM MECHANICKÝCH ČI JINÝCH RIZIK.

ADVARSEL (DA)

- DENNE SERVICEMANUAL FINDES KUN PÅ ENGELSK.
- HVIS EN KUNDES TEKNIKER HAR BRUG FOR ET ANDET SPROG END ENGELSK, ER DET KUNDENS ANSVAR AT SØRGE FOR OVERSÆTTELSE.
- FORSØG IKKE AT SERVICERE UDSTYRET MEDMINDRE DENNE SERVICEMANUAL HAR VÆRET KONSULTERET OG ER FORSTÅET.
- MANGLENDE OVERHOLDELSE AF DENNE ADVARSEL KAN MEDFØRE SKADE PÅ GRUND AF ELEKTRISK, MEKANISK ELLER ANDEN FARE FOR TEKNIKEREN, OPERATØREN ELLER PATIENTEN.

WAARSCHUWING (NL)

- DEZE ONDERHOUDSHANDLEIDING IS ENKEL IN HET ENGELS VERKRIJGBAAR.
- ALS HET ONDERHOUDSPERSONEEL EEN ANDERE TAAL VEREIST, DAN IS DE KLANT VERANTWOORDELIJK VOOR DE VERTALING ERVAN.
- PROBEER DE APPARATUUR NIET TE ONDERHOUDEN VOORDAT DEZE ONDERHOUDSHANDLEIDING WERD GERAADPLEEGD EN BEGREPEN IS.
- INDIEN DEZE WAARSCHUWING NIET WORDT OPGEVOLGD, ZOU HET ONDERHOUDSPERSONEEL, DE OPERATOR OF EEN PATIËNT GEWOND KUNNEN RAKEN ALS GEVOLG VAN EEN ELEKTRISCHE SCHOK, MECHANISCHE OF ANDERE GEVAREN.

WARNING (EN)

- THIS SERVICE MANUAL IS AVAILABLE IN ENGLISH ONLY.
- IF A CUSTOMER'S SERVICE PROVIDER REQUIRES A LANGUAGE OTHER THAN ENGLISH, IT IS THE CUSTOMER'S RESPONSIBILITY TO PROVIDE TRANSLATION SERVICES.
- DO NOT ATTEMPT TO SERVICE THE EQUIPMENT UNLESS THIS SERVICE MANUAL HAS BEEN CONSULTED AND IS UNDERSTOOD.
- FAILURE TO HEED THIS WARNING MAY RESULT IN INJURY TO THE SERVICE PROVIDER, OPERATOR, OR PATIENT FROM ELECTRIC SHOCK, OR FROM MECHANICAL OR OTHER HAZARDS.

HOIATUS (ET)

- KÄESOLEV TEENINDUSJUHEND ON SAADAVAL AINULT INGLISE KEELES.
- KUI KLIENDITEENINDUSE OSUTAJA NÕUAB JUHENDIT INGLISE KEELEST ERINEVAS KEELES, VASTUTAB KLIENT TÕLKETEENUSE OSUTAMISE EEST.
- ÄRGE ÜRITAGE SEADMEID TEENINDADA ENNE EELNEVALT KÄESOLEVA TEENINDUSJUHENDIGA TUTVUMIST JA SELLEST ARU SAAMIST.
- KÄESOLEVA HOIATUSE EIRAMINE VÕIB PÕHJUSTADA TEENUSEOSUTAJA, OPERAATORI VÕI PATSIENDI VIGASTAMIST ELEKTRILÖÖGI, MEHAANILISE VÕI MUU OHU TAGAJÄRJEL.

VAROITUS (FI)

- TÄMÄ HUOLTO-OHJE ON SAATAVILLA VAIN ENGLANNIKSI.
- JOS ASIAKKAAN HUOLTOHENKILÖSTÖ VAATII MUUTA KUIN ENGLANNINKIELISTÄ MATERIAALIA, TARVITTAVAN KÄÄNNÖKSEN HANKKIMINEN ON ASIAKKAAN VASTUULLA.
- ÄLÄ YRITÄ KORJATA LAITTEISTOA ENNEN KUIN OLET VARMASTI LUKENUT JA YMMÄRTÄNYT TÄMÄN HUOLTO-OHJEEN.
- MIKÄLI TÄTÄ VAROITUSTA EI NOUDATETA, SEURAUKSENA VOI OLLA HUOLTOHENKILÖSTÖN, LAITTEISTON KÄYTTÄJÄN TAI POTILAAN VAHINGOITTUMINEN SÄHKÖISKUN, MEKAANISEN VIAN TAI MUUN VAARATILANTEEN VUOKSI.

ATTENTION (FR)

- CE MANUEL DE SERVICE N'EST DISPONIBLE QU'EN ANGLAIS.
- SI LE TECHNICIEN DU CLIENT A BESOIN DE CE MANUEL DANS UNE AUTRE LANGUE QUE L'ANGLAIS, C'EST AU CLIENT QU'IL INCOMBE DE LE FAIRE TRADUIRE.
- NE PAS TENTER D'INTERVENIR SUR LES EQUIPEMENTS TANT QUE LE MANUEL SERVICE N'A PAS ETE CONSULTE ET COMPRIS.
- LE NON-RESPECT DE CET AVERTISSEMENT PEUT ENTRAÎNER CHEZ LE TECHNICIEN, L'OPÉRATEUR OU LE PATIENT DES BLESSURES DUES À DES DANGERS ÉLECTRIQUES, MÉCANIQUES OU AUTRES.

WARNUNG (DE)

- DIESE SERVICEANLEITUNG EXISTIERT NUR IN ENGLISCHER SPRACHE.
- FALLS EIN FREMDER KUNDENDIENST EINE ANDERE SPRACHE BENÖTIGT, IST ES AUFGABE DES KUNDEN FÜR EINE ENTSPRECHENDE ÜBERSETZUNG ZU SORGEN.
- VERSUCHEN SIE NICHT DIESE ANLAGE ZU WARTEN, OHNE DIESE SERVICEANLEITUNG GELESEN UND VERSTANDEN ZU HABEN.
- WIRD DIESE WARNUNG NICHT BEACHTET, SO KANN ES ZU VERLETZUNGEN DES KUNDENDIENSTTECHNIKERS, DES BEDIENERS ODER DES PATIENTEN DURCH STROMSCHLÄGE, MECHANISCHE ODER SONSTIGE GEFAHREN KOMMEN.

ΠΡΟΕΙΔΟΠΟΙΗΣΗ (EL)

- ΤΟ ΠΑΡΟΝ ΕΓΧΕΙΡΙΔΙΟ ΣΕΡΒΙΣ ΔΙΑΤΙΘΕΤΑΙ ΣΤΑ ΑΓΓΛΙΚΑ ΜΟΝΟ.
- ΕΑΝ ΤΟ ΑΤΟΜΟ ΠΑΡΟΧΗΣ ΣΕΡΒΙΣ ΕΝΟΣ ΠΕΛΑΤΗ ΑΠΑΙΤΕΙ ΤΟ ΠΑΡΟΝ ΕΓΧΕΙΡΙΔΙΟ ΣΕ ΓΛΩΣΣΑ ΕΚΤΟΣ ΤΩΝ ΑΓΓΛΙΚΩΝ, ΑΠΟΤΕΛΕΙ ΕΥΘΥΝΗ ΤΟΥ ΠΕΛΑΤΗ ΝΑ ΠΑΡΕΧΕΙ ΥΠΗΡΕΣΙΕΣ ΜΕΤΑΦΡΑΣΗΣ.
- ΜΗΝ ΕΠΙΧΕΙΡΗΣΕΤΕ ΤΗΝ ΕΚΤΕΛΕΣΗ ΕΡΓΑΣΙΩΝ ΣΕΡΒΙΣ ΣΤΟΝ ΕΞΟΠΛΙΣΜΟ ΕΚΤΟΣ ΕΑΝ ΕΧΕΤΕ ΣΥΜΒΟΥΛΕΥΤΕΙ ΚΑΙ ΕΧΕΤΕ ΚΑΤΑΝΟΗΣΕΙ ΤΟ ΠΑΡΟΝ ΕΓΧΕΙΡΙΔΙΟ ΣΕΡΒΙΣ.
- ΕΑΝ ΔΕ ΛΑΒΕΤΕ ΥΠΟΨΗ ΤΗΝ ΠΡΟΕΙΔΟΠΟΙΗΣΗ ΑΥΤΗ, ΕΝΔΕΧΕΤΑΙ ΝΑ ΠΡΟΚΛΗΘΕΙ ΤΡΑΥΜΑΤΙΣΜΟΣ ΣΤΟ ΑΤΟΜΟ ΠΑΡΟΧΗΣ ΣΕΡΒΙΣ, ΣΤΟ ΧΕΙΡΙΣΤΗ Ή ΣΤΟΝ ΑΣΘΕΝΗ ΑΠΟ ΗΛΕΚΤΡΟΠΛΗΞΙΑ, ΜΗΧΑΝΙΚΟΥΣ Ή ΑΛΛΟΥΣ ΚΙΝΔΥΝΟΥΣ.

FIGYELMEZTETÉS (HU)

- EZEN KARBANTARTÁSI KÉZIKÖNYV KIZÁRÓLAG ANGOL NYELVEN ÉRHETŐ EL.
- HA A VEVŐ SZOLGÁLTATÓJA ANGOLTÓL ELTÉRŐ NYELVRE TART IGÉNYT, AKKOR A VEVŐ FELELŐSSÉGE A FORDÍTÁS ELKÉSZÍTTETÉSE.
- NE PRÓBÁLJA ELKEZDENI HASZNÁLNI A BERENDEZÉST, AMÍG A KARBANTARTÁSI KÉZIKÖNYVBEN LEÍRTAKAT NEM ÉRTELMEZTÉK.
- EZEN FIGYELMEZTETÉS FIGYELMEN KÍVÜL HAGYÁSA A SZOLGÁLTATÓ, MŰKÖDTETŐ VAGY A BETEG ÁRAMÜTÉS, MECHANIKAI VAGY EGYÉB VESZÉLYHELYZET MIATTI SÉRÜLÉSÉT EREDMÉNYEZHETI.

AÐVÖRUN (IS)	• ÞESSI ÞJÓNUSTUHANDBÓK ER EINGÖNGU FÁANLEG Á ENSKU. • EF AÐ ÞJÓNUSTUVEITANDI VIÐSKIPTAMANNS ÞARFNAST ANNAS TUNGUMÁLS EN ENSKU, ER ÞAÐ SKYLDA VIÐSKIPTAMANNS AÐ SKAFFA TUNGUMÁLAÞJÓNUSTU. • REYNIÐ EKKI AÐ AFGREIÐA TÆKIÐ NEMA AÐ ÞESSI ÞJÓNUSTUHANDBÓKHEFUR VERIÐ SKOÐUÐ OG SKILIN. • BROT Á SINNA ÞESSARI AÐVÖRUN GETUR LEITT TIL MEIÐSLA Á ÞJÓNUSTUVEITANDA, STJÓRNANDA EÐA SJÚKLINGS FRÁ RAFLOSTI, VÉLRÆNU EÐA ÖÐRUM ÁHÆTTUM.
AVVERTENZA (IT)	• IL PRESENTE MANUALE DI MANUTENZIONE E DISPONIBILE SOLTANTO IN INGLESE. • SE UN ADDETTO ALLA MANUTENZIONE ESTERNO ALLA GEMS RICHIEDE IL MANUALE IN UNA LINGUA DIVERSA, IL CLIENTE E TENUTO A PROVVEDERE DIRETTAMENTE ALLA TRADUZIONE. • SI PROCEDA ALLA MANUTENZIONE DELL'APPARECCHIATURA SOLO DOPO AVER CONSULTATO IL PRESENTE MANUALE ED AVERNE COMPRESO IL CONTENUTO. • IL NON RISPETTO DELLA PRESENTE AVVERTENZA POTREBBE FAR COMPIERE OPERAZIONI DA CUI DERIVINO LESIONI ALL'ADDETTO ALLA MANUTENZIONE, ALL'UTILIZZATORE ED AL PAZIENTE PER FOLGORAZIONE ELETTRICA, PER URTI MECCANICI OD ALTRI RISCHI.
警告 (JA)	• このサービスマニュアルには英語版しかありません。 • サービスを担当される業者が英語以外の言語を要求される場合、翻訳作業はその業者の責任で行うものとさせていただきます。 • このサービスマニュアルを熟読し理解せずに、装置のサービスを行わないでください。 • この警告に従わない場合、サービスを担当される方、操作員あるいは患者さんが、感電や機械的又はその他の危険により負傷する可能性があります。
경고 (KO)	• 본 서비스 지침서는 영어로만 이용하실 수 있습니다. • 고객의 서비스 제공자가 영어 이외의 언어를 요구할 경우, 번역 서비스를 제공하는 것은 고객의 책임입니다. • 본 서비스 지침서를 참고했고 이해하지 않는 한은 해당 장비를 수리하려고 시도하지 마십시오. • 이 경고에 유의하지 않으면 전기 쇼크, 기계상의 혹은 다른 위험으로부터 서비스 제공자, 운영자 혹은 환자에게 위해를 가할 수 있습니다.

BRĪDINĀJUMS (LV)

- ŠĪ APKALPES ROKASGRĀMATA IR PIEEJAMA TIKAI ANGĻU VALODĀ.
- JA KLIENTA APKALPES SNIEDZĒJAM NEPIECIEŠAMA INFORMĀCIJA CITĀ VALODĀ, NEVIS ANGĻU, KLIENTA PIENĀKUMS IR NODROŠINĀT TULKOŠANU.
- NEVEICIET APRĪKOJUMA APKALPI BEZ APKALPES ROKASGRĀMATAS IZLASĪŠANAS UN SAPRAŠANAS.
- ŠĪ BRĪDINĀJUMA NEIEVĒROŠANA VAR RADĪT ELEKTRISKĀS STRĀVAS TRIECIENA, MEHĀNISKU VAI CITU RISKU IZRAISĪTU TRAUMU APKALPES SNIEDZĒJAM, OPERATORAM VAI PACIENTAM.

ĮSPĖJIMAS (LT)

- ŠIS EKSPLOATAVIMO VADOVAS YRA PRIEINAMAS TIK ANGLŲ KALBA.
- JEI KLIENTO PASLAUGŲ TIEKĖJAS REIKALAUJA VADOVO KITA KALBA – NE ANGLŲ, NUMATYTI VERTIMO PASLAUGAS YRA KLIENTO ATSAKOMYBĖ.
- NEMĖGINKITE ATLIKTI ĮRANGOS TECHNINĖS PRIEŽIŪROS, NEBENT ATSIŽVELGĖTE Į ŠĮ EKSPLOATAVIMO VADOVĄ IR JĮ SUPRATOTE.
- JEI NEATKREIPSITE DĖMESIO Į ŠĮ PERSPĖJIMĄ, GALIMI SUŽALOJIMAI DĖL ELEKTROS ŠOKO, MECHANINIŲ AR KITŲ PAVOJŲ PASLAUGŲ TIEKĖJUI, OPERATORIUI AR PACIENTUI.

ADVARSEL (NO)

- DENNE SERVICEHÅNDBOKEN FINNES BARE PÅ ENGELSK.
- HVIS KUNDENS SERVICELEVERANDØR TRENGER ET ANNET SPRÅK, ER DET KUNDENS ANSVAR Å SØRGE FOR OVERSETTELSE.
- IKKE FORSØK Å REPARERE UTSTYRET UTEN AT DENNE SERVICEHÅNDBOKEN ER LEST OG FORSTÅTT.
- MANGLENDE HENSYN TIL DENNE ADVARSELEN KAN FØRE TIL AT SERVICELEVERANDØREN, OPERATØREN ELLER PASIENTEN SKADES PÅ GRUNN AV ELEKTRISK STØT, MEKANISKE ELLER ANDRE FARER.

OSTRZEŻENIE (PL)

- NINIEJSZY PODRĘCZNIK SERWISOWY DOSTĘPNY JEST JEDYNIE W JĘZYKU ANGIELSKIM.
- JEŚLI DOSTAWCA USŁUG KLIENTA WYMAGA JĘZYKA INNEGO NIŻ ANGIELSKI, ZAPEWNIENIE USŁUGI TŁUMACZENIA JEST OBOWIĄZKIEM KLIENTA.
- NIE PRÓBOWAĆ SERWISOWAĆ WYPOSAŻENIA BEZ ZAPOZNANIA SIĘ I ZROZUMIENIA NINIEJSZEGO PODRĘCZNIKA SERWISOWEGO.
- NIEZASTOSOWANIE SIĘ DO TEGO OSTRZEŻENIA MOŻE SPOWODOWAĆ URAZY DOSTAWCY USŁUG, OPERATORA LUB PACJENTA W WYNIKU PORAŻENIA ELEKTRYCZNEGO, ZAGROŻENIA MECHANICZNEGO BĄDŹ INNEGO.

ATENÇÃO (PT)

- ESTE MANUAL DE ASSISTÊNCIA TÉCNICA SÓ SE ENCONTRA DISPONÍVEL EM INGLÊS.
- SE QUALQUER OUTRO SERVIÇO DE ASSISTÊNCIA TÉCNICA, QUE NÃO A GEMS, SOLICITAR ESTES MANUAIS NOUTRO IDIOMA, É DA RESPONSABILIDADE DO CLIENTE FORNECER OS SERVIÇOS DE TRADUÇÃO.
- NÃO TENTE REPARAR O EQUIPAMENTO SEM TER CONSULTADO E COMPREENDIDO ESTE MANUAL DE ASSISTÉNCIA TÉCNICA.
- O NÃO CUMPRIMENTO DESTE AVISO PODE POR EM PERIGO A SEGURANÇA DO TÉCNICO, OPERADOR OU PACIENTE DEVIDO A CHOQUES ELÉTRICOS, MECÂNICOS OU OUTROS.

ATENȚIE (RO)

- ACEST MANUAL DE SERVICE ESTE DISPONIBIL NUMAI ÎN LIMBA ENGLEZĂ.
- DACĂ UN FURNIZOR DE SERVICII PENTRU CLIENȚI NECESITĂ O ALTĂ LIMBĂ DECÂT CEA ENGLEZĂ, ESTE DE DATORIA CLIENTULUI SĂ FURNIZEZE O TRADUCERE.
- NU ÎNCERCAȚI SĂ REPARAȚI ECHIPAMENTUL DECÂT ULTERIOR CONSULTĂRII ȘI ÎNȚELEGERII ACESTUI MANUAL DE SERVICE.
- IGNORAREA ACESTUI AVERTISMENT AR PUTEA DUCE LA RĂNIREA DEPANATORULUI, OPERATORULUI SAU PACIENTULUI ÎN URMA PERICOLELOR DE ELECTROCUTARE, MECANICE SAU DE ALTĂ NATURĂ.

ОСТОРОЖНО! (RU)

- ДАННОЕ РУКОВОДСТВО ПО ОБСЛУЖИВАНИЮ ПРЕДЛАГАЕТСЯ ТОЛЬКО НА АНГЛИЙСКОМ ЯЗЫКЕ.
- ЕСЛИ СЕРВИСНОМУ ПЕРСОНАЛУ КЛИЕНТА НЕОБХОДИМО РУКОВОДСТВО НЕ НА АНГЛИЙСКОМ, А НА КАКОМ-ТО ДРУГОМ ЯЗЫКЕ, КЛИЕНТУ СЛЕДУЕТ САМОСТОЯТЕЛЬНО ОБЕСПЕЧИТЬ ПЕРЕВОД.
- ПЕРЕД ОБСЛУЖИВАНИЕМ ОБОРУДОВАНИЯ ОБЯЗАТЕЛЬНО ОБРАТИТЕСЬ К ДАННОМУ РУКОВОДСТВУ И ПОЙМИТЕ ИЗЛОЖЕННЫЕ В НЕМ СВЕДЕНИЯ.
- НЕСОБЛЮДЕНИЕ ТРЕБОВАНИЙ ДАННОГО ПРЕДУПРЕЖДЕНИЯ МОЖЕТ ПРИВЕСТИ К ТОМУ, ЧТО СПЕЦИАЛИСТ ПО ОБСЛУЖИВАНИЮ, ОПЕРАТОР ИЛИ ПАЦИЕНТ ПОЛУЧАТ УДАР ЭЛЕКТРИЧЕСКИМ ТОКОМ, МЕХАНИЧЕСКУЮ ТРАВМУ ИЛИ ДРУГОЕ ПОВРЕЖДЕНИЕ.

UPOZORNENIE (SK)

- TENTO NÁVOD NA OBSLUHU JE K DISPOZÍCII LEN V ANGLIČTINE.
- AK ZÁKAZNÍKOV POSKYTOVATEĽ SLUŽIEB VYŽADUJE INÝ JAZYK AKO ANGLIČTINU, POSKYTNUTIE PREKLADATEĽSKÝCH SLUŽIEB JE ZODPOVEDNOSŤOU ZÁKAZNÍKA.
- NEPOKÚŠAJTE SA O OBSLUHU ZARIADENIA SKÔR, AKO SI NEPREČÍTATE NÁVOD NA OBLUHU A NEPOROZUMIETE MU.
- ZANEDBANIE TOHTO UPOZORNENIA MÔŽE VYÚSTIŤ DO ZRANENIA POSKYTOVATEĽA SLUŽIEB, OBSLUHUJÚCEJ OSOBY ALEBO PACIENTA ELEKTRICKÝM PRÚDOM, DO MECHANICKÉHO ALEBO INÉHO NEBEZPEČENSTVA.

ATENCION (ES)

- ESTE MANUAL DE SERVICIO SOLO EXISTE EN INGLES.
- SI ALGUN PROVEEDOR DE SERVICIOS AJENO A GEMS SOLICITA UN IDIOMA QUE NO SEA EL INGLES, ES RESPONSABILIDAD DEL CLIENTE OFRECER UN SERVICIO DE TRADUCCION.
- NO SE DEBERA DAR SERVICIO TECNICO AL EQUIPO, SIN HABER CONSULTADO Y COMPRENDIDO ESTE MANUAL DE SERVICIO.
- LA NO OBSERVANCIA DEL PRESENTE AVISO PUEDE DAR LUGAR A QUE EL PROVEEDOR DE SERVICIOS, EL OPERADOR O EL PACIENTE SUFRAN LESIONES PROVOCADAS POR CAUSAS ELÉCTRICAS, MECÁNICAS O DE OTRA NATURALEZA.

VARNING (SV)

- DEN HÄR SERVICEHANDBOKEN FINNS BARA TILLGÄNGLIG PÅ ENGELSKA.
- OM EN KUNDS SERVICETEKNIKER HAR BEHOV AV ETT ANNAT SPRÅK ÄN ENGELSKA ANSVARAR KUNDEN FÖR ATT TILLHANDAHÅLLA ÖVERSÄTTNINGSTJÄNSTER.
- FÖRSÖK INTE UTFÖRA SERVICE PÅ UTRUSTNINGEN OM DU INTE HAR LÄST OCH FÖRSTÅR DEN HÄR SERVICEHANDBOKEN.
- OM DU INTE TAR HÄNSYN TILL DEN HÄR VARNINGEN KAN DET RESULTERA I SKADOR PÅ SERVICETEKNIKERN, OPERATÖREN ELLER PATIENTEN TILL FÖLJD AV ELEKTRISKA STÖTAR, MEKANISKA FAROR ELLER ANDRA FAROR.

DİKKAT (TR)

- BU SERVİS KILAVUZUNUN SADECE İNGİLİZCESİ MEVCUTTUR.
- EĞER MÜŞTERİ TEKNİSYENİ BU KILAVUZU İNGİLİZCE DIŞINDA BİR BAŞKA LİSANDAN TALEP EDERSE, BUNU TERCÜME ETTİRMEK MÜŞTERİYE DÜŞER.
- SERVİS KILAVUZUNU OKUYUP ANLAMADAN EKİPMANLARA MÜDAHALE ETMEYİNİZ.
- BU UYARIYA UYULMAMASI, ELEKTRİK, MEKANİK VEYA DİĞER TEHLİKELERDEN DOLAYI TEKNİSYEN, OPERATÖR VEYA HASTANIN YARALANMASINA YOL AÇABİLİR.

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Revision History

Rev.	Date	Primary Reason(s) for Change
1	Oct. 13, 2008	Initial Release for New Program

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Table of Contents

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Chapter 1 Introduction

1 Introduction (Readme First)

1.1 Product Overview

The Signa Oncology Table is a purchased option which provides the customer the ability to acquire image sets with the patient situated in a proposed radiation therapy position. See Illustration 1-1 .

The Signa Oncology Table option is compatible with both 1.5T and 3.0T field strengths and is available to customers that have any of the following system hardware configurations:

- Signa HD
- Signa HDx

This option is used for Oncology imaging only and cannot be used for system calibrations because the flat surface cradle is not compatible with the calibration fixtures and phantoms. The Signa system must be fully calibrated using the primary Patient Transport Table prior to the installation and setup of the Signa Oncology Table.

Illustration 1-1: SIGNA ONCOLOGY TABLE AND CRADLE

1.2 Safety

CAUTION

Potential for injury to service personnel.
The cradles used on the Signa Oncology Table and the Signa OR-Compatible Table weigh approximately 74 lbs (34 kgs).
Use your judgment as a second person may be required to assist in the cradle positioning process.

NOTICE

Follow ALL required safety and PPE procedures customary for your organization when working on this product.

- **Wear gloves for hand protection.**
- **Wear safety glasses for eye protection.**

1.3 Damage In Transportation

All packages should be closely examined at time of delivery. If damage is apparent, have notation **Damage in Shipment** written on **all** copies of the freight or express bill before delivery is accepted or signed for by a General Electric representative or a hospital receiving agent. Whether noted or concealed, damage **MUST** be reported to the carrier **immediately** upon discovery, or in any event, within **14 days** after receipt, and the contents and containers held for inspection by the carrier. A transportation company will not pay a claim for damage if an inspection is not requested within this **14-day** period.

File a report by calling 1-800-548-3366, using Option 8. Contact your local service coordinator for more information on this process.

Rev. 08/15/2003

1.4 Product Locator

The Signa Oncology Table has two rating plates so there are two sets of Product Locator cards included with the shipment.

- The first product locator card matches the rating plate for the table, which is manufactured in India.
- The second product locator matches the modification rating plate for the changes made to the table and the cradle.
  Information from both product locator cards must be entered into the Global Install Base (GIB) database.

The Global Install Base Database (also known as Product Locator System) tracks shipment, trans–shipment, and field location of the serialized models. There are now two methods for submitting Product Locator information.

1. At this time, for **U.S. ONLY**, the preferred method of submitting information is the FE Site Verification Web Site at: http://gib.gehealthcare.com
   The FE Site Verification consists of three components that are available on the web from the main menu. They are:
   - Install/deinstall product locator model and serial numbers
   - Add/modify ship to address information
   - Update CARES FE data for primary/secondary FEs

   To obtain a copy of the FE training tutorial for using this website, a downloadable copy is available at the following: Product Locator Support Central Page – http://supportcentral.ge.com/15563

NOTE: One Shipping Card is filled out and submitted when shipped (extra cards are supplied for trans–shipments between storage and distribution points), and the Installation Card and extra shipment cards are attached.

2. Verify that serial and model number on each rating plate matches the installation card numbers before removing installation cards.

NOTE: There may be one or more shipment cards and bar code labels with the installation card. These shipment cards are used to trace the transfer of serialized units between various inventory storage and distribution points until the product reaches its final installation destination. Process just the installation card and discard any extra shipment cards and labels.

3. Enter the information from the 2 installation cards into the GIB database for this customer.

1.5 Installation And Setup Steering Guide

The Signa Oncology Table utilizes the same hardware interface to the MRI scanner for table docking and cradle attachment to the LPCA, so the majority of the service procedures will be the same. For simplicity the procedures that are required to be performed on this table are included in this document.

The following steering guide is intended to assist you for the installation, setup and service maintenance for this table.

1.5.1 Scanner Interface Prerequisites

Make sure the scanner interface hardware and adjustments have been setup using the customer's primary Signa Lightweight Patient Transport Table. Refer to the Service Methods DVD that was shipped with the system.

- Leveling the Patient Transport
- Table Top Height Adjustment
- Dock Centering
- Dock Hook Tension Adjustment
- Dock Pedal Stroke Adjustment
- Longitudinal Drive Motor Alignment to Field
- Longitudinal Drive Clutch Tension Adjustment
- Longitudinal Drive Calibration
- Cradle Release from Patient Transport Adjustment
- Cradle Release from Carriage Adjustment

1.5.2 Signa Oncology Table Process

For simplicity, the setup procedures that are required for this table are included in this document.

1. Perform the following setup procedures on the Signa Oncology Table.
   - Chapter 2, Section 2, Leveling the Patient Transport
   - Chapter 2, Section 3, Table Top Height Adjustment

- Chapter 2, Section 4, Removing Cradle Assembly
- Chapter 2, Section 5, Dual Table Alignment Procedure
- Chapter 2, Section 6, Dock Hook Tension Adjustment
- Chapter 2, Section 7, Dock Pedal Stroke Adjustment
- Chapter 2, Section 8, Cradle Release from Patient Transport Adjustment
- Chapter 2, Section 9, Cradle Release from Carriage Adjustment
- Chapter 2, Section 10, Cradle Guide Rail Latch Adjustment
- Chapter 2, Section 11, Cradle Home Table Down Interlock Adjustment
- Chapter 2, Section 12, Cradle Release Block Adjustment
- Chapter 2, Section 13, Cradle Release Pin Height Adjustment

2. Perform the following Functional Check procedures:
   - Chapter 3, Section 2, Cradle Motion Test
   - Chapter 3, Section 3, Patient Transport Movement Verification

Chapter 2 Adjustment and Calibration

1 Introduction

These procedures are applicable to the following table products:

- Signa Lightweight Patient Transport Table
- Signa Oncology Patient Transport Table Option
- Signa OR-Compatible Patient Transport Table Option

2 Leveling Patient Transport

2.1 Personnel Requirements

Required Persons	Preliminary Reqs	Procedure	Finalization
1	Not Applicable	30 mins	Not Applicable

2.2 Overview

This procedure is applicable to the following table products:

- Signa Lightweight Patient Transport Table
- Signa Oncology Patient Transport Table Option
- Signa OR-Compatible Patient Transport Table Option

The Patient Transport is leveled to the magnet by adjusting caster height; however, caster height must also be checked for uniform loading.

2.3 Preliminary Requirements

2.3.1 Tools and Test Equipment

Item	Qty	Effectivity	Part#	Manufacturer
Torque Wrench, ratchet type, 3/8 inch drive, 30-200 in-lb.	1	-	46-194427P255	-
Socket Driver, hex Allen head, 5/32 inch bit, 3/8 inch drive.	1	-	46-307811p1	-
Straightedge/Level 2 ft long, non-magnetic	1	-		-
4 mm Allen Wrench	1	-	-	-

2.3.2 Safety

WARNING

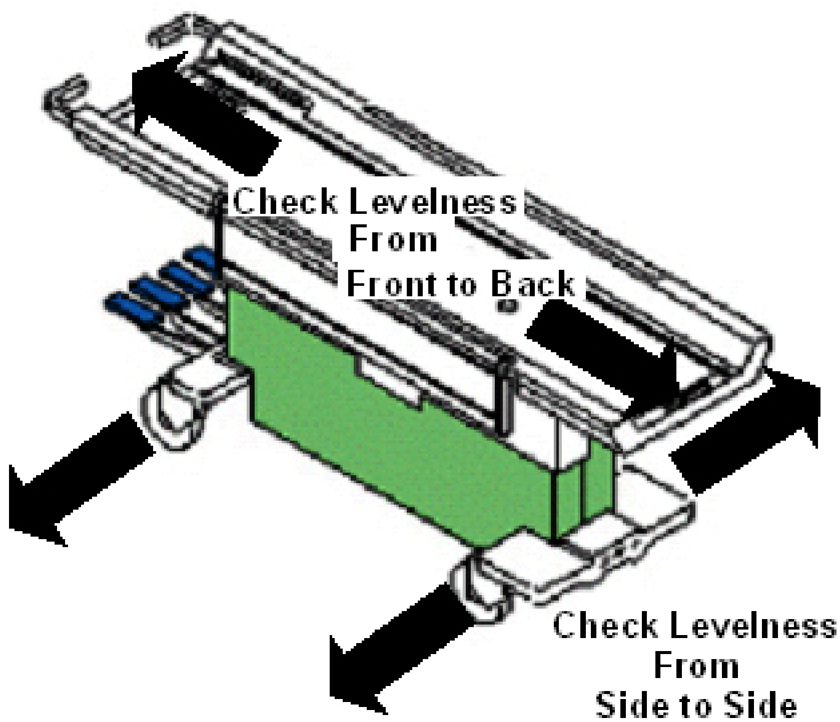
FLYING PROJECTILE DANGER!
FERROUS MATERIAL HAZARD !! TORQUE WRENCH AND SOCKET DRIVE CONTAIN FERROUS MATERIAL AND CAN BECOME FORCIBLY ATTRACTED TO THE MAGNET.
DO NOT BRING THESE TOOLS INTO THE MAGNET ROOM.

2.4 Procedure

2.4.1 Adjusting Patient Transport for Floor Levelness

1. The first step in leveling Patient Transport is to adjust casters to compensate for an uneven floor at the front of the magnet.
2. Dock the Patient Transport. If docking mechanism is not properly adjusted yet, position the Patient Transport in the location it will be in when it is docked.
3. Lock the caster brakes.
4. With table top in fully raised position, check Patient Transport for levelness front-to-back at table top level and left-to-right at the front and rear base castings. See Illustration 2-1 .

Illustration 2-1: CHECKING LEVELNESS OF PATIENT TRANSPORT



5. Check that all casters are sharing the load approximately equally by applying a small lifting force at each corner of the table and seeing if the caster easily lifts off the floor.
6. If one caster, or two diagonally mounted casters, seem to not be loaded as much as the other casters, they will require adjustment.

2.4.2 Checking Caster Height Adjustment

1. To help keep track of measurement data, temporarily label each caster as 1, 2, 3, and 4. Suggestion: label "steering" caster (green colored locking tab) as 1, and proceed labeling in a clockwise direction.

2. At each of the four caster wheel assemblies, measure the height from floor to top of aluminum support casting (see Illustration 2-2 , measurement A). Record this measurement in Table 2-1 , in MEASUREMENT A column labeled 1ST HEIGHT. This height should be 11.0 inches ±0.250 inches (279.4 mm ±6.4 mm). If 1ST HEIGHT MEASUREMENT A is not within tolerance, calculate amount by which it is out of tolerance ("-" if caster height is too small; "+" if height is too large) and record this value in 1ST DELTA column under MEASUREMENT A.

Illustration 2-2: CASTER MEASUREMENTS

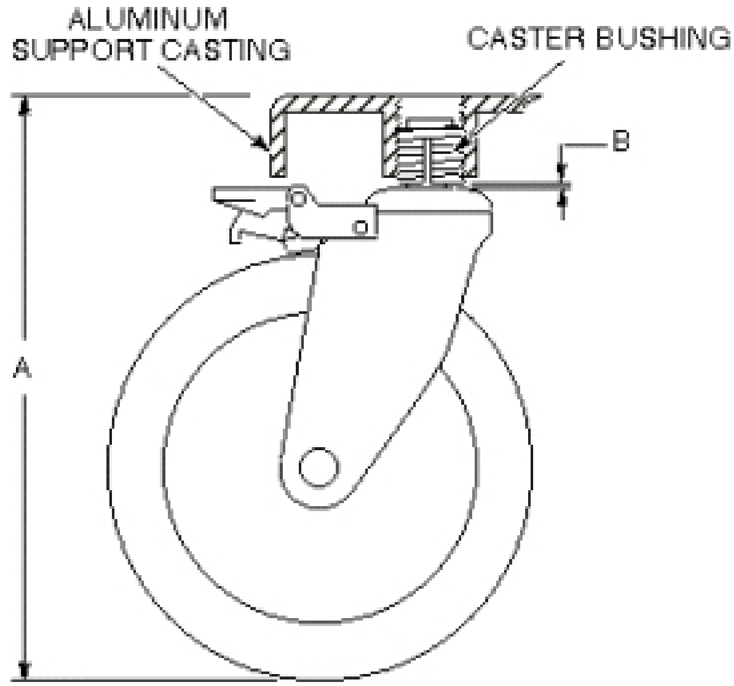


Table 2-1: MEASUREMENTS

	MEASUREMENT A				MEASUREMENT B	
	1ST		2ND			
CASTER	HEIGHT	DELTA +/-	HEIGHT	DELTA +/-	GAP	DELTA +/-
1						
2						
3						
4						

NOTE: Tolerance for MEASUREMENT A: 11.0 ± 0.250 inch

Tolerance for MEASUREMENT B: <0.250 inch

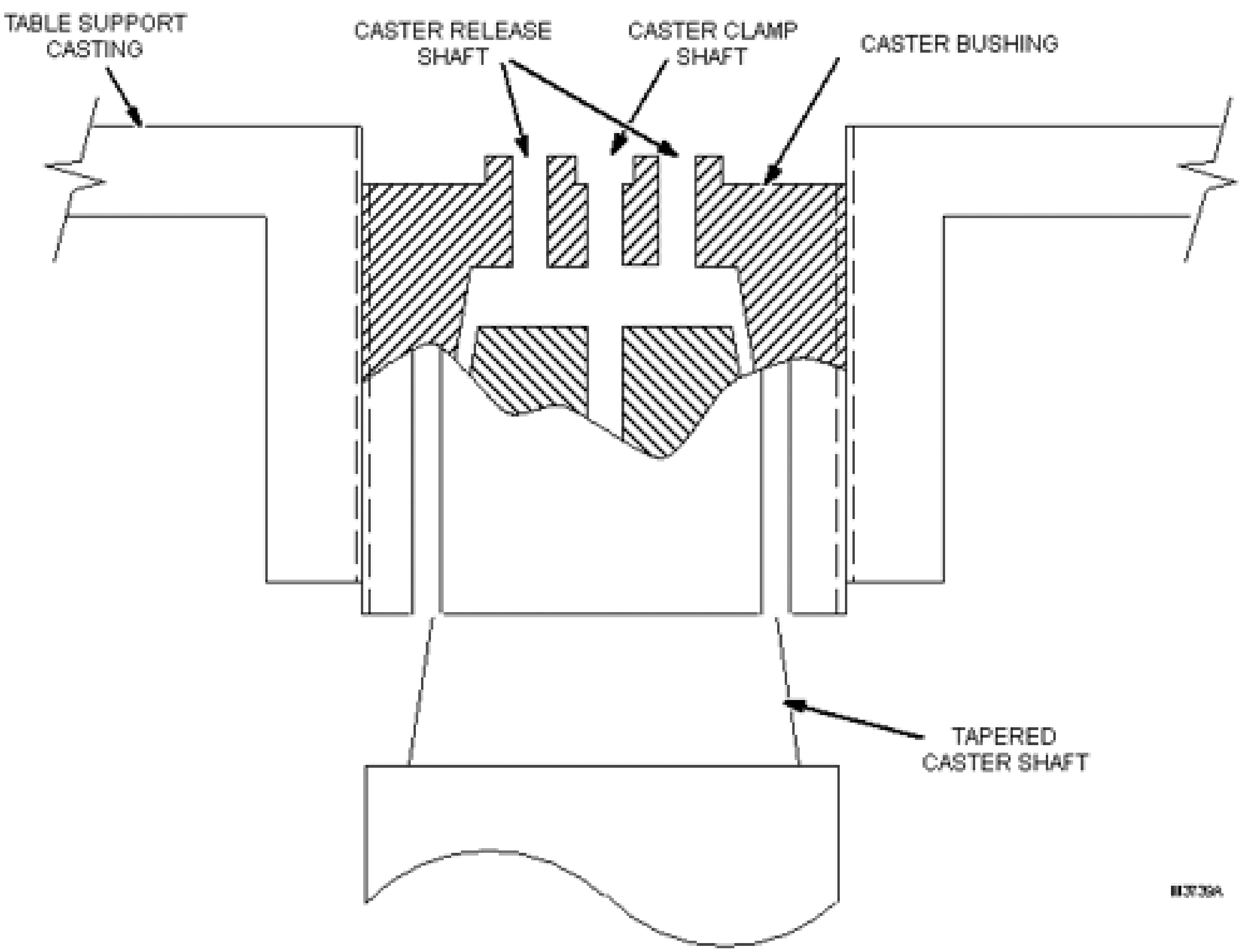
3. At each of the four caster wheel assemblies, measure the gap between bottom of caster bushing and base of caster shaft (see Illustration 2-2 , measurement B). Record this measurement in Table 2, in MEASUREMENT B column labeled GAP. The acceptable measurement for this gap is less than 0.250 inches (6.4 mm). If gap is not within tolerance, calculate amount by which it is out of tolerance ("-" if gap is too small; "+" if gap is too large, and record this value in DELTA column under MEASUREMENT B.

2.4.3 Caster Mounting Design

When the caster bushing can turn freely inside the aluminum support casting, the bushing provides a means of adjusting caster height.

1. The Caster Bushing is slotted and has a tapered hole to accept the Tapered Caster Shaft (see Illustration 2-3).

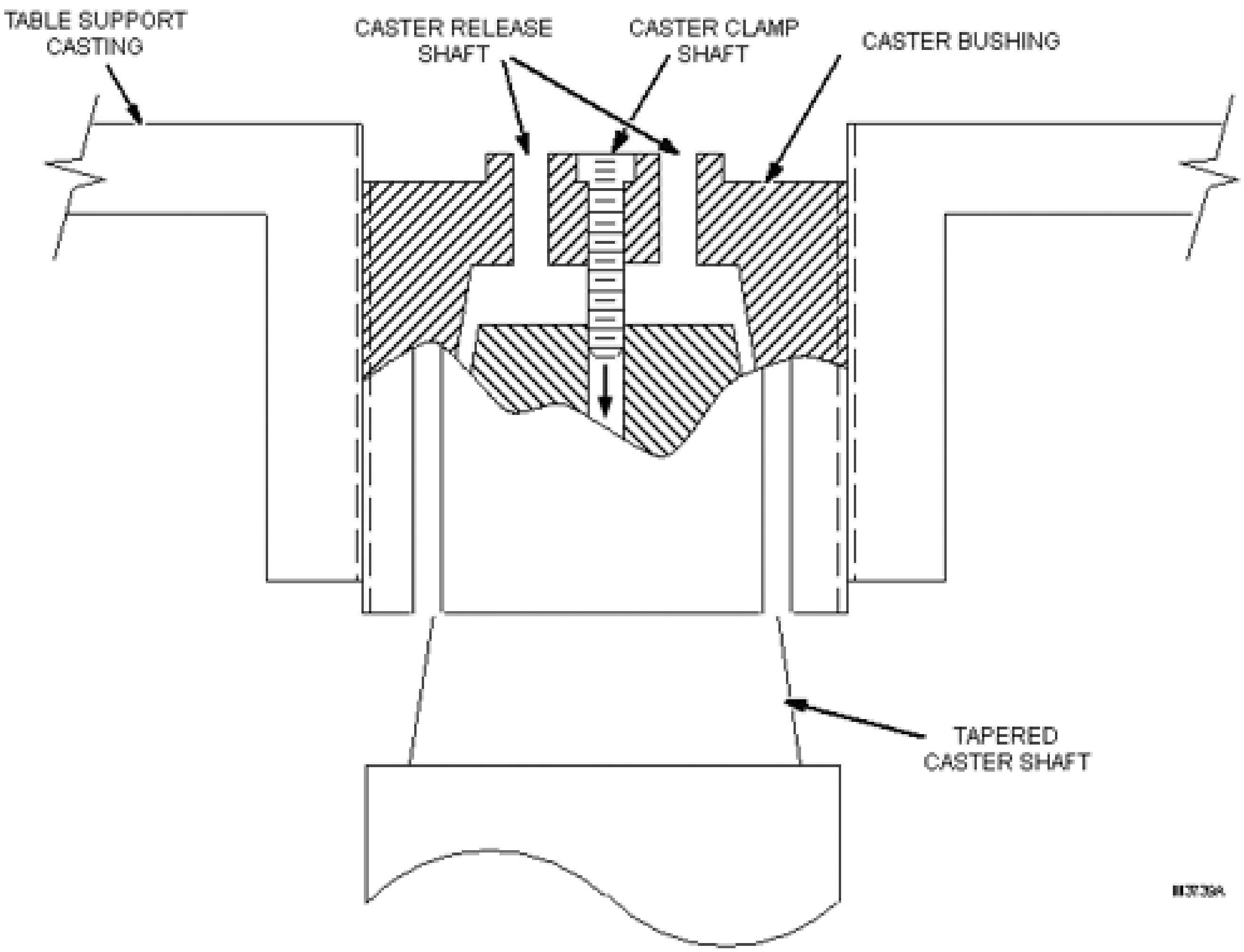
Illustration 2-3: CASTER BUSHING AND TAPERED CASTER SHAFT



2. When a caster screw is inserted into the Caster Clamp Shaft and tightened securely, the screw allows the weight of the table to draw the shaft up into the Caster Bushing.

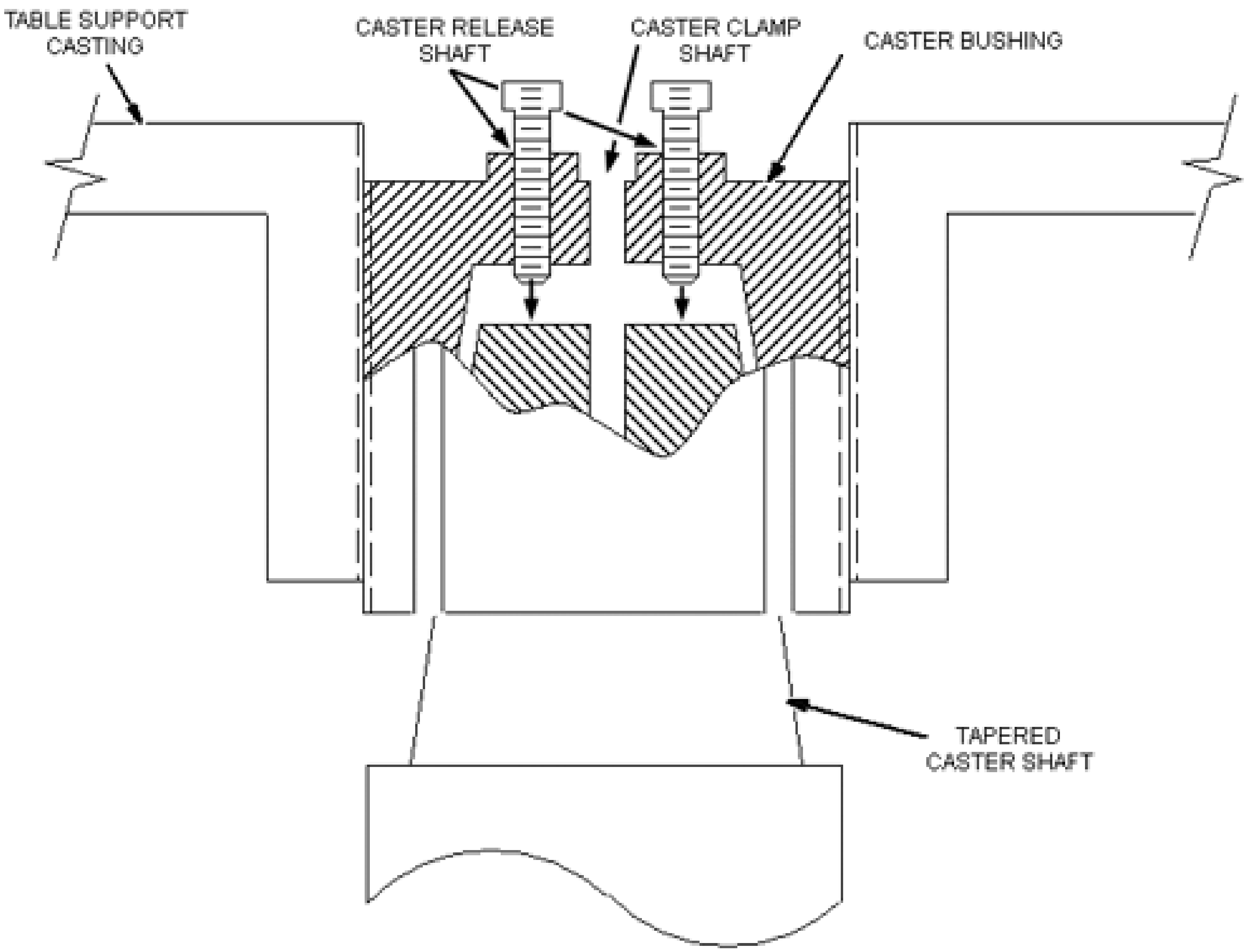
This causes the bushing to spread tightly against the aluminum support casting threads, preventing the bushing from turning and changing the height adjustment (see Illustration 2-4).

Illustration 2-4: SECURING SCREW IN CASTER CLAMP SHAFT



3. To adjust caster height, the screw is removed from the Caster Clamp Shaft and placed in one of the other Caster Release shafts. To fill the second Caster Release shaft, remove the screw from the Caster Clamp shaft of one of the adjacent casters and place that screw into the other Caster Release shaft (see Illustration 2-5).

Illustration 2-5: SCREWING SCREWS IN CASTER RELEASE SHAFTS



4. As the screws are screwed into the Caster Release shafts, they push the caster shaft out of its tapered fit in the caster bushing, leaving the bushing loose enough to turn and provide caster height adjustment.

5. After adjustment, the screws must be removed from the Caster Release shafts and returned to their original Caster Clamp shaft to prevent contact with the caster shaft. Once in the Caster Clamp shaft, the screw can be adequately tightened, securing the caster shaft and immobilizing the bushing (see Illustration 2-4).

2.4.4 Problem Conditions, Causes, and Solutions

There are two conditions that could cause the caster assembly to come loose and possibly fall off the Patient Transport. These two conditions, their causes, and solutions are described below.

1. Problem Conditions and Causes

a. The tapered caster shaft can become loose and back out of the caster bushing. This condition can be caused if a screw was left in the Caster Release shaft, the screw could be pushing against the Tapered Caster shaft, preventing the shaft from seating securely into the bushing.

b. The caster bushing can back out (partially or completely) of its threaded hole in the support casting. If the tapered caster shaft becomes loose (as described in condition 1), or if the caster bushing has been improperly positioned during caster height adjustment (correct bushing positioning will be detailed in - Patient Transport Leveling Procedure) the bushing can back out of the aluminum support casting.

2. Solutions:

a. When it becomes necessary to force the tapered caster shaft out of the caster bushing during an adjustment procedure, two screws will be used in the vacant caster release shafts. One of the caster clamp screws will come from the caster being adjusted, and one will be borrowed from another caster assembly (see Section 2.4.3, Step 3).

b. A 50 in-lb torque requirement for the caster clamp screw. This torque on the caster clamp screw will result in sufficient tightness of the caster bushing inside the aluminum support casting to prevent the bushing from backing out when the caster assembly swivels.

2.4.5 Patient Transport Leveling Procedure

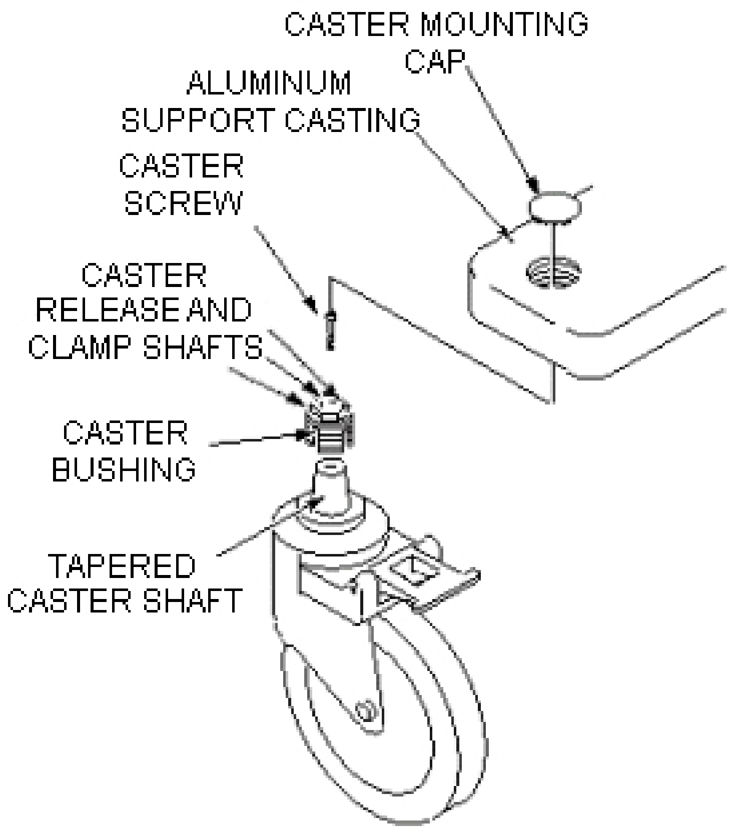
WARNING

FLYING PROJECTILE DANGER!
FERROUS MATERIAL HAZARD !! TORQUE WRENCH AND SOCKET DRIVE CONTAIN FERROUS MATERIAL AND CAN BECOME FORCIBLY ATTRACTED TO THE MAGNET.
DO NOT BRING THESE TOOLS INTO THE MAGNET ROOM.

1. The torque wrench and socket driver cannot be brought into magnet room. Undock Patient Transport and move it out of magnet room.

2. On first caster to be adjusted, remove Caster Mounting Cap by prying it off with a small screwdriver. See Illustration 2-6 .

Illustration 2-6: CASTER WHEEL ASSEMBLY



NOTE: Adjustments which exceed tolerances given in the steps below leave caster bushing vulnerable to damage, which can result in injury to patient and/or persons near Patient Transport.

3. For each caster whose Measurement B is outside the required tolerance, twist the bushing to adjust Measurement B the amount noted in Table 2-1 .

4. For each caster whose Measurement A is within tolerance, as determined in Table 2-1 proceed to Step 20 . Otherwise, proceed to the next step for Caster Height Adjustment.

5. On first caster to be adjusted, remove the screw from Caster Clamp shaft, and thread the screw into one of the empty Caster Release shafts. Turn Caster Clamp Screw in finger tight only.

6. Go to another caster assembly and remove the screw from the Caster Clamp shaft. Place the screw into the Caster Release shaft of the caster being adjusted.

7. Turn second screw finger tight.

8. Alternately turn the screws one turn at a time until caster shaft is as loose as possible in caster bushing.

9. Using a means of leverage, slightly prop up aluminum support casting at caster being adjusted only enough to take Patient Transport weight off caster wheel. This will

disengage caster shaft from caster bushing and prevent excessive wear on threads during adjustment.

10. Remove both screws from the two Caster Release shafts.

11. Adjust caster height an amount equal to the value recorded in 1ST DELTA column of MEASUREMENT A or MEASUREMENT B, whichever pertains, in Table 2-1 . While holding the caster wheel, use a 19 mm socket to turn caster bushing counter-clockwise to raise table corner (for "-" DELTA value), or clockwise to lower table corner (for "+" DELTA value). Each complete revolution of caster bushing raises or lowers caster 0.083 inches (2.1 mm).

12. Remove prop from under aluminum support casting.

13. Return one screw to the Caster Clamp shaft in its original caster assembly.

14. At caster being adjusted, insert remaining screw into Caster Clamp shaft.

15. Using the Torque Wrench (46-194427P255) and Socket Driver (46-307811P1), torque Caster Clamp Screw to 50 inch-pounds. If necessary, refer to operating instructions supplied with the torque wrench.

16. Verify that caster shaft is seated well into caster bushing and gap B is still within tolerance defined in Table 2-1 .

17. Move Patient Transport back into magnet room and align to dock but do not actuate the docking pedal.

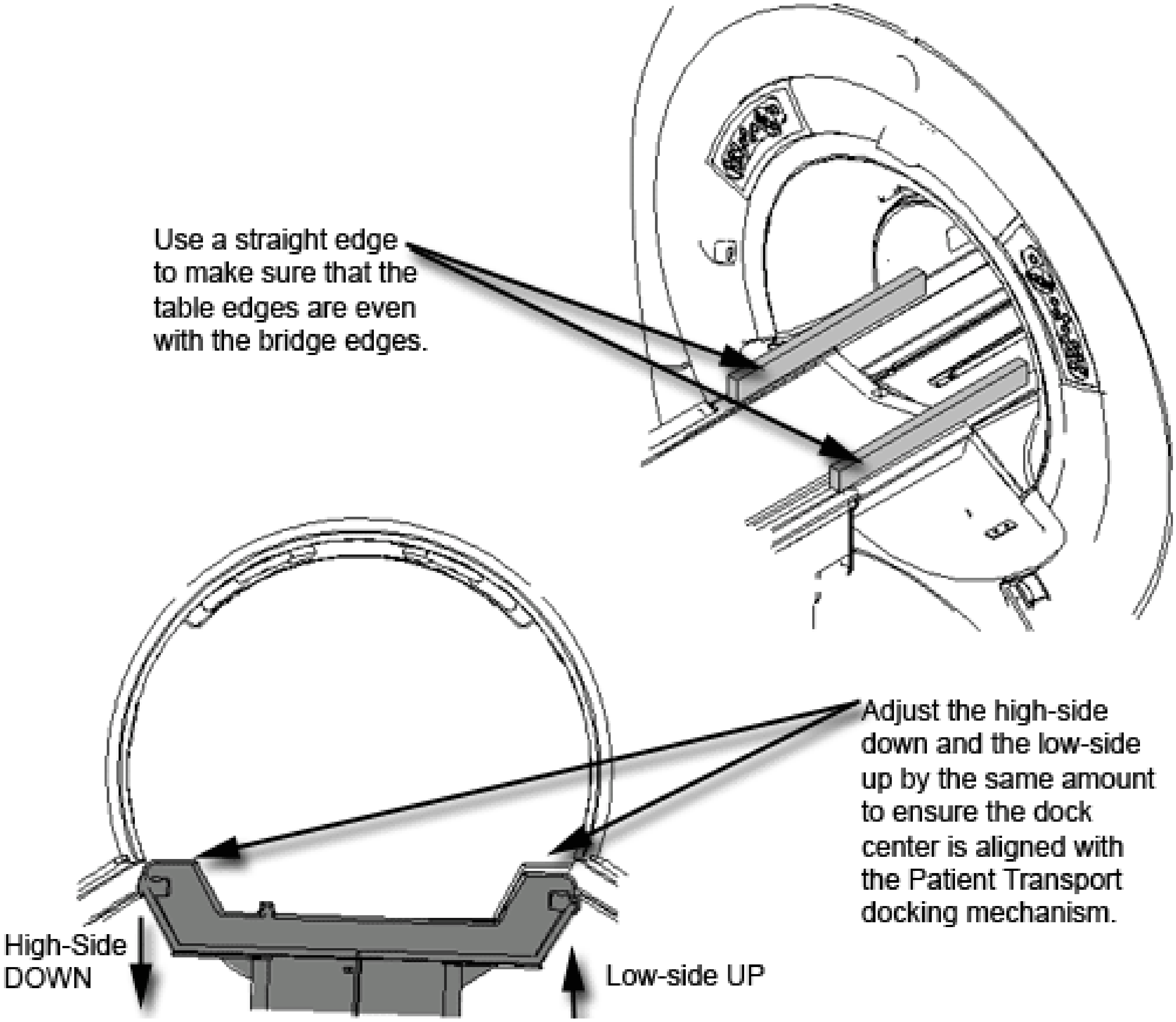
18. For each caster, re-measure height A and record its value in Table 2-1 , in MEASUREMENT A column labeled 2ND HEIGHT. If this 2ND value of A is out of tolerance, calculate amount by which it is out of tolerance, and record this amount in 2ND DELTA column under MEASUREMENT A.

19. For each caster height still out of tolerance, adjust height as determined by value of 2ND DELTA in Table 2-1 , by performing procedure beginning with .

20. Move Patient Transport back into the magnet room. Align to dock, but do NOT actuate the docking pedal.

21. Align Patient Transport surface (rotationally) to bridge (which should be flush to 1 mm below end bell surfaces per Bridge Removal and Replacement procedure) by raising the lower side casters. Be sure to also lower the higher side casters by the same amount, as this will ensure the Patient Transport top will not be shifted to one side of the bridge when docked and the dock center will align with the Patient Transport docking mechanism. See Illustration 2-7 .

Illustration 2-7: ALIGNING PATIENT TRANSPORT SURFACE TO BRIDGE



22. When all caster height measurements are within tolerance and Patient Transport top is aligned to bridge surface, verify each caster has one screw in caster clamp shaft and is torqued to 50 inch-pounds. Then, replace all caster mounting caps.

23. If necessary, now complete Dock Centering procedure.

24. Do not use casters to adjust Patient Transport top height. Follow the Table Top Height Adjustment procedure.

3 Table Top Height Adjustment

3.1 Personnel Requirements

Required Persons	Preliminary Reqs	Procedure	Finalization
1	Not Applicable	15 mins	Not Applicable

3.2 Overview

This procedure is applicable to the following table products:

- Signa Lightweight Patient Transport Table
- Signa Oncology Patient Transport Table Option
- Signa OR-Compatible Patient Transport Table Option

3.3 Preliminary Requirements

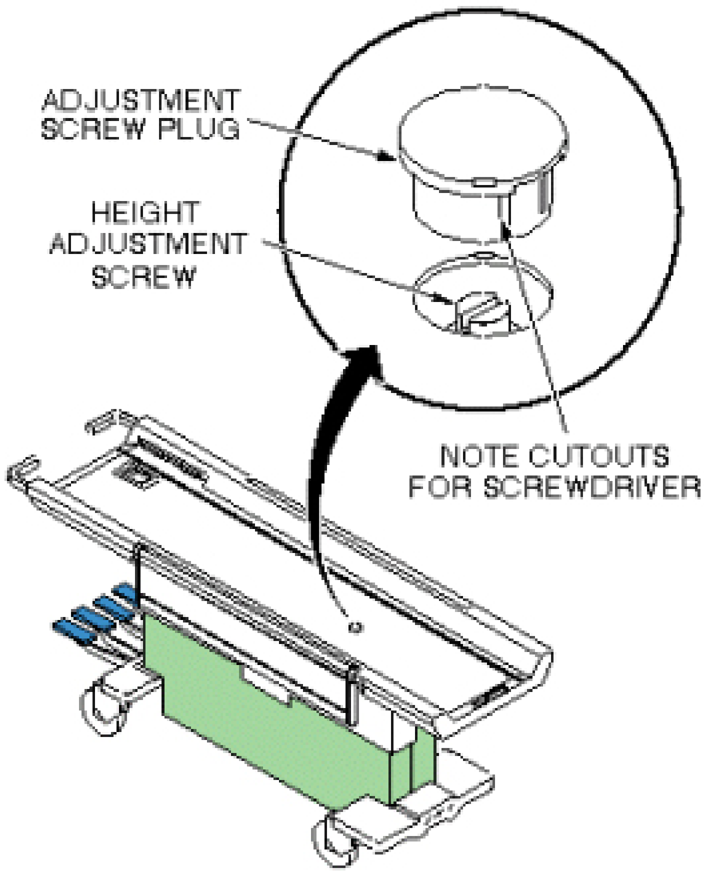
3.3.1 Tools and Test Equipment

Item	Qty	Effectivity	Part#	Manufacturer
Non Magnetic Tool Kit (Flat Head Screwdriver)	1	-	46-320273G4	-

3.4 Procedure

1. Dock Patient Transport. If Dock has not yet been mounted to magnet, complete Dock Centering procedure.
2. Pump table top to fully raised position and move the cradle completely into the magnet bore to expose the table top height adjusting screw.
3. Use a small screwdriver to carefully remove cover plug on table top. See Illustration 2-8 .

Illustration 2-8: REMOVING COVER PLUG



4. Adjust screw in table top to set table top height equal to bridge height.
5. Replace cover plug.

4 Removing Cradle Assembly

4.1 Personnel Requirements

Required Persons	Preliminary Reqs	Procedure	Finalization
1 or 2, depending on system	Not Applicable	10 mins	Not Applicable

4.2 Overview

This procedure is applicable to the following table products:

- Signa Lightweight Patient Transport Table
- Signa Oncology Patient Transport Table Option
- Signa OR-Compatible Patient Transport Table Option

The cradle removal procedure requires:

- One person for a Signa Lightweight Patient Transport Table
- Two people for a Signa Oncology or Signa OR-Compatible Patient Transport Table Option

4.3 Preliminary Requirements

4.3.1 Safety

DANGER

STRONG MAGNETIC FIELD!
FERROUS MATERIALS CAN BECOME DANGEROUS PROJECTILES IN THE PRESENCE OF THE MAGNETIC FIELD PRODUCED BY THE SIGNA MAGNET. DO NOT BRING ANY FERROMAGNETIC TOOLS OR EQUIPMENT INTO THE MAGNET ROOM.

WARNING

RISK OF PERSONAL INJURY!
LIFTING MR PARTS, SUCH AS FRUS, TOOLKITS AND PARTS OF THE SYSTEM WITHOUT ASSISTANCE (MECHANICAL OR ADDITIONAL PERSONNEL), CAN RESULT IN PERSONAL INJURY.
THE USE OF EXTRA PERSONNEL IS RECOMMENDED TO LIFT AN OBJECT WEIGHING MORE THAN 35 POUNDS AND LESS THAN 50 POUNDS, BUT REQUIRED FOR OBJECTS WEIGHING MORE THAN 50 POUNDS. THE USE OF MECHANICAL LIFT ASSISTANCE DEVICES IS REQUIRED FOR OBJECTS WEIGHING MORE THAN 90 POUNDS.

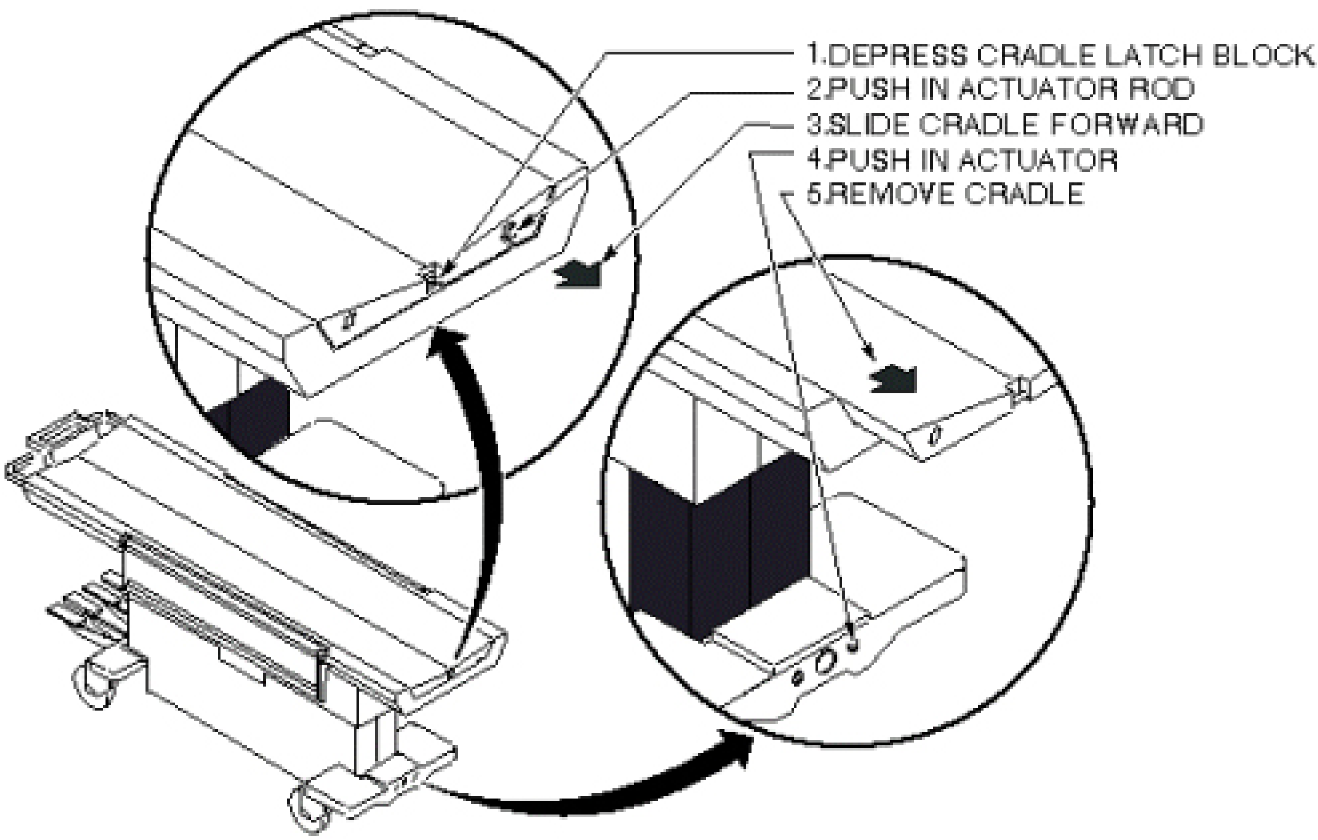
WARNING

RISK OF PERSONAL INJURY!
LIFTING MR PARTS, SUCH AS FRUS, TOOLKITS AND PARTS OF THE SYSTEM WITHOUT ASSISTANCE (MECHANICAL OR ADDITIONAL PERSONNEL), CAN RESULT IN PERSONAL INJURY.
THE USE OF EXTRA PERSONNEL IS RECOMMENDED TO LIFT AN OBJECT WEIGHING MORE THAN 35 POUNDS AND LESS THAN 50 POUNDS, BUT REQUIRED FOR OBJECTS WEIGHING MORE THAN 50 POUNDS. THE USE OF MECHANICAL LIFT ASSISTANCE DEVICES IS REQUIRED FOR OBJECTS WEIGHING MORE THAN 90 POUNDS.

4.4 Procedure

1. Undock patient transport.
2. At the same time, depress cradle latch block at front end of transport and push in actuator rod at front of cradle. See Illustration 2-9 .

Illustration 2-9: PATIENT TRANSPORT CRADLE REMOVAL



3. While holding these actuators, slide cradle forward on patient transport approximately three inches. The actuators can now be released.
4. Push in right actuator in base at front end of transport and slide cradle the rest of the way off the patient transport. See Illustration 2-9 .

5 Dual Table Alignment Procedure

5.1 Personnel Requirements

Required Persons	Preliminary Reqs	Procedure	Finalization
1 or 2, depending on system	Not Applicable	2 hours	Not Applicable

5.2 Overview

These procedures are applicable to the following table products:

- Signa Lightweight Patient Transport Table
- Signa Oncology Patient Transport Table Option
- Signa OR-Compatible Patient Transport Table Option

The alignment procedure requires:

- One person for a Signa Lightweight Patient Transport Table
- Two people for a Signa Oncology or Signa OR-Compatible Patient Transport Table Option

5.3 Preliminary Requirements

5.3.1 Tools and Test Equipment

Item	Qty	Effectivity	Part#	Manufacturer
Non Magnetic Tool Kit	1	-	-	-
Non-ferrous plumb bob	1	-	-	-
Non-ferrous measuring tool. (minimum 12” [300 mm])	1	-	-	-
Non-ferrous level	1	-	-	-

5.3.2 Consumables

Item	Qty	Effectivity	Part#	Manufacturer
Marking tape - 12” (300 mm)	1	-	-	-

5.3.3 Safety

WARNING

STRONG MAGNETIC FIELDS!
FERROUS HAND TOOLS CAN BECOME DANGEROUS PROJECTILES IN THE VICINITY OF THE SYSTEM MAGNET.
DO NOT USE ANY MAGNETIC/FERROUS TOOLS OR EQUIPMENT INSIDE THE MAGNET ROOM.

WARNING

RISK OF PERSONAL INJURY!
LIFTING MR PARTS, SUCH AS FRUS, TOOLKITS AND PARTS OF THE SYSTEM WITHOUT ASSISTANCE (MECHANICAL OR ADDITIONAL PERSONNEL), CAN RESULT IN PERSONAL INJURY.
THE USE OF EXTRA PERSONNEL IS RECOMMENDED TO LIFT AN OBJECT WEIGHING MORE THAN 35 POUNDS AND LESS THAN 50 POUNDS. THE USE OF EXTRA PERSONNEL IS REQUIRED FOR OBJECTS WEIGHING MORE THAN 90 POUNDS.

WARNING

RISK OF PERSONAL INJURY!
LIFTING MR PARTS, SUCH AS FRUS, TOOLKITS AND PARTS OF THE SYSTEM WITHOUT ASSISTANCE (MECHANICAL OR ADDITIONAL PERSONNEL), CAN RESULT IN PERSONAL INJURY.
THE USE OF EXTRA PERSONNEL IS RECOMMENDED TO LIFT AN OBJECT WEIGHING MORE THAN 35 POUNDS AND LESS THAN 50 POUNDS. THE USE OF EXTRA PERSONNEL IS REQUIRED FOR OBJECTS WEIGHING MORE THAN 90 POUNDS.

5.3.4 Required Conditions

Condition	Reference	Effectivity
Review the following procedures prior to completing this procedure: Section 2, Aligning Patient Transport, Dock Centering, and Section 3, Table Top Height Adjustment.	-	-
Review the following procedures prior to completing this procedure: Dock Centering, Table Top Height Adjustment	-	-

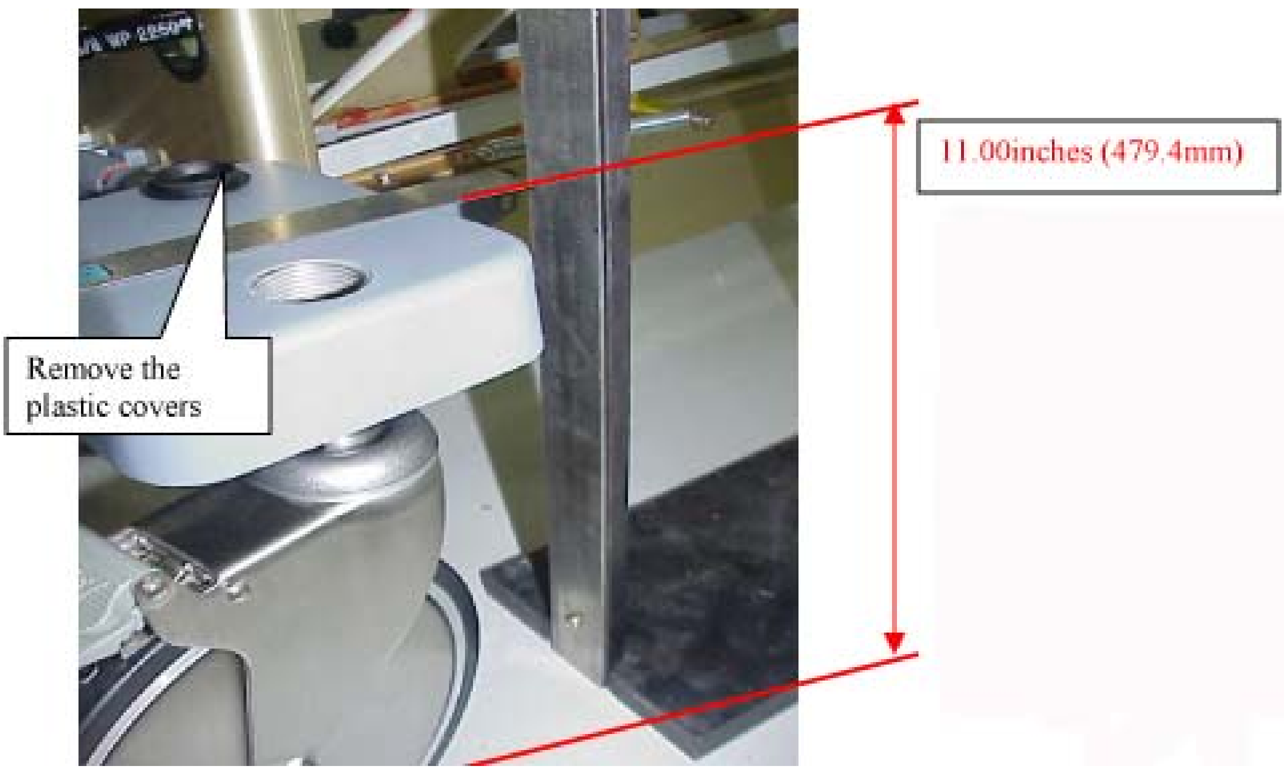
5.4 Procedure

1. Remove the cradle from both tables, see Section 4, Removing Cradle Assembly. (see Illustration 2-10).

Illustration 2-10: REMOVE CRADLES FROM TABLES

2. Remove the plastic covers from the castings. (see Illustration 2-11).

Illustration 2-11: CASTER COVER AND HEIGHT



3. Center the dock assembly with the bridge, at the front of the magnet "Loosely attach the Dock assembly to the front of the magnet. Do not secure the Dock to the magnet or floor. Centering will be accomplished in Step 6 ." (see Illustration 2-12).

Illustration 2-12: ATTACH TABLE DOCK

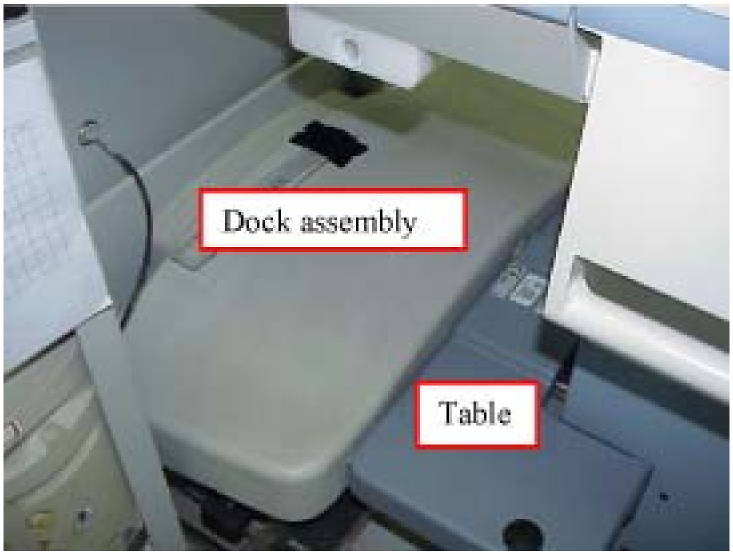
4. Center the Dock assembly by using a Plumb Bob (Non magnetic material). (see Illustration 2-13).

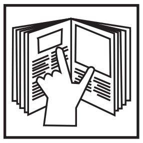
Illustration 2-13: CENTER THE TABLE

5. Use the $^{3}/_{4}$” socket to adjust both front casters of table such that the top of the casting above the wheel is 11.00 (279.4mm) inches from the floor. (see Illustration 2-11)

6. Position the table to the dock, taking care not to move the dock. DO NOT actuate the docking pedal. (see Illustration 2-14).

Illustration 2-14: DOCK TABLE





NOTICE

Over adjusting the caster adjustment will lead to caster being damaged. Do not exceed the +/- 0.25” (6.35mm) tolerance.

7. With table #1 fully elevated, align the table top to the bridge (which should be flush to 1 mm below the end bells per the Bridge Removal and Replacement procedure) by adjusting the front two casters, see Illustration 2-15 . Raise and lower the casters as needed while keeping the center (between the two casters) close to the 11” (279.4mm) height. REMEMBER, the casters should never be adjusted more than +/- .25” (6.35mm) from the 11” (279.4mm) height. (see Illustration 2-16). (Adjusting the high side down and the low side up by the same amount ensures the dock center will stay aligned with the Patient Transport docking mechanism.)

Illustration 2-15: ALIGNING PATIENT TRANSPORT SURFACE TO BRIDGE

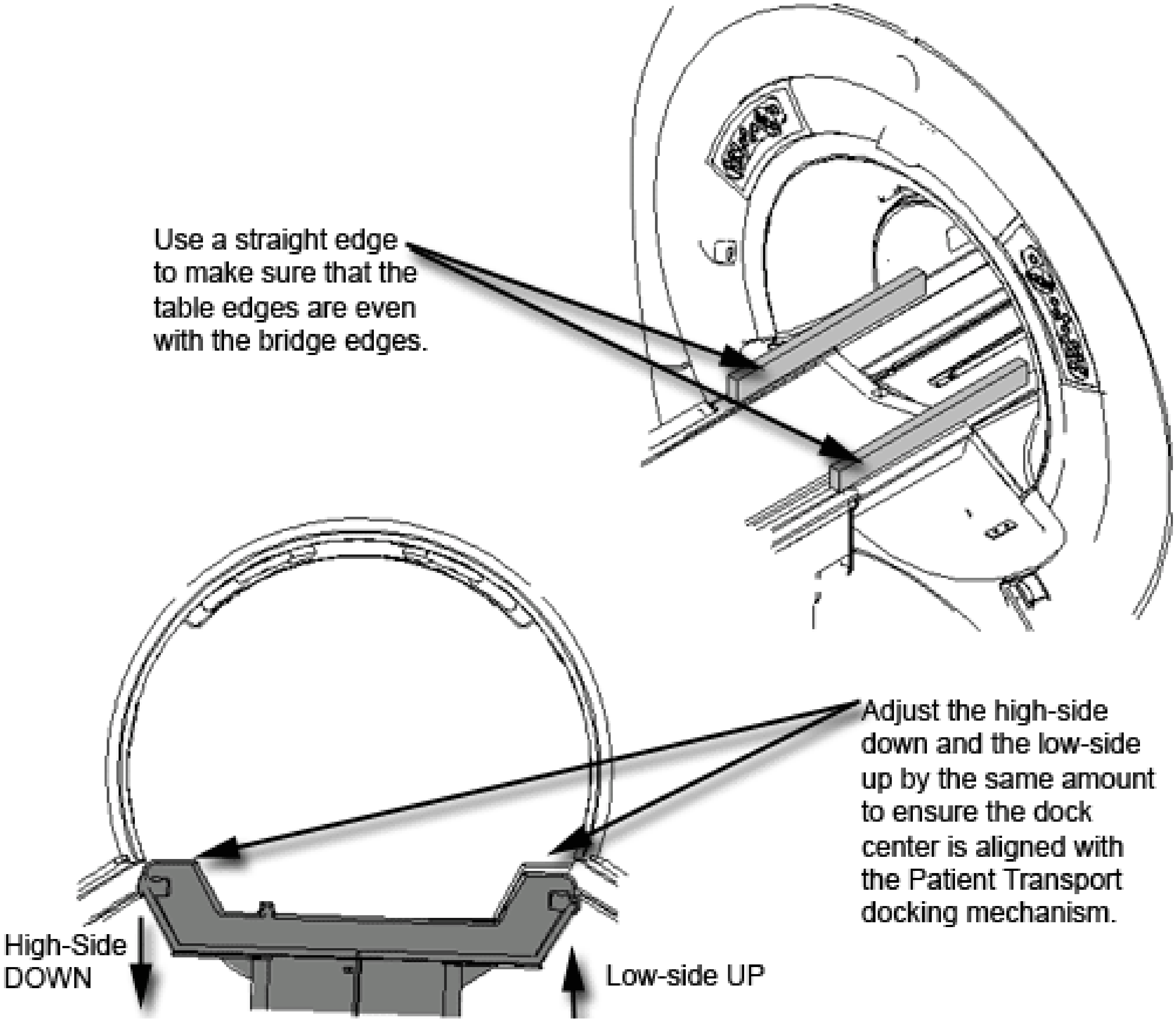
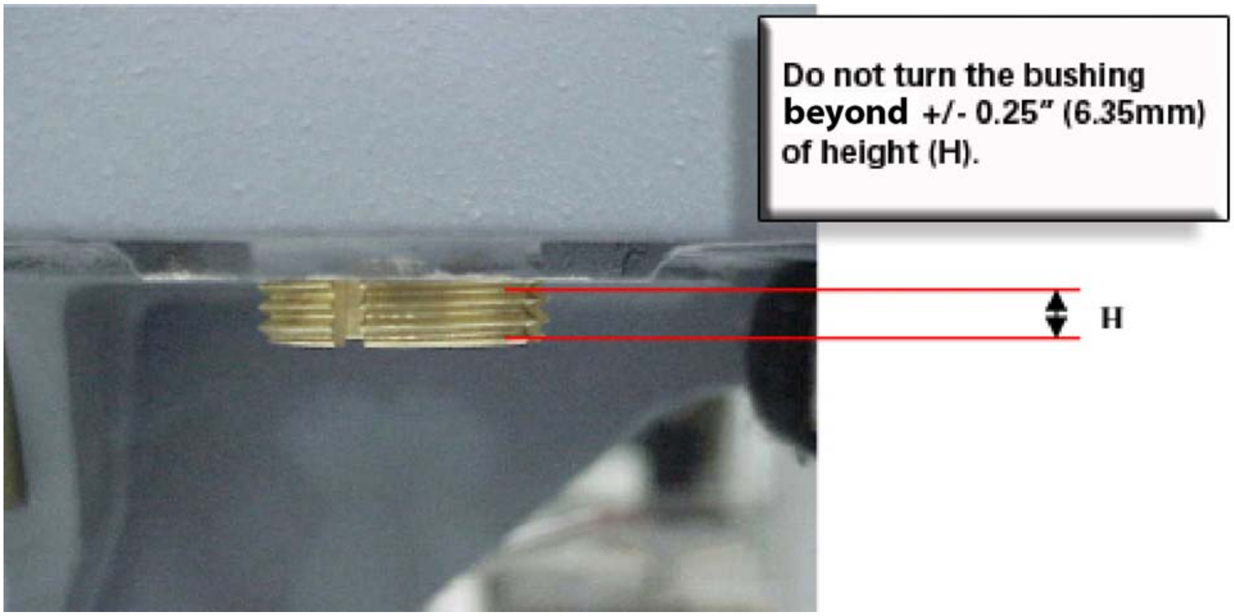
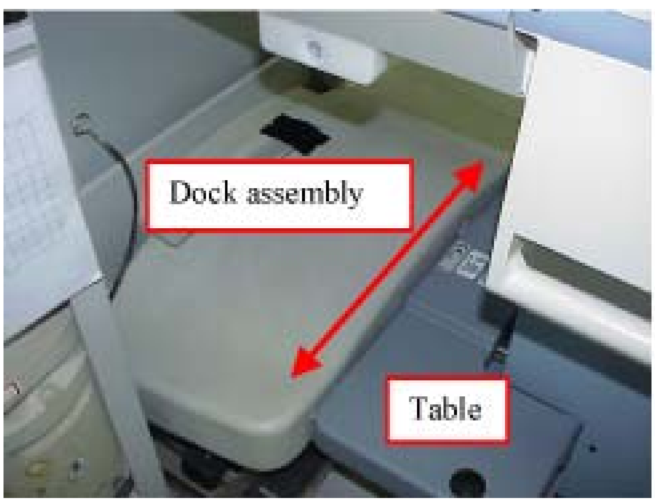


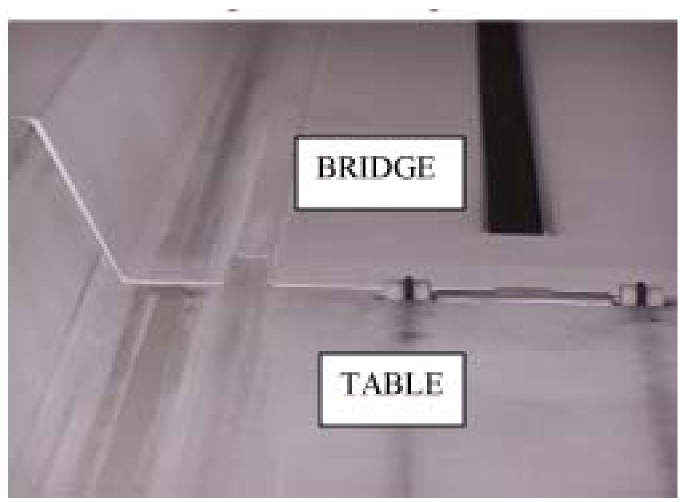
Illustration 2-16: CASTER ADJUSTMENT WARNING



8. Move the table and dock left or right until the table rail and the bridge rail are in alignment. See Illustration 2-17 .

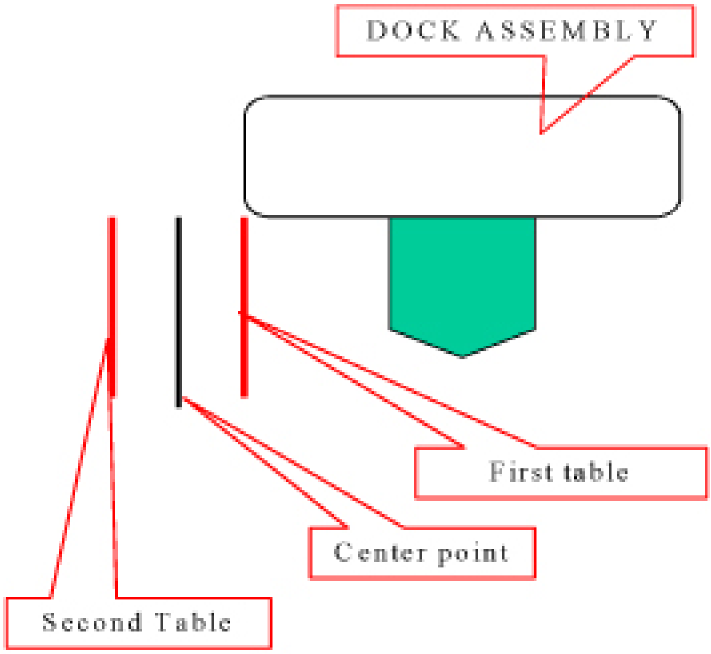
Illustration 2-17: ALIGN BRIDGE AND CRADLE





9. Place a mark on the floor adjacent to the side of the dock. (tape)
10. Repeat steps Step 6 to Step 9 for table #2 and then continue with Step 10.a .
    a. Mark the center point between the 2 marks. (see Illustration 2-18).
    b. Shift the dock assembly to align with the center point mark.

Illustration 2-18: EXAMPLE OF RESULTS FROM Step 6 THROUGH Step 9



11. Secure the dock assembly in position.

12. Complete the final lateral alignment by adjusting the front two casters if necessary. Raise and lower the casters as needed while keeping the center (between the two casters) close to the 11” (279.4mm) height. REMEMBER, the casters should not be adjusted more than +/- .25” (6.35mm). (see Illustration 2-16).

13. Tighten the Bushing Screw at on each caster.

6 Dock Hook Tension Adjust

6.1 Personnel Requirements

Required Persons	Preliminary Reqs	Procedure	Finalization
1	Not Applicable	30 mins	Not Applicable

6.2 Overview

This procedure is applicable to the following table products:

- Signa Lightweight Patient Transport Table
- Signa Oncology Patient Transport Table Option
- Signa OR-Compatible Patient Transport Table Option

6.3 Preliminary Requirements

6.3.1 Tools and Test Equipment

Item	Qty	Effectivity	Part#	Manufacturer
Standard SAE Allen Wrench Set	1	-	-	-
SAE Socket Set or Open-end Wrench	1	-	-	-

6.4 Procedure

1. Attempt to dock patient transport, by means of dock pedal, paying close attention to the force required to engage the docking mechanism. Initial docking should not require any force to depress docking pedal.

2. Observe contact between patient transport front casting and the rubber bumpers on dock casting. Adjust dock hook until there is no gap or compression of bumpers. (see Illustration 2-20). Note: Patient transport must be un-docked for all hook adjustments. Loosen lock nut first then loosen set screw, tighten in reverse order.

3. Once Step 2 is complete, un-dock patient transport, adjust hook to proper tension by turning hook 3 full turns clockwise. Tighten setscrew and the lock nut.

4. Attempt to dock patient transport paying close attention to force required to depress docking pedal. If force seems excessive, STOP and adjust hook 1 full turn counter clockwise and repeat this step. Dock should have a distinctive snap when hook fully engages docking mechanism. There should not be a spring like resistance to the foot pedal, DO NOT FORCE pedal to engage. See Illustration 2-19 .

5. When table is properly adjusted, there shall be no separation between the bumpers and patient transport front casting while table is fully elevated by means of depressing up pedal, on the dock, enabling dock motor to run beyond the maximum table height (motor runs and table stops).

6. If the force to dock seems excessive, STOP and loosen hook. Verify that the problem symptoms indicated in Table 2-2 is present.

Table 2-2: Dock Hook Mis-Adjustment Symptoms

SYMPTOM	CAUSE	ACTION
Excessive force is required todepress dock pedal and fullyengage dock latch. See Illustration 2-19 .	Hook tension too tight	Loosen hook one turn and retest

NOTE: Hook tension adjustment is very sensitive. Never adjust hook more than one turn at a time without retesting. Table should be undocked when making Hook Adjustments. Locknut must be tightened before attempting to dock.

Illustration 2-19: DOCK HOOK TENSION CHECK

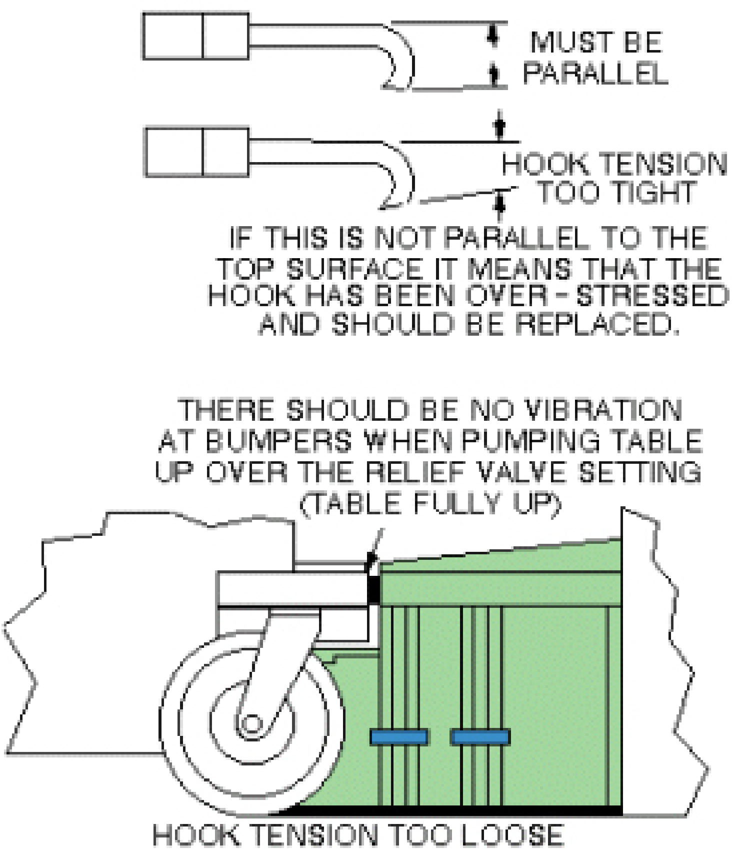
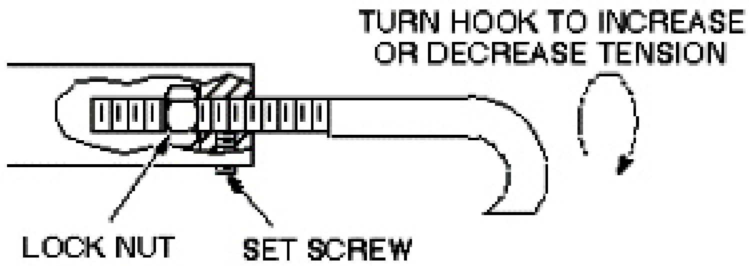


Illustration 2-20: DOCK HOOK TENSION ADJUST



NOTE: Do not adjust the dock hook tension more than one turn at a time. Adjusting more than one turn at a time may cause damage to internal Patient Transport mechanisms.

7 Dock Pedal Stroke Adjustment

7.1 Personnel Requirements

Required Persons	Preliminary Reqs	Procedure	Finalization
1	Not Applicable	30 mins	Not Applicable

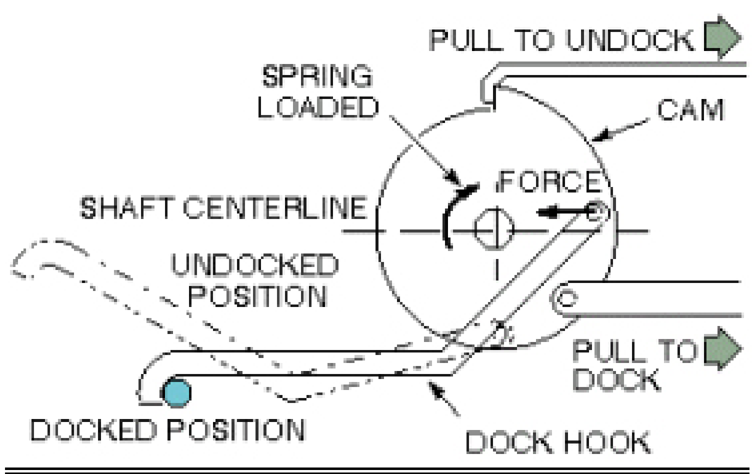
7.2 Overview

This procedure is applicable to the following table products:

- Signa Lightweight Patient Transport Table
- Signa Oncology Patient Transport Table Option
- Signa OR-Compatible Patient Transport Table Option

The table dock hook operates on the over-center principle. This is implemented by fixing one end of the hook to a cam on a shaft. When the force on the hook at its point of attachment to the cam is on the opposite side of the shaft as the hook end, this force will tend to keep the hook in the docked position. When the point of force is moved over-center of the shaft centerline, the force will no longer act to keep the hook retracted, and the spring loaded cam will extend the hook to the undocked position. See Illustration 2-21 . Two style table adjustments are currently in the field. The type table you have is determined by manufacturing date. Choose the procedure appropriate for your table style.

Illustration 2-21: DOCKING MECHANISM DIAGRAM



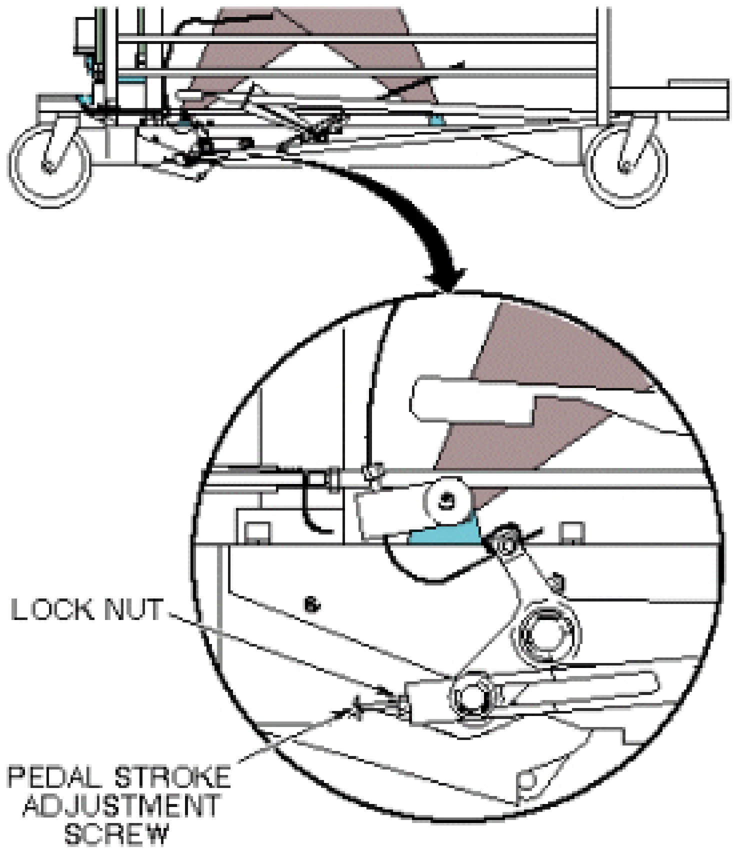
7.3 Procedure

7.3.1 Tables Manufactured Before April 2002 (Signa Lightweight Patient Transport Table ONLY)

1. Remove all outer and inner trim covers from sides of Patient Transport.

2. Tighten or loosen the stroke adjustment screw as required so when the dock pedal is fully depressed during the docking operation, the hook cam mechanism just goes over-center. See Illustration 2-22 .

Illustration 2-22: DOCK PEDAL STROKE ADJUSTMENT



7.3.2 Tables Manufactured After April 2002 (Signa Lightweight, Signa Oncology, and Signa OR-Compatible Patient Transport Tables)

1. Remove the hatch at the rear of the table for adjusting the undock cable. See Illustration 2-23 .

Illustration 2-23: HATCH REMOVAL

2. Loosen the Screw (Counter clock wise) until the table undocks easily when the pedal is pushed. (This will tighten the cable pulling the lever down). See Illustration 2-24 .

Illustration 2-24: CABLE TENSION ADJUSTMENT

3. When finished, tighten the lock nut and reattach the cover. See Illustration 2-23 .

8 Cradle Release from Patient Transport Adjustment

8.1 Personnel Requirements

Required Persons	Preliminary Reqs	Procedure	Finalization
1	Not Applicable	30 mins	Not Applicable

8.2 Overview

This procedure is applicable to the following table products:

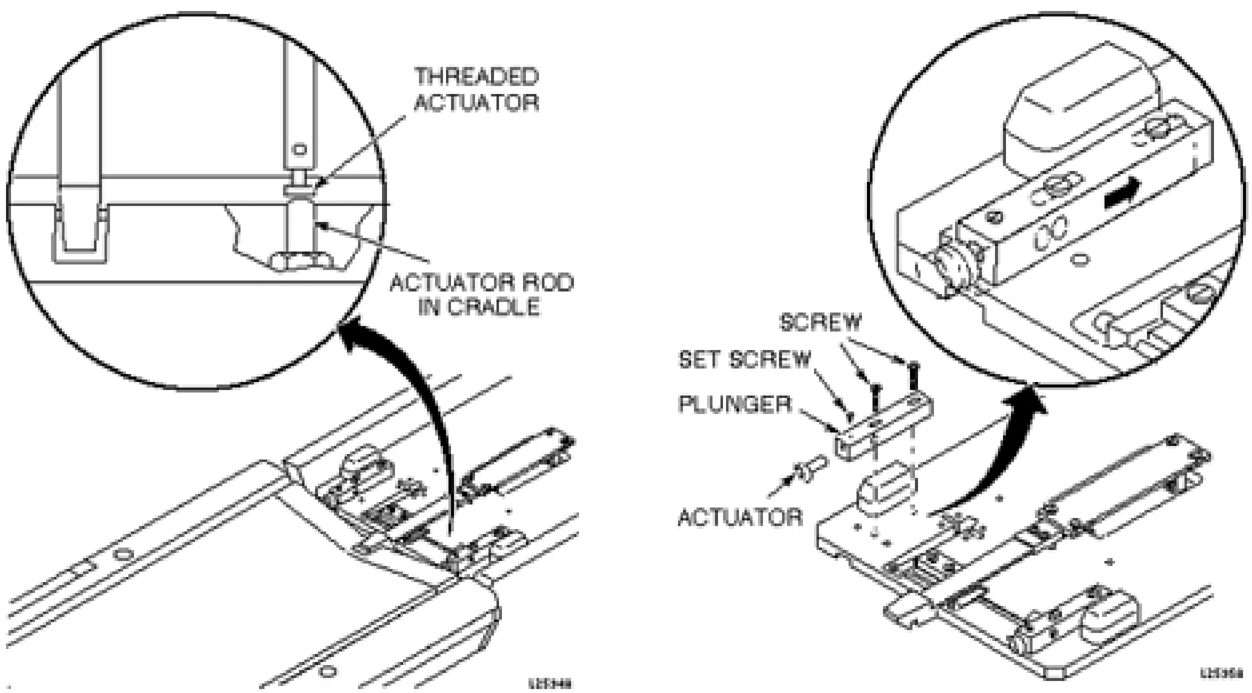
- Signa Lightweight Patient Transport Table
- Signa Oncology Patient Transport Table Option
- Signa OR-Compatible Patient Transport Table Option

For systems with a Lightweight Patient Transport, the actuator on the longitudinal drive carriage must be set so that it presses a button on the cradle to mechanically release the cradle from the patient transport, allowing it to move forward into the magnet bore.

8.3 Procedure

1. To properly adjust the cradle release, slide the plunger back as far as it can go from the front of the carriage. (See Illustration 2-25 .)

Illustration 2-25: CRADLE RELEASE FROM TRANSPORT ADJUSTMENT



2. Loosen the set screw and rotate the threaded actuator in or out as needed, then tighten the set screw.

9 Cradle Release from Carriage Adjustment

9.1 Personnel Requirements

Required Persons	Preliminary Reqs	Procedure	Finalization
1	Not Applicable	Not Applicable	Not Applicable

9.2 Overview

This procedure is applicable to the following table products:

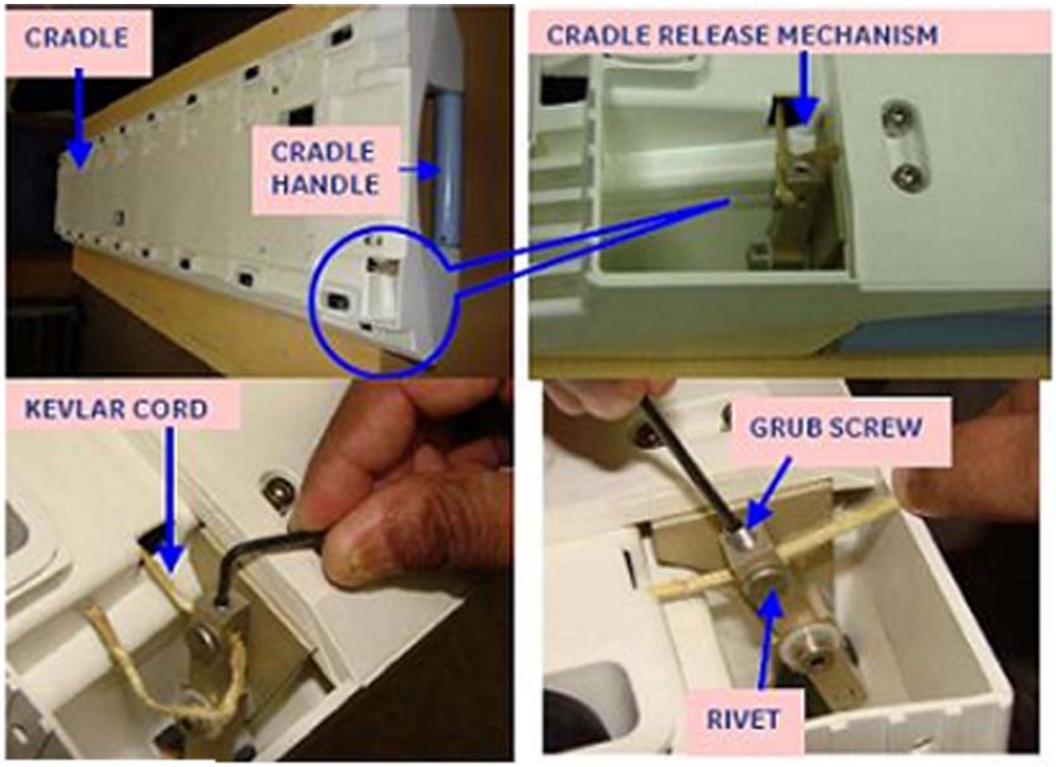
- Signa Lightweight Patient Transport Table
- Signa Oncology Patient Transport Table Option
- Signa OR-Compatible Patient Transport Table Option

The head coil carriage mechanism allows the cradle to be released from the head coil carriage by squeezing the handle of the patient cradle. The threaded actuator on the carriage must be adjusted so that the carriage latch hook engages cradle sufficiently, yet allows hook to be raised to release cradle.

9.3 Procedure

1. Remove head coil carriage cover.
2. Dock patient transport.
3. Latch carriage to cradle.
4. Check that the carriage meets the following criteria:
   a. Carriage latch hook must be low enough to positively engage cradle.
   b. Carriage latch hook must be high enough to be disengaged from cradle when cradle handle is squeezed or twisted.
5. If needed, adjust Cradle Release Mechanism Tension (see Illustration 2-26). Refer to Cradle Release Tension Adjustment and Front Guide Roller Height Adjustment procedure.

Illustration 2-26: CRADLE RELEASE FROM CARRIAGE



10 Cradle Guide Rail Latch Adjustment

10.1 Personnel Requirements

Required Persons	Preliminary Reqs	Procedure	Finalization
1	Not Applicable	15 mins	Not Applicable

10.2 Overview

This procedure is applicable to the following table products:

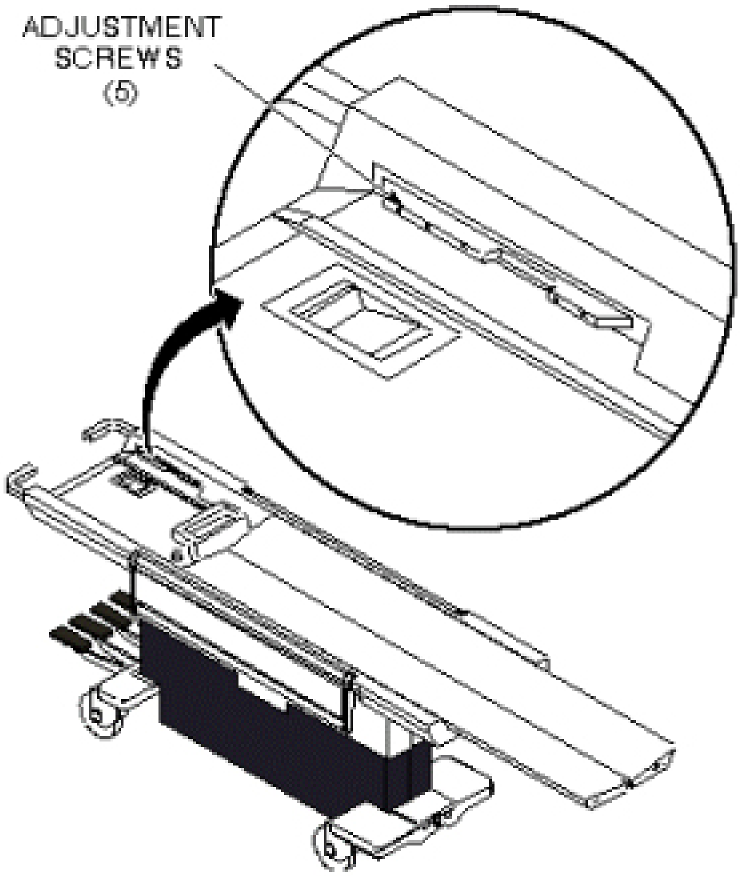
- Signa Lightweight Patient Transport Table
- Signa Oncology Patient Transport Table Option
- Signa OR-Compatible Patient Transport Table Option

The cradle is locked in home position on the Patient Transport by a latch in the cradle which engages a notch in the cradle guide strip near the end of the Patient Transport. The cradle should be able to just engage this notch, without extra force, when pushed to home position on the table top, yet it must engage it tight enough so the table top will not move on the Patient Transport when it is undocked.

10.3 Procedure

1. Refer to procedure in Section 4, Removing Cradle Assembly, to partially remove cradle from table top.
2. Loosen five screws securing guide rail and slide guide rail forward or backward as required. Tighten screws. See Illustration 2-27 .

Illustration 2-27: CRADLE GUIDE RAIL LATCH MOD



11 Cradle Home Table Down Interlock Adjustment

11.1 Personnel Requirements

Required Persons	Preliminary Reqs	Procedure	Finalization
1	Not Applicable	15 mins	Not Applicable

11.2 Overview

This procedure is applicable to the following table products:

- Signa Lightweight Patient Transport Table
- Signa Oncology Patient Transport Table Option
- Signa OR-Compatible Patient Transport Table Option

This adjustment is for the mechanical interlock which prevents the Patient Transport from being undocked or lowered while the cradle is in the magnet bore.

11.3 Procedure

1. Remove all upper and lower trim covers from sides of Patient Transport.
2. With cradle NOT HOME (removed), check the interlock mechanisms for tolerances shown in Illustration 2-28 . To adjust:
    a. 1/16 to 1/32 inch (1.59 to 0.79 mm) gap - carefully bend interlock rod. See Illustration 2-28 .
    b. 1/8 inch (3.18 mm) gap - loosen lock nut and tighten or loosen the cable adjustment screw to achieve the proper gap. Tighten lock nut. See Illustration 2-29 .

Illustration 2-28: CRADLE HOME/TABLE DOWN INTERLOCK

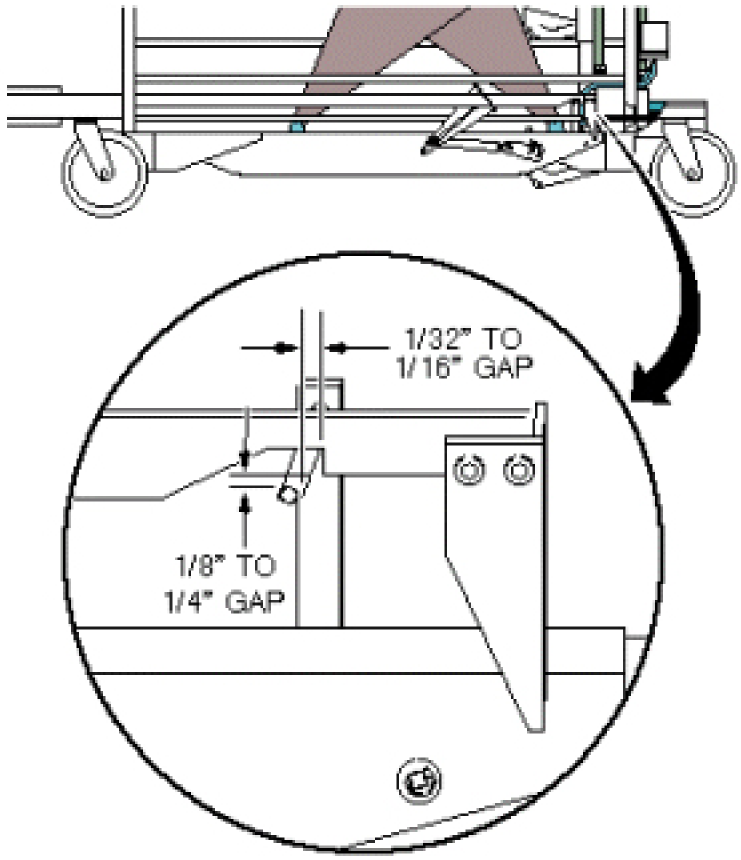
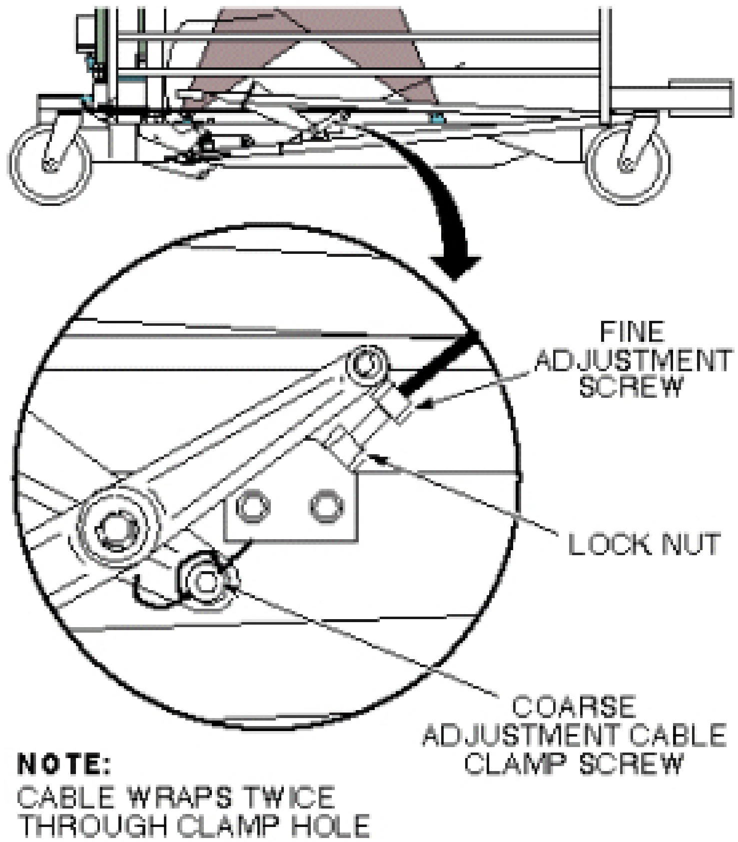


Illustration 2-29: CRADLE HOME/TABLE DOWN INTERLOCK



NOTE: If the adjustment screw does not provide enough range for adjustment, coarse adjustment may be made to the cable length where the cable end is clamped into actuator bracket. Fine adjustment will be required after performing a coarse adjustment.

12 Cradle Release Block Adjustment

12.1 Personnel Requirements

Required Persons	Preliminary Reqs	Procedure	Finalization
1	Not Applicable	Not Applicable	Not Applicable

12.2 Overview

This procedure is applicable to the following table products:

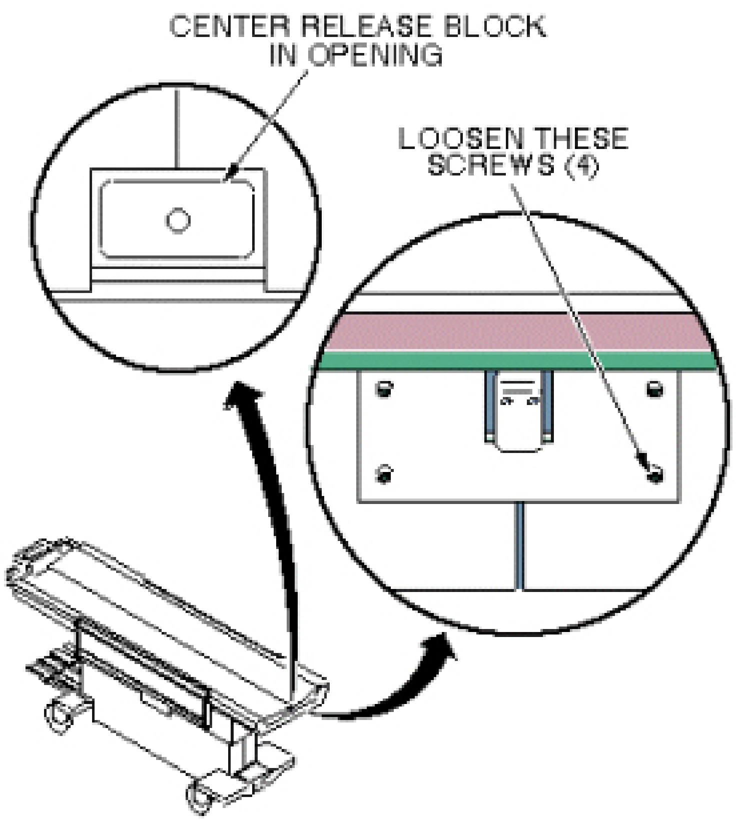
- Signa Lightweight Patient Transport Table
- Signa Oncology Patient Transport Table Option
- Signa OR-Compatible Patient Transport Table Option

12.3 Procedure

12.3.1 Cradle Release Block Position Adjustment

1. Loosen four (4) hex socket head screws securing the release block assembly. See Illustration 2-30 .

Illustration 2-30: CRADLE RELEASE BLOCK POSITION ADJUSTMENT

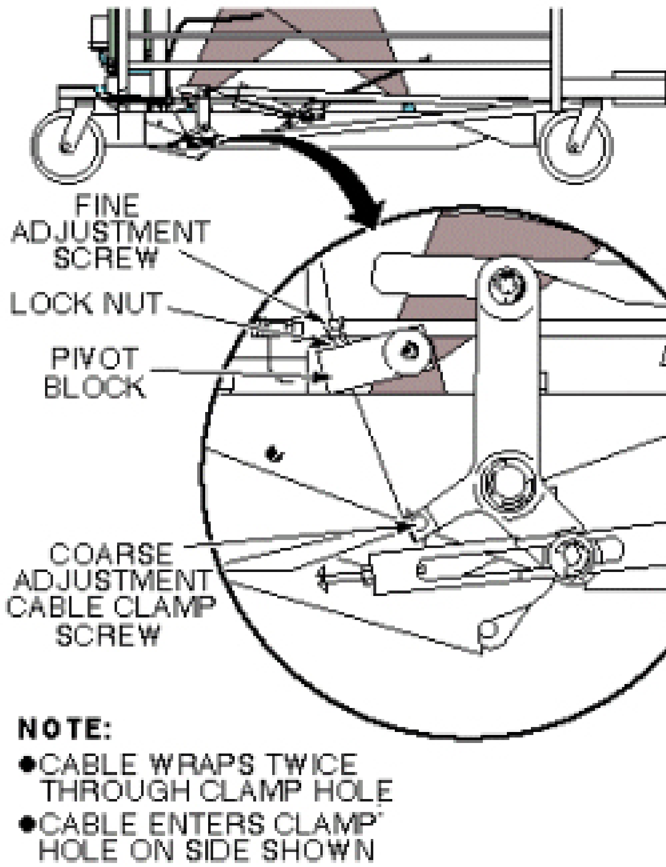


2. Adjust the position of the assembly so the release block is centered in the cutout on the front of the cradle assembly. Tighten hex socket screws. See Illustration 2-30 .

12.3.2 Cradle Release Block Retraction Adjustment

1. Dock Patient Transport.
2. Remove all upper and lower trim covers from sides of Patient Transport.
3. Loosen the lock nut, then tighten or loosen the cable adjustment screw in the pivot block so that the cradle release block is flush or just below the level of the table top. See Illustration 2-31 .

Illustration 2-31: CRADLE RELEASE BLOCK RETRACTION ADJUSTMENT



4. Tighten lock nut.

NOTE: If the adjustment screw does not provide enough range for adjustment, coarse adjustment may be made to the cable length where the cable end is clamped into actuator bracket. Fine adjustment will be required after performing a coarse adjustment.

13 Cradle Release Pin Height Adjustment

13.1 Personnel Requirements

Required Persons	Preliminary Reqs	Procedure	Finalization
1	Not Applicable	Not Applicable	Not Applicable

13.2 Overview

This procedure is applicable to the following table products:

- Signa Lightweight Patient Transport Table
- Signa Oncology Patient Transport Table Option
- Signa OR-Compatible Patient Transport Table Option

13.3 Procedure

1. Remove Cradle Assembly from Patient Transport per procedure in Section 4, Removing Cradle Assembly.

2. Remove all left side covers from sides of Patient Transport.

3. When docked, tighten or loosen the cable adjustment screw so that the cradle release pin is flush to 1/16 inch (1.59 mm) below the level of the table top. See Illustration 2-32 and Illustration 2-33 .

Illustration 2-32: CRADLE RELEASE PIN HEIGHT ADJUSTMENT

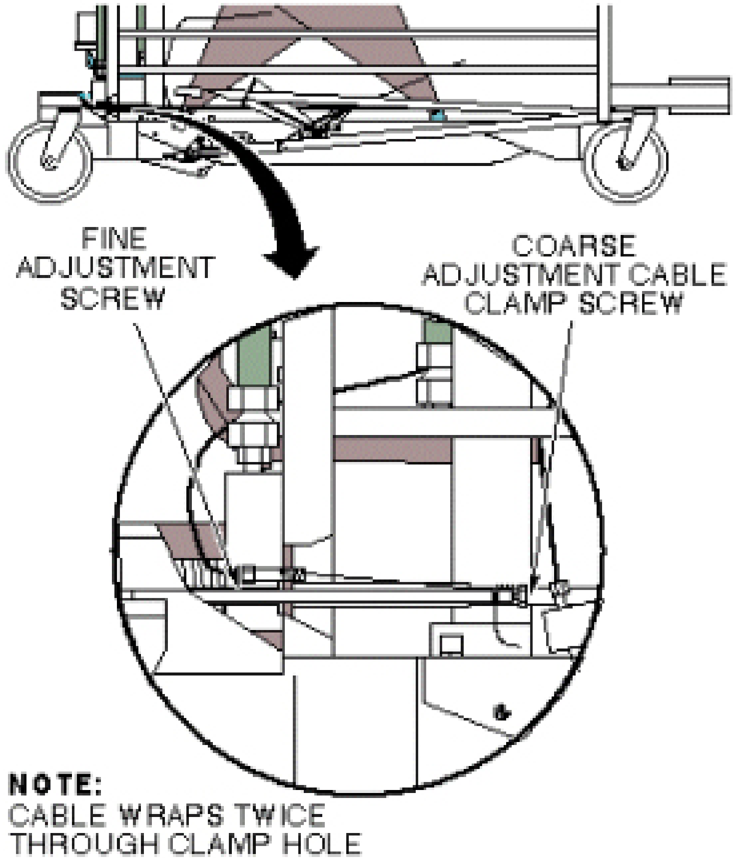
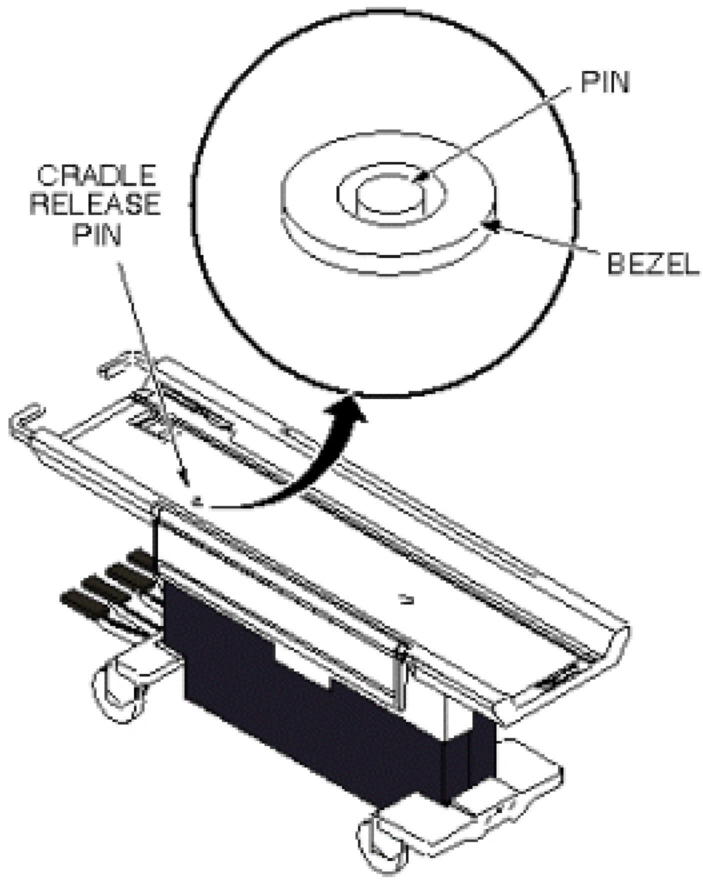


Illustration 2-33: CRADLE RELEASE PIN HEIGHT



13.4 Finalization

1. Replace side covers.
2. Replace cradle.

Chapter 3 Functional Checks

1 Introduction

This procedure is applicable to the following table products:

- Signa Lightweight Patient Transport Table
- Signa Oncology Patient Transport Table Option
- Signa OR-Compatible Patient Transport Table Option

2 Cradle Motion Test

2.1 Personnel Requirements

Required Persons	Preliminary Reqs	Procedure	Finalization
1	Not Applicable	30 mins	Not Applicable

2.2 Overview

This procedure is applicable to the following table products:

- Signa Lightweight Patient Transport Table
- Signa Oncology Patient Transport Table Option
- Signa OR-Compatible Patient Transport Table Option

This test moves the cradle into the magnet bore, and then returns it to its starting position. When this test is run, the cradle is automatically moved into the magnet bore a maximum distance of 600 mm, then is returned to its starting position. Any error messages generated during this test are stored in the Error Message Log. If the cradle starts in the Home position, it is moved about 100 mm into the magnet bore, is paused there, then is advanced a maximum of 600 mm into the bore, then is returned close to the 100 mm position. If the cradle starts in a position beyond 1700 mm into the bore (max. Is 1160 for LCC), it is automatically returned to the Home position, then is advanced a maximum of 600 mm into the bore, then is returned close to the 100 mm position.

2.3 Preliminary Requirements

2.3.1 Safety

DANGER

STRONG MAGNETIC FIELD!
FERROUS MATERIALS CAN BECOME DANGEROUS PROJECTILES IN THE PRESENCE OF THE MAGNETIC FIELD PRODUCED BY THE SIGNA MAGNET.
DO NOT BRING ANY FERROMAGNETIC TOOLS OR EQUIPMENT INTO THE MAGNET ROOM.

CAUTION

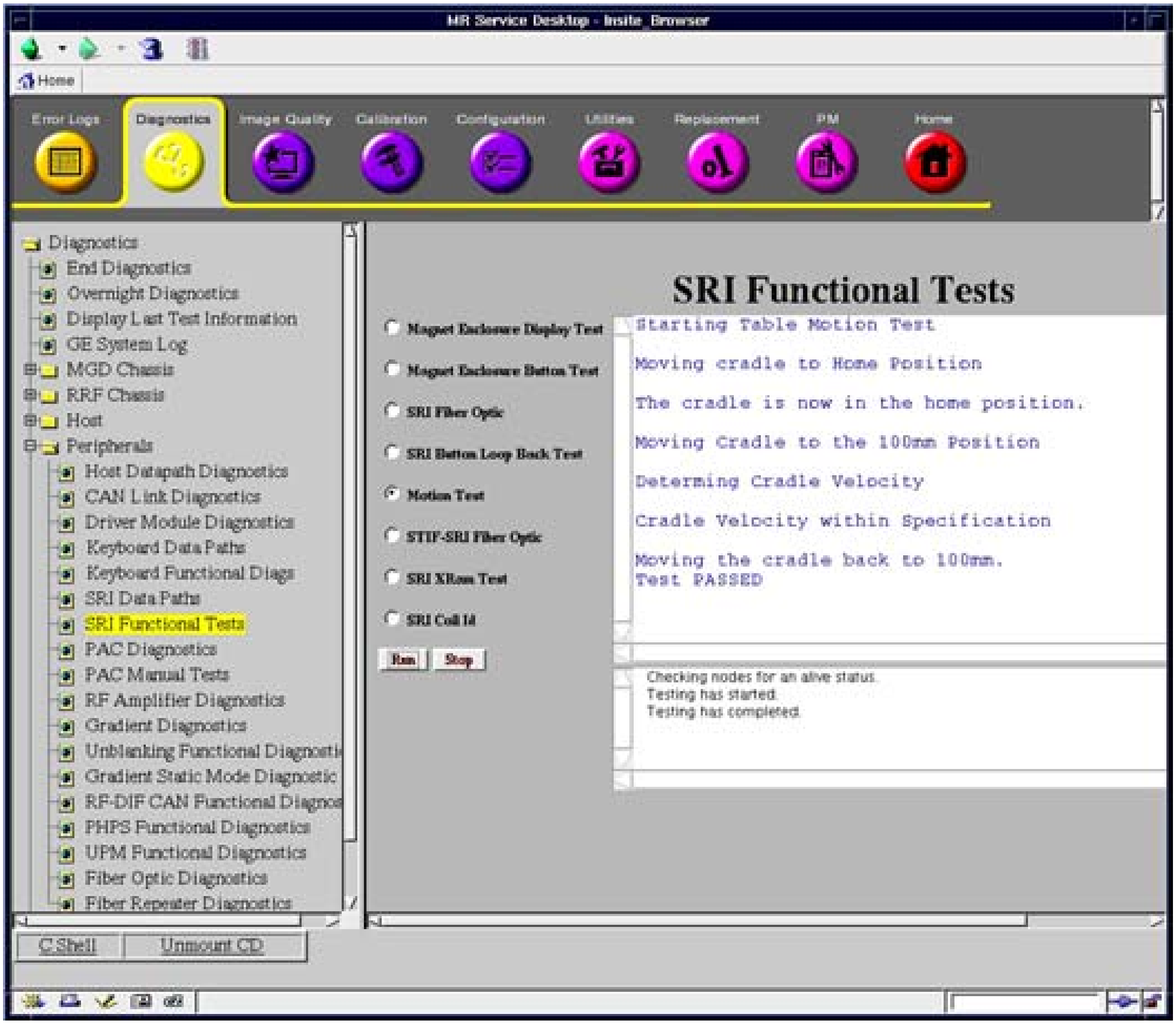
POSSIBLE PERSONAL INJURY AND/OR EQUIPMENT DAMAGE.
OBSTRUCTIONS OR PERSONNEL IN CRADLE PATH MAY CAUSE PERSONAL INJURY OR EQUIPMENT DAMAGE.
VERIFY THAT MAGNET BORE IS CLEAR OF PERSONNEL AND EQUIPMENT BEFORE STARTING CRADLE MOTION TEST.

2.4 Procedure

1. Start the Service Browser and go to the MR Service Desktop on the host computer.

2. Go to the [Diagnostics tab], [Peripherals], [SRI Functional Tests], and select the [Motion Test]. See Illustration 3-1 .

Illustration 3-1: SRI FUNCTIONAL TESTS



3. Select [Motion Test] and then click [Run].
4. A pop-up window will appear warning that the cradle is about to move. Press [Yes] to continue.
5. Verify that the cradle moves smoothly, and that magnet enclosure cradle position display increments, or decrements, corresponding to cradle travel into or out of magnet bore.
6. After the cradle stops, verify the test passed in the status window of the diagnostic tool.
7. Go to [Diagnostics], End Diagnostics and select the [End Diagnostics] Button to exit the diagnostics mode.

3 Patient Transport Movement Verification

3.1 Personnel Requirements

Required Persons	Preliminary Reqs	Procedure	Finalization
1	Not Applicable	15 mins	Not Applicable

3.2 Overview

This procedure is applicable to the following table products:

- Signa Lightweight Patient Transport Table
- Signa Oncology Patient Transport Table Option
- Signa OR-Compatible Patient Transport Table Option

The patient transport may be operated in either the docked or undocked position. Undocked, it may be raised or lowered via the foot pedals located at the end of the transport; in addition, floor locks are available for locking transport wheels during patient transfer.

While the patient transport is docked to the magnet enclosure, the table may be driven up or down by either the motor located in the enclosure dock or the foot pedals.

NOTE: The OR-Compatible Patient Transport Table cannot be driven up or down by the pedals or motor when the table is docked to the magnet enclosure.

3.3 Procedure

1. Undock the Patient Transport table from the magnet.
2. Verify upward movement of table top by pressing Table Up pedal repeatedly.
3. Verify downward movement of table top by pressing Table Down pedal to initiate movement. Downward travel of table top should stop when pedal is released.
4. Verify floor locks on casters inhibit all patient transport motion when locked in position.
5. Verify there is no movement of cradle while patient transport is undocked.
6. Dock patient transport by aligning table with enclosure dock and then pressing the Dock pedal. Successful docking should not require excessive force. If table is fully up when docked, verify that carriage advances forward until it connects with cradle latch immediately after docking.
7. Press Table Down pedal on dock.
   a. **(For the Lightweight Table and Oncology Table)** Verify the table travels from the highest to lowest position in no more than 20 seconds with the table unloaded, then proceed to Step 8 .
   b. **(For the OR-Compatible Table)** Verify the table remains up when docked to the magnet enclosure, then proceed to Step 10 .

8. Press Table Up pedal on dock, and verify that table travels from lowest to highest position in no more than 19 seconds with table unloaded.

9. Starting with table at lowest position, verify that it reaches its highest position in about 17 pedal strokes or fewer, using Table Up pedal. For accurate check of this function to specification given, pedal strokes must be complete (i.e., push pedal all the way down and let pedal come all the way up before starting the next downward thrust).

10. Verify that the emergency release lever (located on the right side of the dock) releases the patient transport from the dock when it is pulled.

11. Redock the Patient Transport.

12. Verify that the cradle release handle releases cradle from carriage when twisted. Refer to procedure for Cradle Release Adjustments (Chapter 2, Section 12, Cradle Release Block Adjustment, Cradle Release From Carriage Adjustment, or Cradle Release From Patient Transport Adjustment.

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Chapter 4 Planned Maintenance

1 Introduction

This procedure is applicable to the following table products:

- Signa Lightweight Patient Transport Table
- Signa Oncology Patient Transport Table Option
- Signa OR-Compatible Patient Transport Table Option

2 Patient Handling Planned Maintenance

This procedure is applicable to the following table products:

- Signa Lightweight Patient Transport Table
- Signa Oncology Patient Transport Table Option
- Signa OR-Compatible Patient Transport Table Option

Each procedure should be completed 6 times a year. This page can be printed for record keeping.

Procedure	Period					
	A	B	C	D	E	F
Emergency Release of Cradle						
Chapter 2, Section 8, Cradle Release from Patient Transport Adjustment						
Unlocking of Table With Red Handle						
Castor Brake Check						
Table Leveling Check						
Delrin Block Centering						
Table Dock Verification						
Chapter 3, Section 2, Cradle Motion Test						
Table Up/Down From Dock						
Table Up/Down Check From Table Pedal						
Delrin Rail Check						
Cradle Release Pin Check						
Delrin Block Height Check						
Table Down Check						
Table Caster Checks						
Bungee Tube Access / Inspection / Replacement (For Signa OR-Compatible Patient Transport Table Option only)						

FE Name: ______________________________

Date Completed: _________________________

Chapter 5 Parts

1 Introduction

This procedure is applicable to the following table products:

- Signa Lightweight Patient Transport Table
- Signa Oncology Patient Transport Table Option
- Signa OR-Compatible Patient Transport Table Option

2 Table and Cradle Assembly Spare Parts

This procedure is applicable to the following table products:

- Signa Lightweight Patient Transport Table
- Signa Oncology Patient Transport Table Option

Table 5-1: Signa Oncology Patient Transport Table New Spare Parts

Item	Name	Part Number	Where Used
1	Cradle Wheel Kit	5179918	Signa Oncology Cradle/HDx Cradle
2	Oncology Cradle Assembly Only (without coil support frames)	5304826	Only used on Signa Oncology Table 5189593-3
3	Coil Support Frame (utilized in two locations)	5214662-2	Attaches to the cradle assembly; left and right sides
4	Pinch Point Decal (utilized in four locations)	5142065	Located on table arm boards
5	Table Spacer Kit (utilized in four locations and attaches with adhesive tape)	5308567	Located on table top surface; front and rear; left and right sides
6	Roll of adhesive tape	46-170106P1	For adhering Front and Rear Spacers
7	Sheet of Decals which includes the "Heavy Object" Decal	5306483	Located on the top surface of the cradle near the release handle

Illustration 5-1: SIGNA ONCOLOGY PATIENT TRANSPORT TABLE AND CRADLE ASSEMBLY

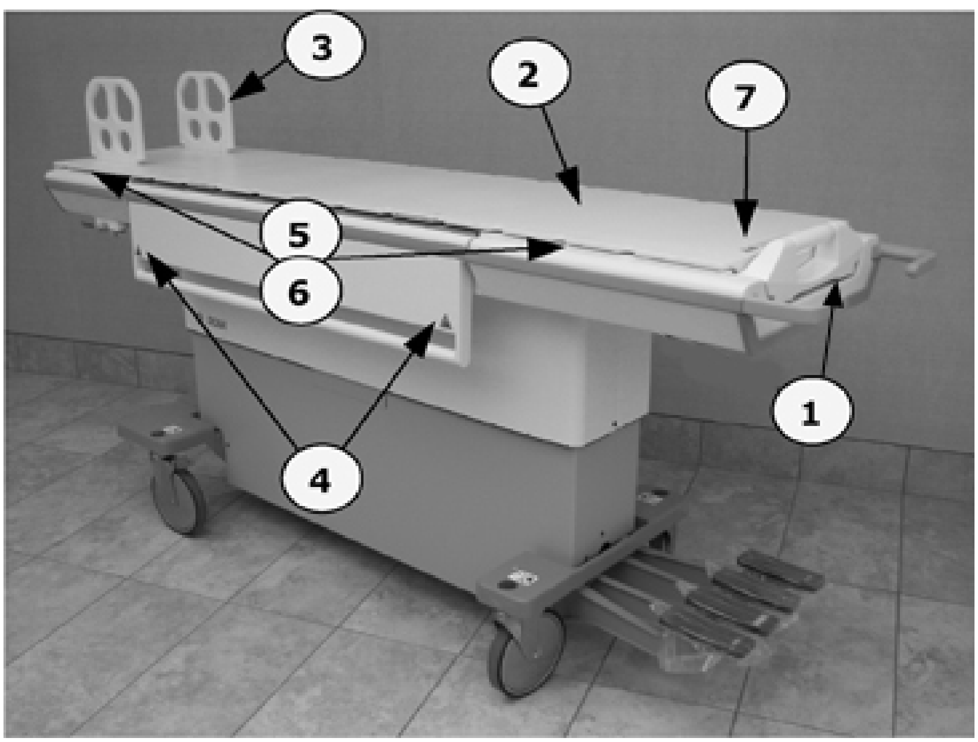


Table 5-2: Signa HDx Table Existing Spare Parts

Name	Part Number
Wheel FRU Kit, contains 6 Wheel and housing assemblies for the Oncology Cradle Assembly	5179918
Hydraulic Pump assembly with packaging	5132699
Hydraulic Cylinder assembly with packaging	5132700
Hydraulic Oil Filter and Gasket	46-307878G1
Catis Castor assembly with packaging	136226
Inner Cover	5145934
Head End Bumper	5145935
Pinch Guard	5145936
Table Handle	5145937
22-Side Bumper	5145939
16.25 Side Bumper	5145939-2
Foot End Bumper	5145941
Trim Cover	5145942
Bumper Arm Board	5145943
Table Top Panel-Rh	5145946
Table Top Panel-Lh	5145947
Outer Cover	5145952
Arm board Latch Bar	5145954
Flipper Housing	5145964
Flipper Bottom Cover	5145965
Flipper	5145966
Cradle Handle	5145972
Cover	5152089
Outer Cover	5159989
Outer Cover	5159989-2
Link Undock	2297500
Plastic	46-265898P1
Foot Pedal Pad, Rubber	46-265905P1
Casting	46-265907P1
.625 Dia	46-265908P1
Arm board	46-271007P1
Table Head End Bumper, Accent Grey	46-271026P2
Pedal Asm, Pump Up	46-271031G1
Molded Polycarbonate	46-271129P1
.375~& .312~ Dia, 1.17~Lg, Sst	46-271161P1
Up Pedal	46-271195P5
Dock Pedal	46-271195P7
71~ (Trim Ar); Swaged Ball One End	46-271511P1

Table 5-2: Signa HDx Table Existing Spare Parts (cont'd)

Right Actuator	46-271516G1
14.13~ Long Aluminum Rod W/Bumper	46-271517G1
Mechanical Linkage	46-271576G1
Up Indicator Rod Assembly	46-271916G1
Table Latch Assembly	46-317191G1
Arm Board Assembly	5161493

Chapter 6 Replacements

1 Introduction

This procedure is applicable to the following table products:

- Signa Lightweight Patient Transport Table
- Signa Oncology Patient Transport Table Option
- Signa OR-Compatible Patient Transport Table Option

2 Hydraulic Filter Replacement

2.1 Personnel Requirements

Required Persons	Preliminary Reqs	Procedure	Finalization
1	Not Applicable	Not Applicable	Not Applicable

2.2 Overview

This procedure is applicable to the following table products:

- Signa Lightweight Patient Transport Table
- Signa Oncology Patient Transport Table Option
- Signa OR-Compatible Patient Transport Table Option

Perform the following procedure once each year to maintain smooth, responsive operation of the patient transport hydraulics.

2.3 Preliminary Requirements

2.3.1 Tools and Test Equipment

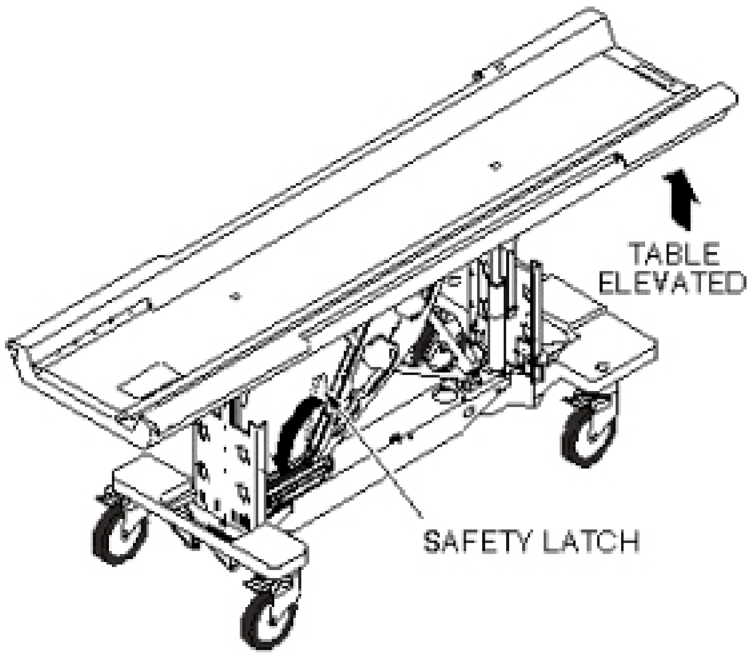
Item	Qty	Effectivity	Part#	Manufacturer
Filter Kit	1	-	46-307878G1	-
Hydraulic Oil	1	-	46-230006P2	-
Absorbent Toweling	As required	-	-	-
13/16” open end wrench	1	-	-	-
$^{3}/_{4}$” open end wrench	2	-	-	-
Ear Syringe	1	-	-	-

2.4 Procedure

2.4.1 Remove Hydraulic Filter

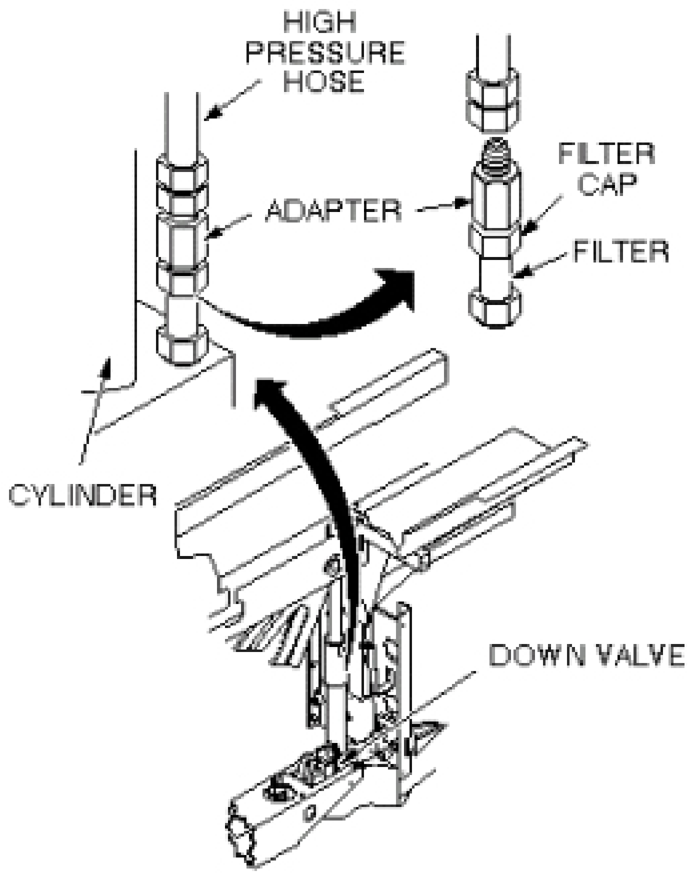
1. Undock table and remove from magnet room.
2. Raise patient transport table to highest elevation with Up foot pedal.
3. Remove patient transport side covers on right side (when facing magnet enclosure).
4. Lower the safety latch by disconnecting spring from scissors assembly (see Illustration 6-1).

Illustration 6-1: LOWERING SAFETY LATCH



5. Release hydraulic pressure by gently pressing Down foot pedal until table is supported by safety latch.
6. Place absorbent material or a drip pan on the floor under the cylinder block to catch any dripping of oil.
7. Raise patient transport table with UP foot pedal (three (3) or four (4) pumps) until table support bar is out of the way. This provides enough clearance to get a 13/16" wrench onto high pressure hose fitting (see Illustration 6-2).

Illustration 6-2: LOCATION OF HYDRAULIC FILTER

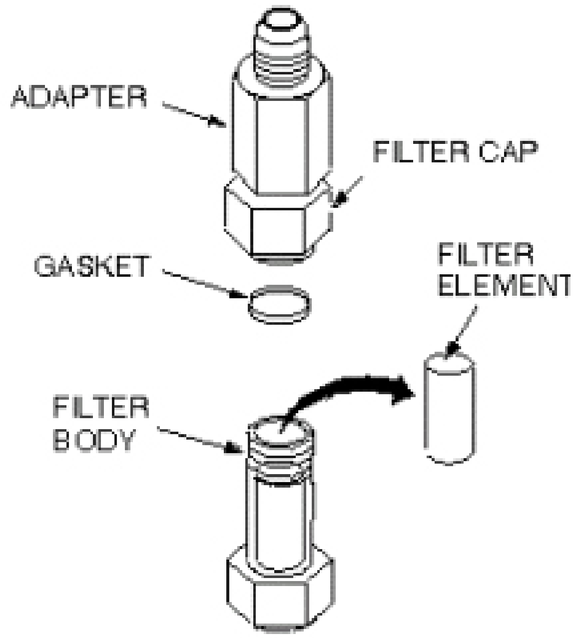


8. Turn wrench only enough to loosen the hose fitting initially, but not to let any oil leak out. Then lower the table using the Down foot pedal until table is again supported by safety latch.
9. Remove high-pressure hose from adapter. A small amount of oil will spill out of hose during removal; raise hose as high as possible to limit oil spillage.

NOTE: Equipment damage possibility. Do not damage gasket with wrench during disassembly, if reusing old gasket. It is better to replace gasket with a new one, to prevent oil from leaking.

10. Remove filter cap from filter using two 3/4" wrenches (see Illustration 6-3).

Illustration 6-3: REMOVE FILTER ELEMENT



11. Remove dirty oil from top of filter.

12. Remove gasket and filter element from filter body and throw away.

2.4.2 Install Replacement Hydraulic Filter

1. Clean outside threads (both ends) of filter body with a clean cloth.

2. Clean interior threads of filter cap with a clean cloth.

3. Install new filter element (Part # 46-230889P1), hole end down, in filter body.

NOTE: Equipment damage possibility. When reassembling filter cap, use care not to damage gasket; ensure that gasket is properly seated before tightening with wrench. If gasket is deformed, oil will leak out of this connection when reassembled.

4. Install filter cap and new gasket (Part # 46-230889P2) onto filter body. Ensure that new gasket seats properly before tightening with wrench.

5. Reconnect high-pressure hose to cylinder block. Tighten by hand, then raise the table with Up foot pedal to allow clearance for wrench; tighten hose fitting securely.

6. Raise table with several pumps on the Up foot pedal and check for oil leaks. If leaks occur, tighten connections snugly, but do not over tighten; brass fittings will deform if over tightened.

7. Proceed to Patient Transport Hydraulics Bleeding.

3 Main Cradle Latch Cable Replacement

3.1 Personnel Requirements

Required Persons	Preliminary Reqs	Procedure	Finalization
1 or 2, depending on system	Not Applicable	60-90	Not Applicable

3.2 Overview

This procedure is applicable to the following table products:

- Signa Lightweight Patient Transport Table
- Signa Oncology Patient Transport Table Option
- Signa OR-Compatible Patient Transport Table Option

This procedure requires:

- One person for a Signa Lightweight Patient Transport Table
- Two people for a Signa Oncology or a Signa OR-Compatible Patient Transport Table Option

3.3 Preliminary Requirements

3.3.1 Replacement Parts

Item	Qty	Effectivity	Part#	Manufacturer
Cable	1	-	46-271511p1	-

3.3.2 Safety

WARNING

TO REDUCE THE RISK OF EXPOSING LOOSE FERROUS MATERIAL TO THE MAGNETIC FIELD, IT IS MANDATORY THAT THE PATIENT TRANSPORT IS REMOVED FROM MAGNET ROOM DURING THIS PROCEDURE.

WARNING

RISK OF PERSONAL INJURY!
LIFTING MR PARTS, SUCH AS FRUS, TOOLKITS AND PARTS OF THE SYSTEM WITHOUT ASSISTANCE (MECHANICAL OR ADDITIONAL PERSONNEL), CAN RESULT IN PERSONAL INJURY.
THE USE OF EXTRA PERSONNEL IS RECOMMENDED TO LIFT AN OBJECT WEIGHING MORE THAN 35 POUNDS AND LESS THAN 50 POUNDS. THE USE OF MECHANICAL LIFT ASSISTANCE IS REQUIRED FOR OBJECTS WEIGHING MORE THAN 90 POUNDS.

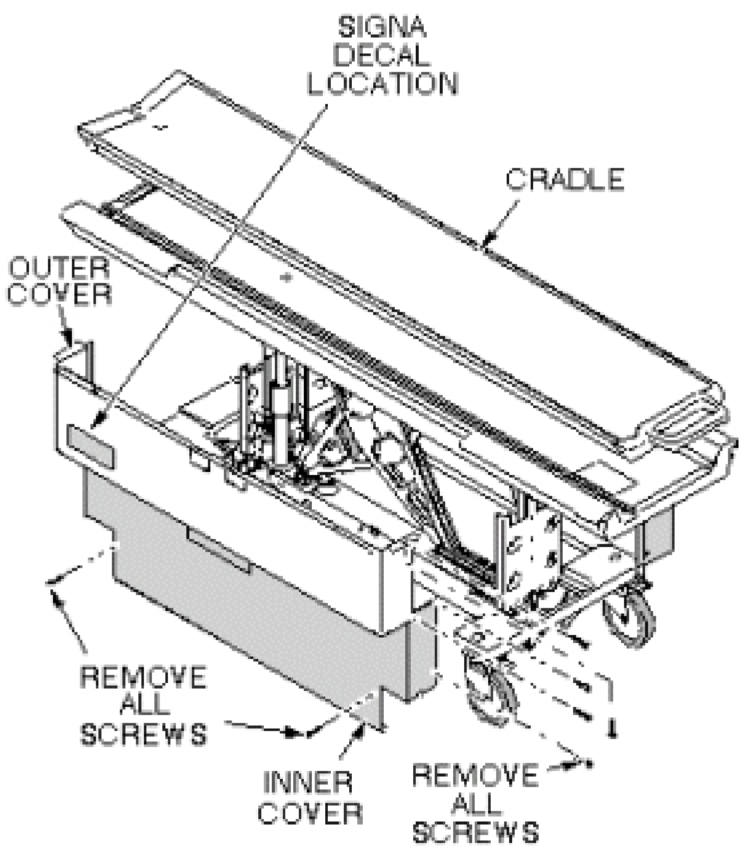
WARNING

RISK OF PERSONAL INJURY!
LIFTING MR PARTS, SUCH AS FRUS, TOOLKITS AND PARTS OF THE SYSTEM WITHOUT ASSISTANCE (MECHANICAL OR ADDITIONAL PERSONNEL), CAN RESULT IN PERSONAL INJURY.
THE USE OF EXTRA PERSONNEL IS RECOMMENDED TO LIFT AN OBJECT WEIGHING MORE THAN 35 POUNDS AND LESS THAN 50 POUNDS. THE USE OF MECHANICAL LIFT ASSISTANCE IS REQUIRED FOR OBJECTS WEIGHING MORE THAN 90 POUNDS.

3.4 Procedure

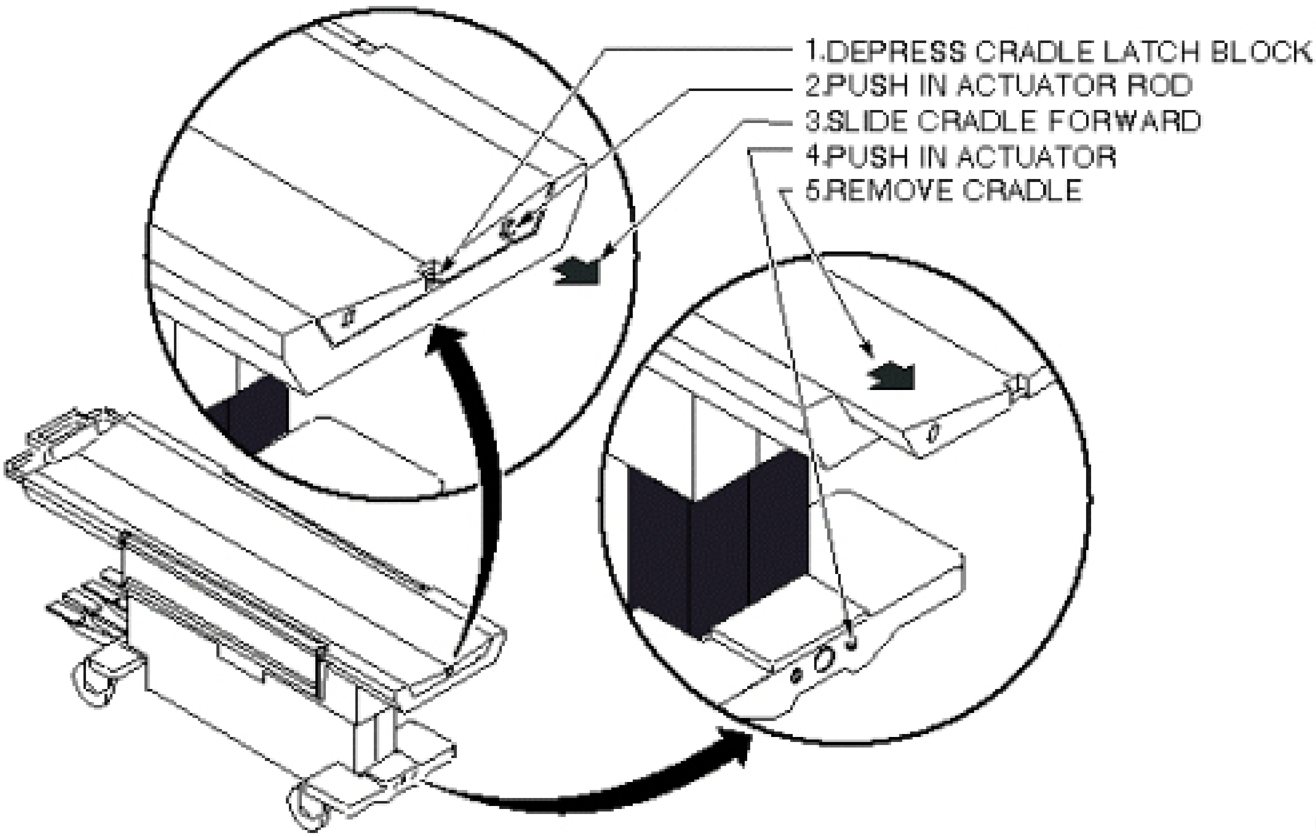
1. Undock patient transport and move it out of exam room.
2. Remove upper and lower left side covers (see Illustration 6-4).

Illustration 6-4: REMOVAL OF PATIENT TABLE COVERS



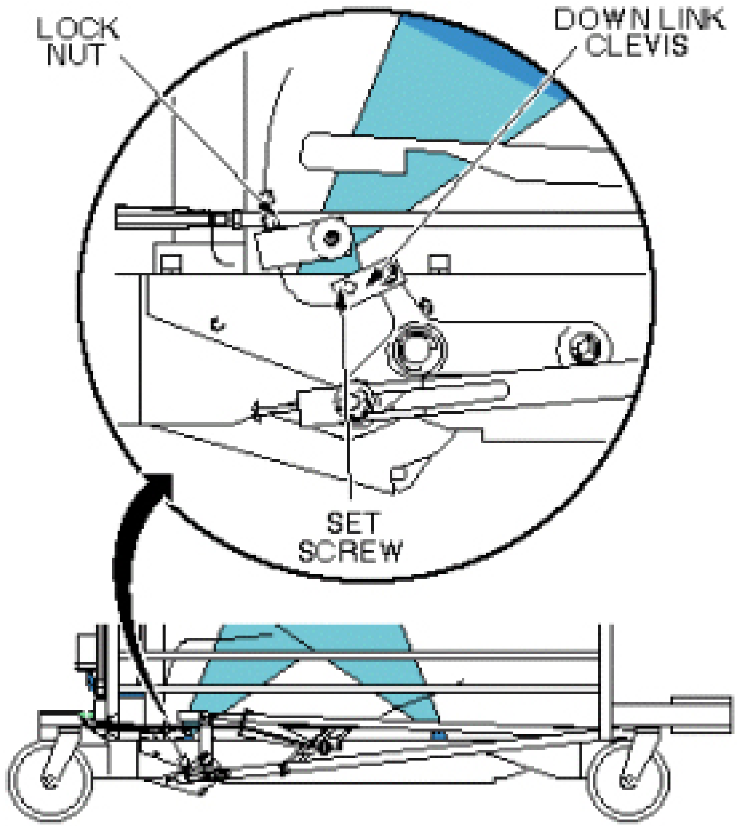
3. Remove cradle (see Illustration 6-5).

Illustration 6-5: PATIENT TRANSPORT CRADLE REMOVAL



4. Using an M2 Allen wrench, loosen set screw at down link clevis. See Illustration 6-6).

Illustration 6-6: MAIN CRADLE LATCH DOWN LINK LEVER

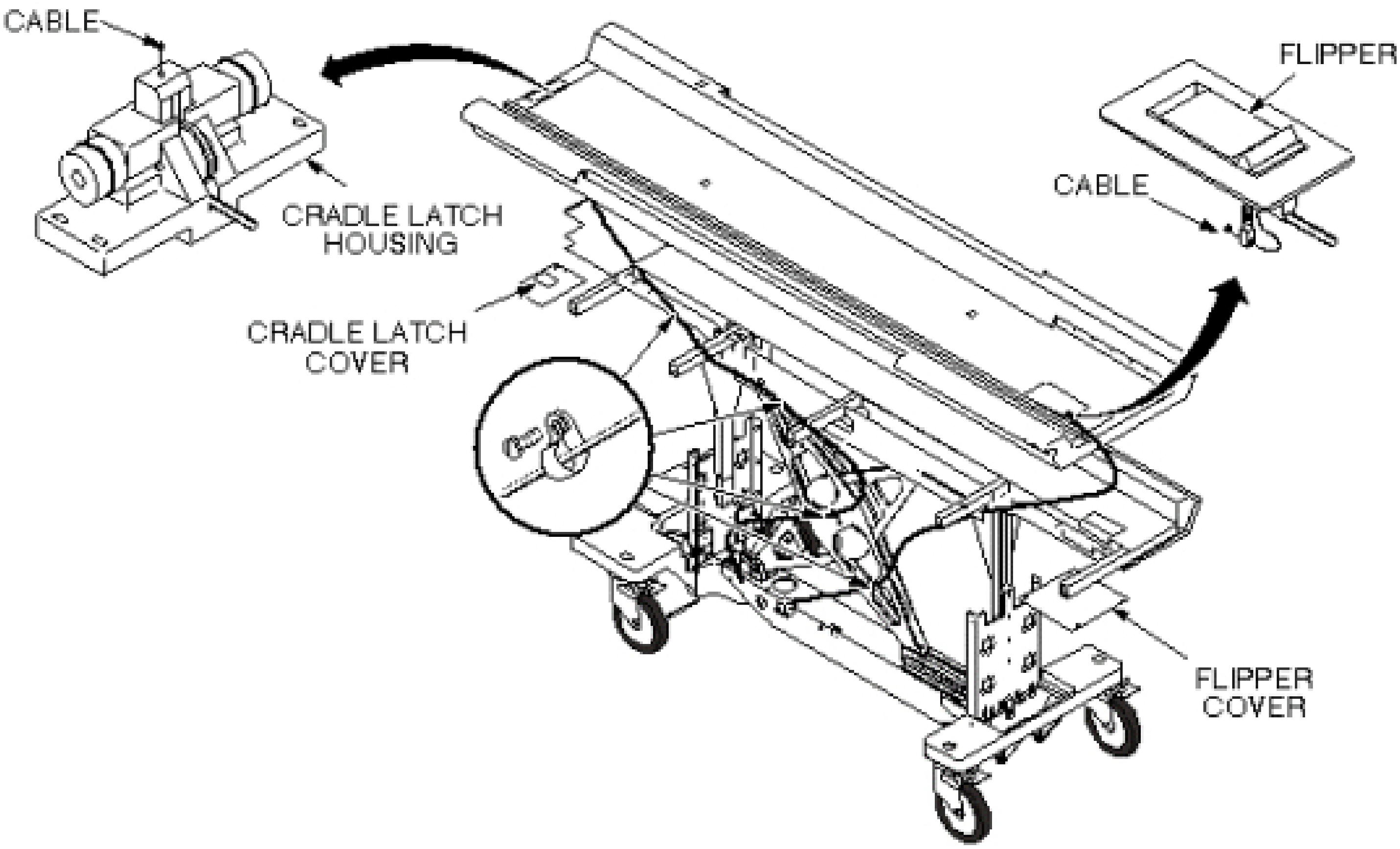


CAUTION

POTENTIAL BODY INJURY AND DAMAGE TO A CRITICAL COMPONENT OF THE PATIENT TABLE.
FALLING OF UNSECURED OBJECT
WHEN REMOVING PLATE THAT SECURES THE CRADLE LATCH HOUSING, MAKE SURE TO HOLD THE PLATE IN PLACE AS THE SCREWS ARE REMOVED. FAILURE TO DO SO WILL RESULT IN THE HOUSING FALLING ONCE NO LONGER SECURED TO THE PATIENT TABLE.

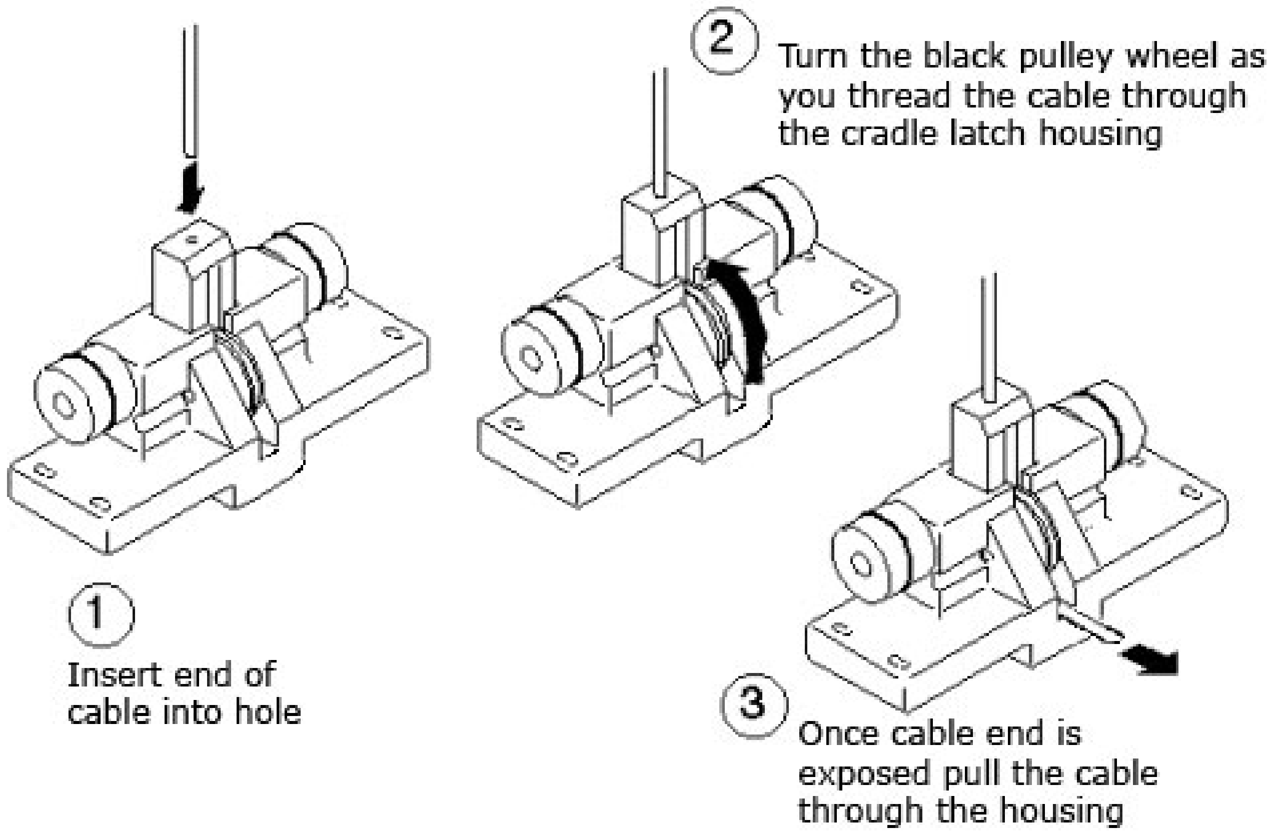
5. Remove four bolts securing cradle latch housing and cradle latch cover (see Illustration 6-7).

Illustration 6-7: INTERLOCK CABLE AND MAIN CRADLE LATCH CABLE ROUTING



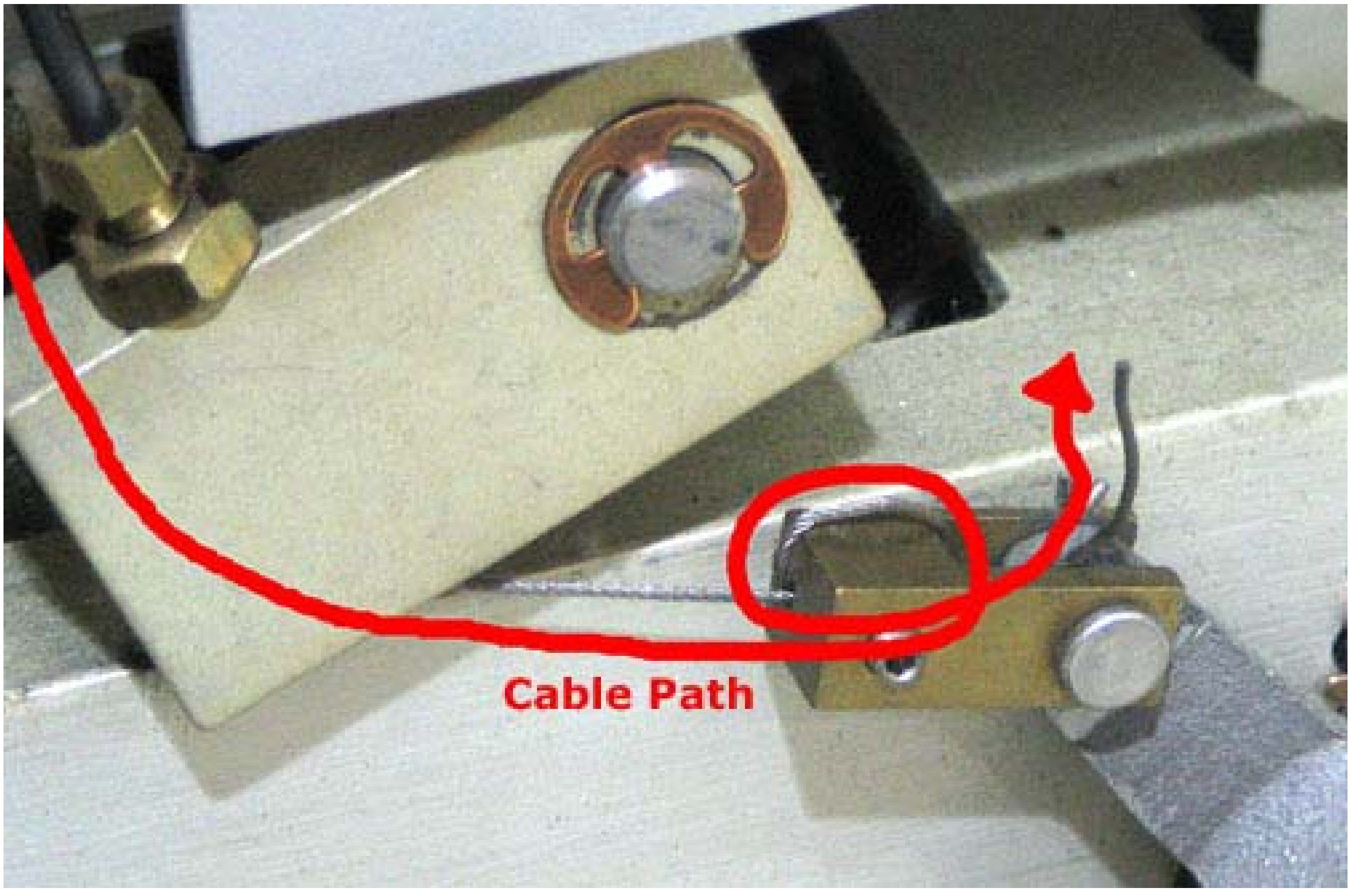
6. Unfasten cable end from Down Link Clevis (see Illustration 6-6).
7. Pull cable out of Cradle Latch Block on front of patient transport (see Illustration 6-7).
8. Insert new cable through head of Cradle Latch Block so once fully threaded, the swaged end fits into hole in latch pin. See Illustration 6-8 for details on installing cable.

Illustration 6-8: INSTALLATION OF MAIN CRADLE LATCH CABLE



9. Route cable through cable casing and insert through cable Down Link Clevis (see Illustration 6-6).
10. Loop cable twice through hole in Down Link Clevis (see Illustration 6-9).

Illustration 6-9: Cable Path through the Down Link Clevis



11. Using an M2 Allen wrench, tighten set screw (see Illustration 6-6).

12. Return patient transport to exam room.

13. Dock patient transport.

14. Adjust cable so that when docked, main latch will be flush with casting. Refer to procedure for Chapter 2, Section 12, Cradle Release Block Adjustment. Cut off excess length after adjustment is complete (see Illustration 6-9 for appropriate excess length). Secure adjustment by tightening set screw after adjustment is complete (see Illustration 6-6).

15. Replace cradle and upper and lower side covers. Note position of Signa logo on outer covers and position toward front of table (see Illustration 6-4).

4 Patient Table Hydraulic Pump Assembly Replacement

4.1 Personnel Requirements

Required Persons	Preliminary Reqs	Procedure	Finalization
1	Not Applicable	Not Applicable	Not Applicable

4.2 Overview

This procedure is applicable to the following table products:

- Signa Lightweight Patient Transport Table
- Signa Oncology Patient Transport Table Option
- Signa OR-Compatible Patient Transport Table Option

4.3 Preliminary Requirements

4.3.1 Tools and Test Equipment

Item	Qty	Effectivity	Part#	Manufacturer
container to catch hydraulic fluid	1	-	-	-

4.3.2 Consumables

Item	Qty	Effectivity	Part#	Manufacturer
Wipes	As required	-	-	-

4.3.3 Safety

WARNING

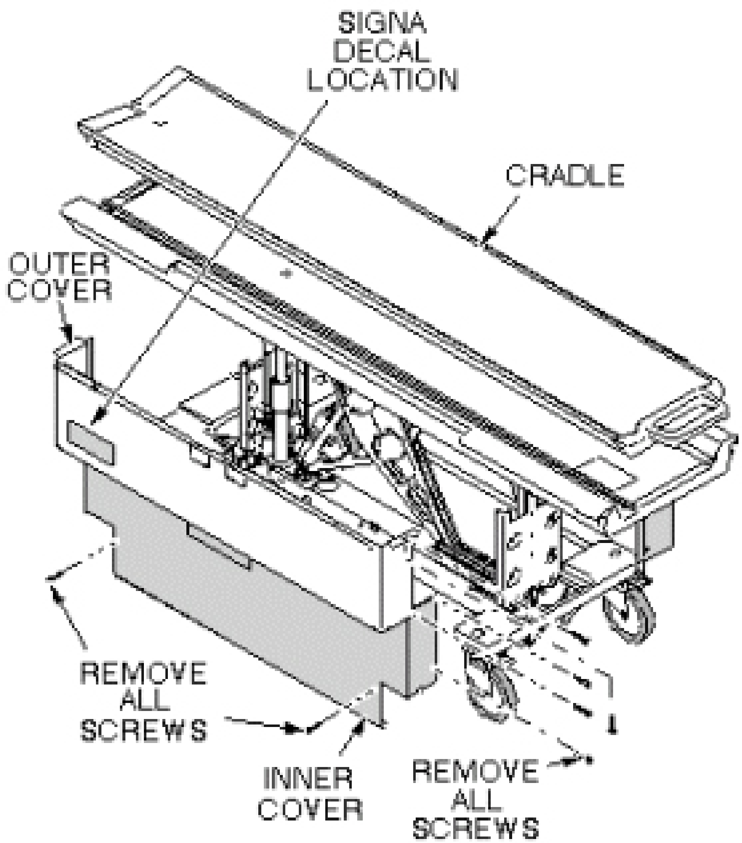
PERSONAL INJURY POSSIBILITY!
SEVERE PERSONAL INJURY CAN OCCUR IF TABLE IS ACCIDENTALLY LOWERED DURING MAINTENANCE PERFORMANCE.
WHEN PERFORMING MAINTENANCE ON THE TABLE IN ITS UP POSITION, ENSURE THAT THE LOCK BAR IS SAFELY IN PLACE.

4.4 Procedure

4.4.1 Remove Hydraulic Pump Assembly

1. Undock patient transport and move it out of exam room.
2. Remove both outer covers by removing mounting screws (see Illustration 6-10).

Illustration 6-10: REMOVAL OF PATIENT TABLE COVERS



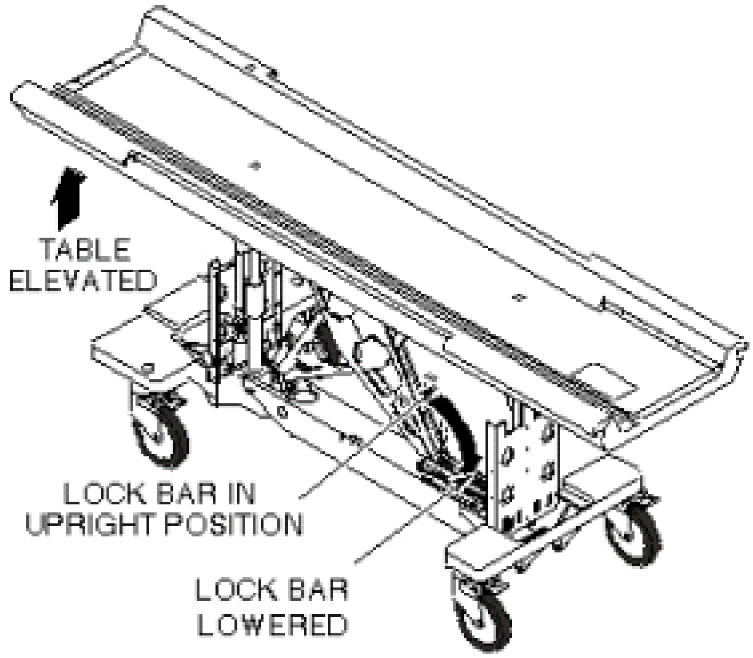
3. Remove inner covers by removing mounting screws.
4. Raise table to up position.

WARNING

PERSONAL INJURY POSSIBILITY!
SEVERE PERSONAL INJURY CAN OCCUR IF TABLE IS ACCIDENTALLY LOWERED DURING MAINTENANCE PERFORMANCE.
WHEN PERFORMING MAINTENANCE ON THE TABLE IN ITS UP POSITION, ENSURE THAT THE LOCK BAR IS SAFELY IN PLACE.

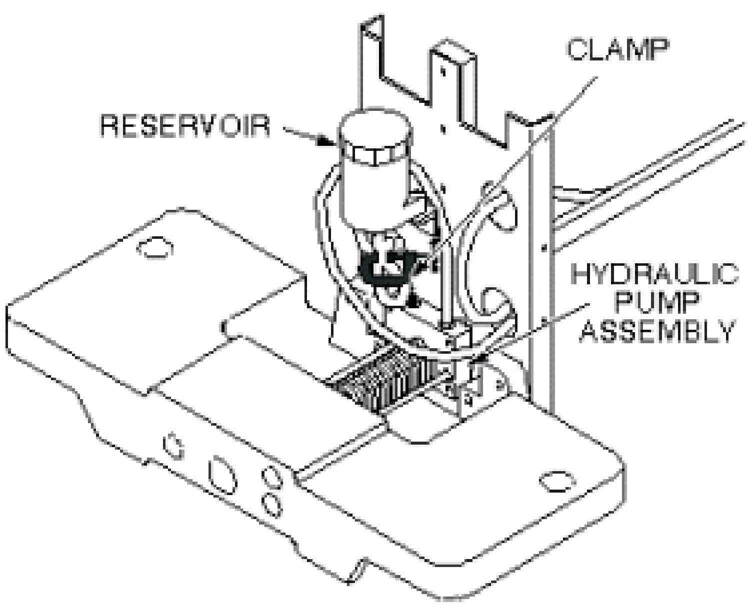
5. Unhook lock bar spring and move lock bar to horizontal position, as shown in Illustration 6-11 , then lower table until lock bar is supporting table in its full up position.

Illustration 6-11: PATIENT TABLE LOCK BAR



6. Disconnect the two hydraulic lines to hydraulic pump assembly and fasten upward and aside to avoid excess spillage. Immediately fold and clamp the translucent (yellow) hose from reservoir to prevent oil from draining from the reservoir (see Illustration 6-12).

Illustration 6-12: HYDRAULIC PUMP ASSEMBLY



7. Loosen two mounting screws attaching upper drip pan to lower drip pan (see Illustration 6-13).

8. Remove upper drip pan (see Illustration 6-13).

Illustration 6-13: UPPER AND LOWER DRIP PANS

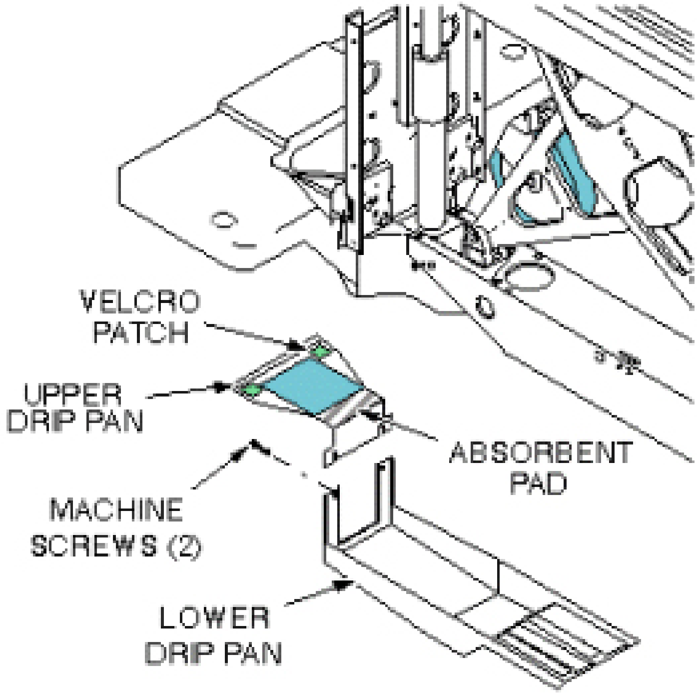
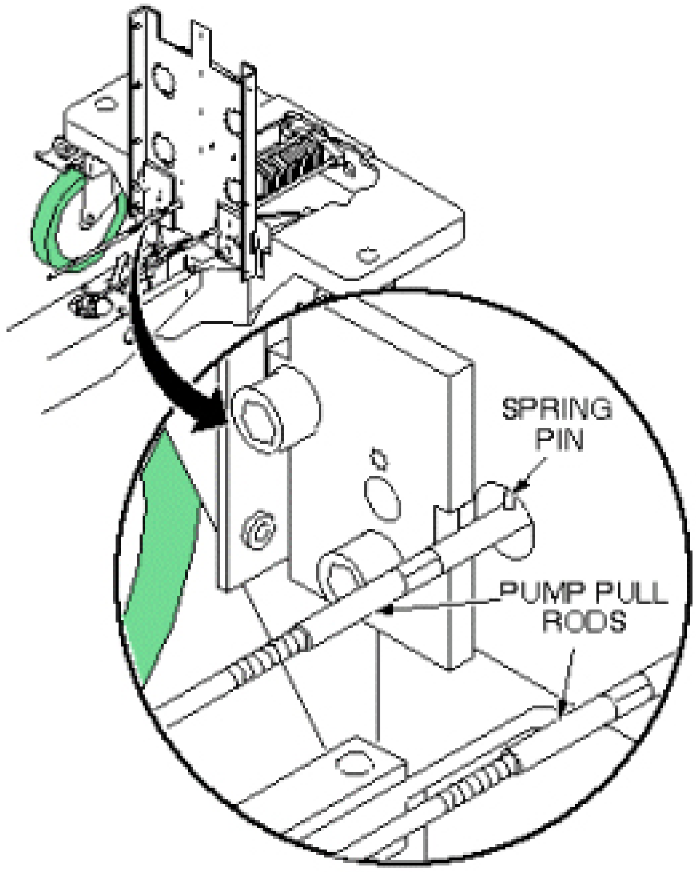


Illustration 6-14: HYDRAULIC PUMP PULL RODS



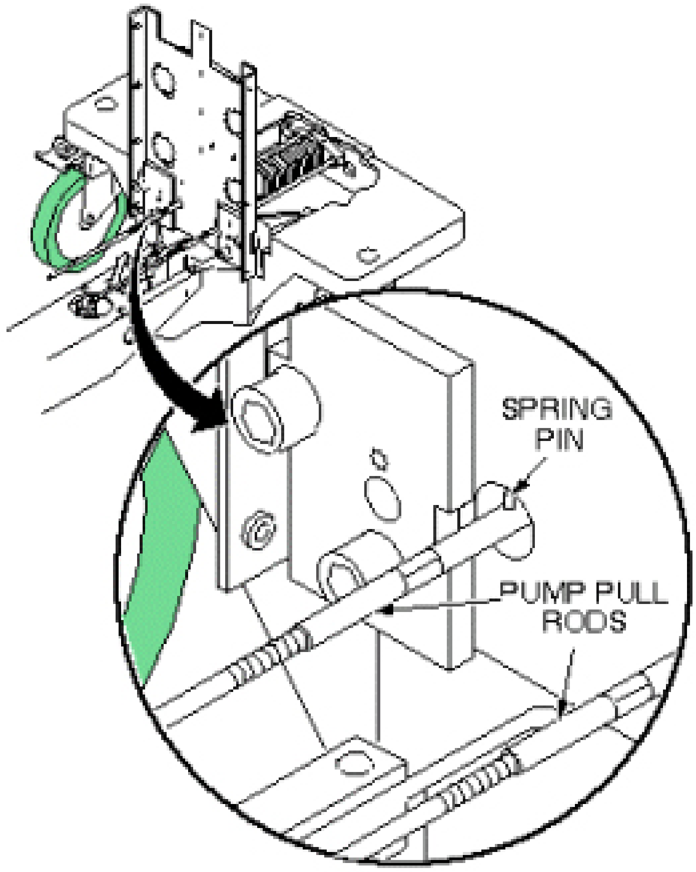
NOTICE

Equipment damage possibility. Do not remove spring pins from either end of pump pull rods; this will cause a side loading on the pump, requiring its replacement. The pump pull rods are disconnected from the pump linkage by turning the pump pull rods in a counterclockwise direction after tension has been removed from the pump springs. This is explained in the following procedure. The pump pull rod spring pins are exposed only when the Table Up pedal is pressed (see Illustration 6-14).

9. To remove tension from pull rods, compress main pump spring, as shown in Illustration 6-16 . Block the pump Up pedal fully up to take its weight off of the pull rods.

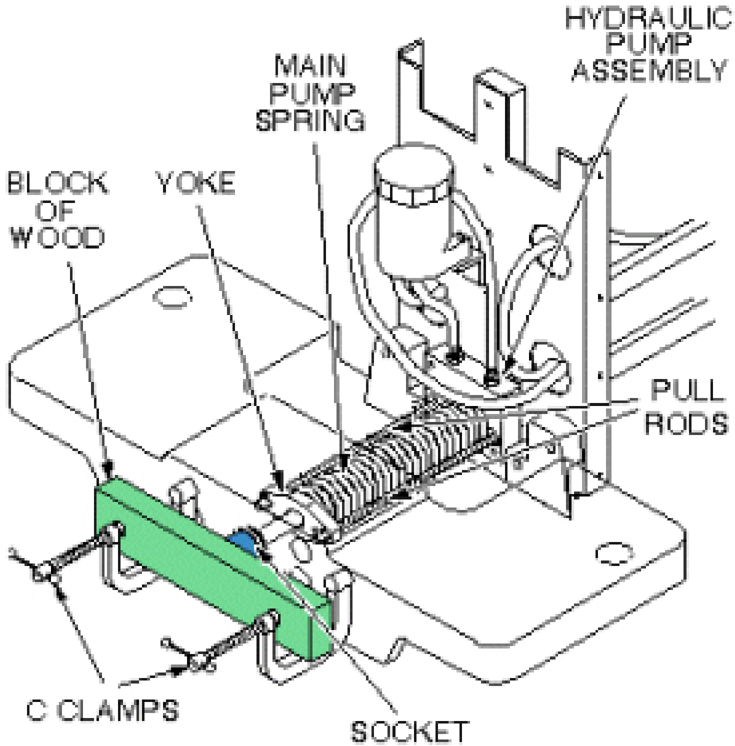
10. Push the yoke back and unscrew the pull rods form the pump linkage (see Illustration 6-15).

Illustration 6-15: HYDRAULIC PUMP PULL RODS



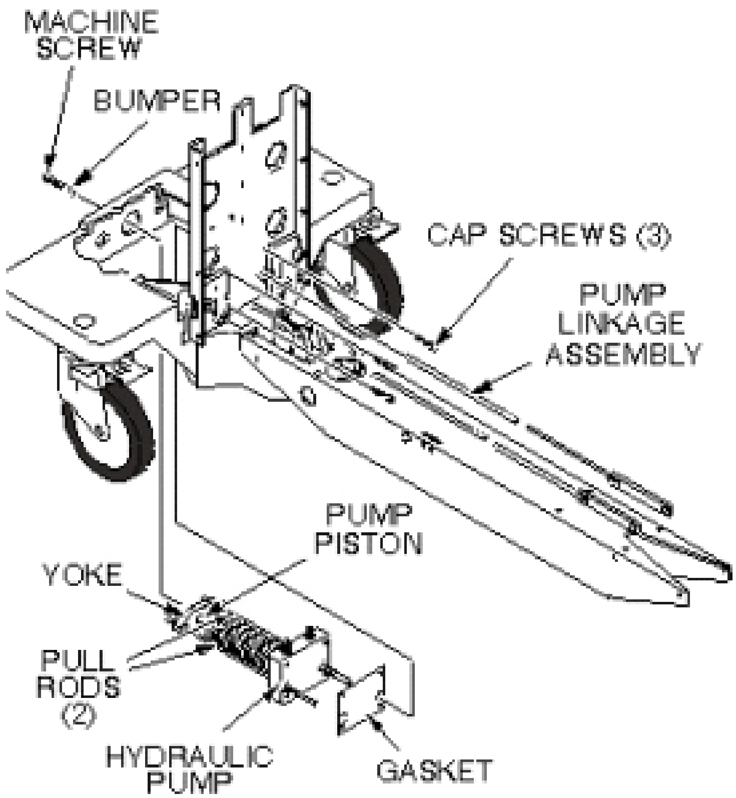
11. Remove C-clamps used to compress main spring, as shown in Illustration 6-16 .

Illustration 6-16: COMPRESS HYDRAULIC PUMP MAIN SPRING



12. Remove machine screw securing bumper to the pump piston (see Illustration 6-17).

Illustration 6-17: HYDRAULIC PUMP REPLACEMENT



13. Remove three mounting cap screws that secure pump assembly to front table support.
14. Remove hydraulic pump assembly and gasket.
15. If absorbent pad attached to upper drip pan appears to be saturated with fluid, remove it and squeeze out fluid.

4.4.2 Install Replacement Hydraulic Pump Assembly

1. Prepare replacement pump for installation by applying Locktite 242 to three (3) mounting screws and to machine screw that secures bumper to pump piston.
2. Set replacement pump assembly on front support of table, making sure that gasket is installed properly, and secure with three mounting screws (see Illustration 6-17).
3. Secure the bumper to the pump piston with machine screw.
4. To compress main pump spring, reinstall C-clamps on front support, as shown in Illustration 6-18 . Screw pump pull rods into pump linkage, alternately tightening each side to avoid side loading the pump. Stop when the rear spring pin on each pull rod is 1/8-inch (3.2 mm) from pump body (flush with back surface of front support) (see Illustration 6-19).

Illustration 6-18: COMPRESS HYDRAULIC PUMP MAIN SPRING

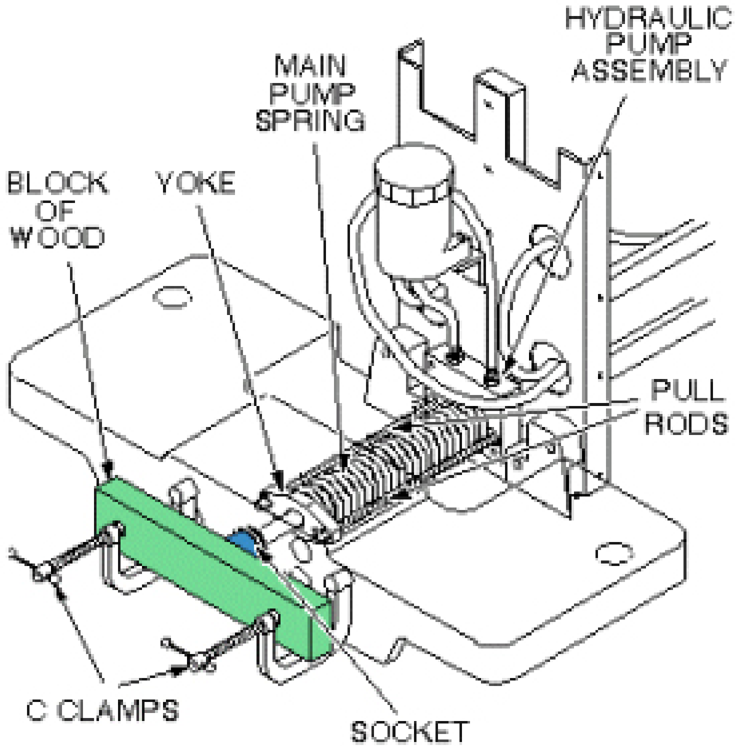
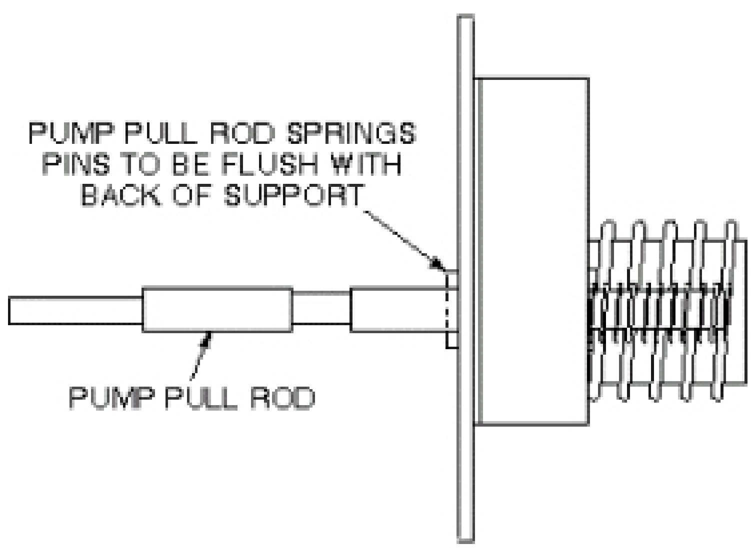
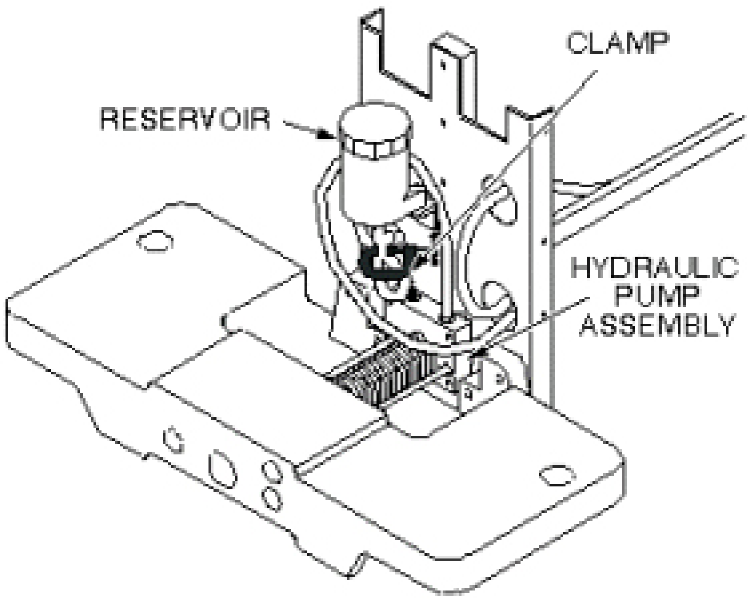


Illustration 6-19: PUMP PULL ROD ADJUSTMENT



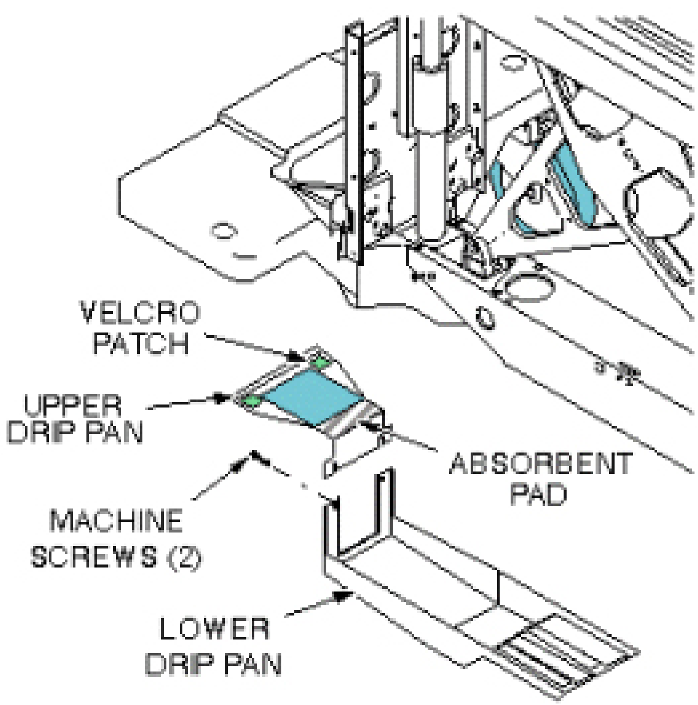
5. Connect two hydraulic lines to hydraulic pump assembly (see Illustration 6-20).

Illustration 6-20: HYDRAULIC PUMP ASSEMBLY



6. Attach upper and lower drip pans (see Illustration 6-21).

Illustration 6-21: UPPER AND LOWER DRIP PANS



7. Block open the Down valve, and pump the Up pedal at least 20 full strokes to purge air from the lines.

8. Close Down valve and raise table.

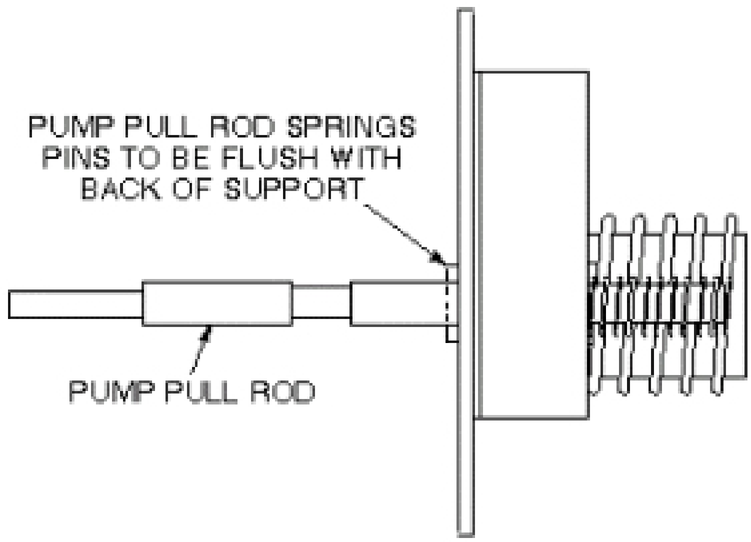
9. Retract lock bar and secure with spring, then lower table.

10. Fill table reservoir so that fluid is $^1/_4$” to $^1/_2$” above full line when table is at its lowest position.

NOTE: If any air gets into the cylinder, open the hydraulic cylinder bleed screw, when the table is raised, until air is removed from the cylinder. Very little fluid loss should occur during a normal bleed procedure.

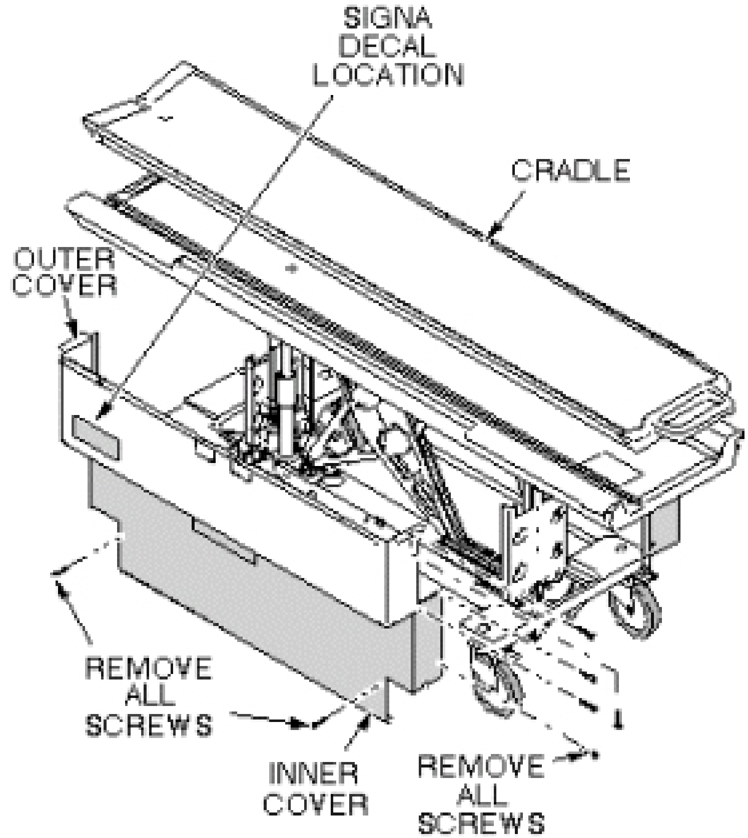
11. Verify that pump pull rods are adjusted equally by comparing position of rear spring pins on pump pull rods (see Illustration 6-22).

Illustration 6-22: PUMP PULL ROD ADJUSTMENT



12. Replace inner and outer covers. Note position of Signa logo (if applicable) on outer covers and the semicircle cut out on the inner covers, and then move that end towards the front of table (see Illustration 6-23).

Illustration 6-23: REMOVAL OF PATIENT TABLE COVERS



5 Patient Table Caster Replacement

5.1 Personnel Requirements

Required Persons	Preliminary Reqs	Procedure	Finalization
1	Not Applicable	15-30 (each)	Not Applicable

5.2 Overview

This procedure is applicable to the following table products:

- Signa Lightweight Patient Transport Table
- Signa Oncology Patient Transport Table Option
- Signa OR-Compatible Patient Transport Table Option

Each patient transport consists of the following: three (3) lock-type casters and one (1) straight-line tracking (steering) caster. The straight-line tracking caster is located on the front right of the patient table. The procedures for replacement of the casters are the same regardless of caster type.

5.3 Preliminary Requirements

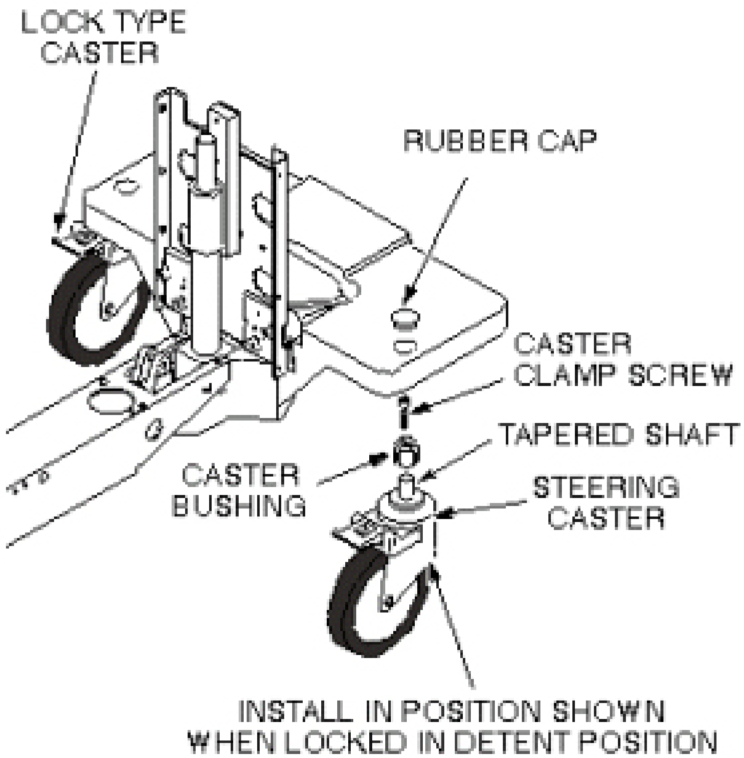
5.3.1 Tools and Test Equipment

Item	Qty	Effectivity	Part#	Manufacturer
Synthetic Lubricating Grease	1	-	46-256047P1	-
Block of wood to support table when casters are being replaced	1	-	-	-
M4 Allen wrench	1	-	-	-

5.4 Procedure

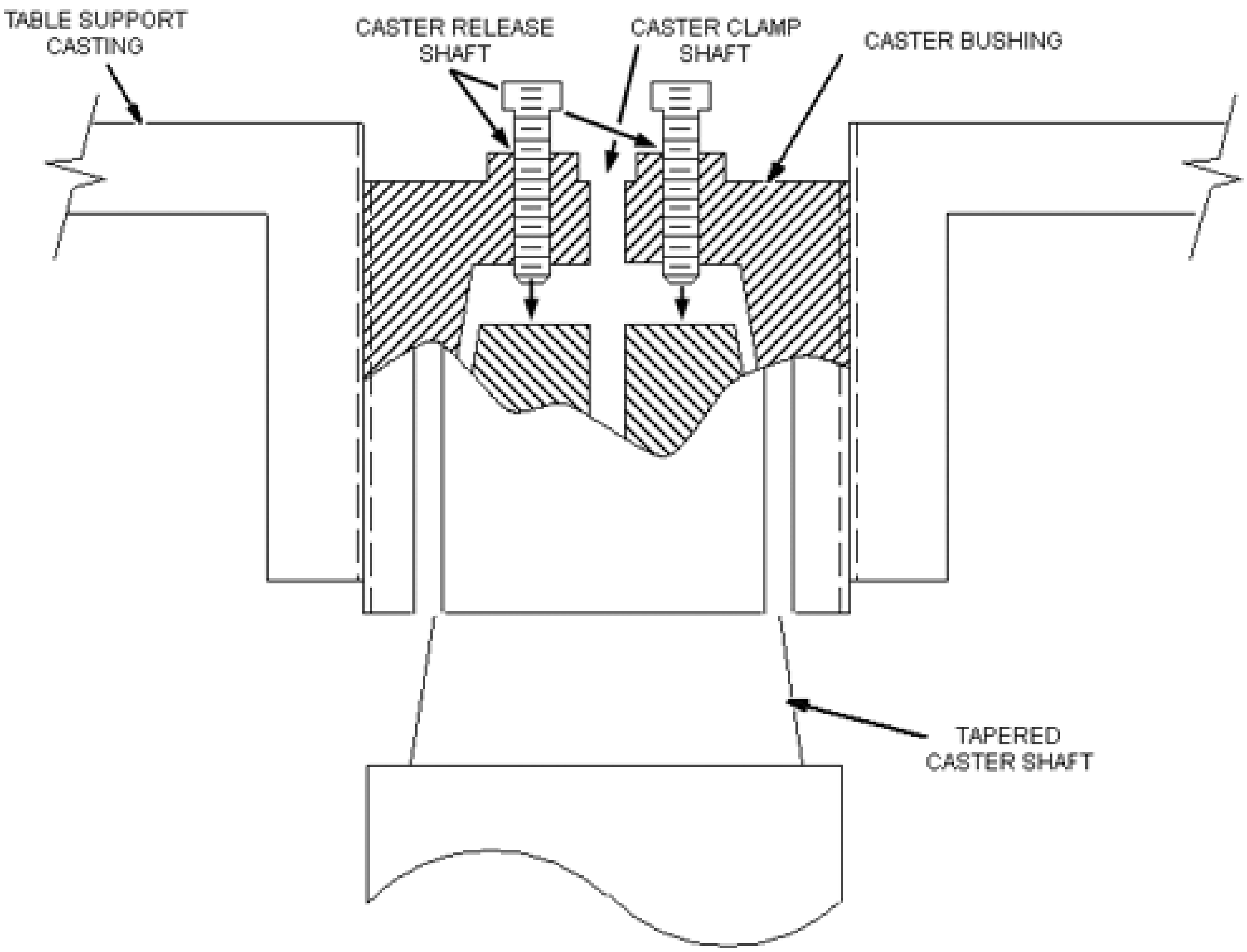
1. Undock patient transport and move from exam room.
2. Prepare block of wood to support table when casters are replaced.
3. Remove rubber cap from the caster to be replaced (see Illustration 6-24).

Illustration 6-24: CASTER WHEEL ASSEMBLY



4. Remove caster screw (center) from the caster to be replaced.
5. Obtain another caster screw from any one of the other casters by repeating steps and .
6. Return to the caster to be replaced. Thread the two caster screws in the vacant Caster Release shafts of the caster bushing (see Illustration 6-25). Force the tapered caster shaft out of the caster bushing by tightening two caster screws, equally, a little at a time, until caster taper fit breaks loose and caster falls off.

Illustration 6-25:



7. Be sure that bushing is not bell-mouthed at bottom. If bushing is damaged, remove by turning out of the bottom of the casting using a $^3/_4$-inch (19 mm) socket.

NOTE: If new bushing is required, apply synthetic lubricating grease to inside taper and threads of bushing before installation.

8. Using synthetic lubricating grease, lubricate tapered stem of caster and threads of caster clamp screw.
9. Install new caster into bushing. Straight line tracking (steering) caster should be installed so that it is oriented in position shown when locked in detent position (see Illustration 6-24).
10. Remove screws from Caster Release shafts and return each screw to the Caster Clamp shaft it originated from.
11. Tighten screws into Caster Clamp shafts.

12. Remove support wood blocks.

13. Perform procedure for checking caster height in Chapter 2, Section 2, Leveling Patient Transport Procedure.

NOTE: Equipment damage possibility. Caster adjustments must be done according to Chapter 2, Section 2, Leveling Patient Transport Procedure. Mis-adjustments leave caster bushing vulnerable to damage, which can result in injury to patient and/or persons near patient transport.

6 Patient Table Hydraulic Cylinder Replacement

6.1 Personnel Requirements

Required Persons	Preliminary Reqs	Procedure	Finalization
1 or 2, depending on system	Not Applicable	Not Applicable	Not Applicable

6.2 Overview

This procedure is applicable to the following table products:

- Signa Lightweight Patient Transport Table
- Signa Oncology Patient Transport Table Option
- Signa OR-Compatible Patient Transport Table Option

The cradle latch cable replacement procedure requires:

- One person for a Signa Lightweight Patient Transport Table
- Two people for a Signa Oncology or a Signa OR-Compatible Patient Transport Table Option

6.3 Preliminary Requirements

6.3.1 Safety

WARNING

PERSONAL INJURY POSSIBILITY!
LIFTING MR PARTS, SUCH AS FRUS, TOOLKITS AND PARTS OF THE SYSTEM WITHOUT ASSISTANCE (MECHANICAL OR ADDITIONAL PERSONNEL), CAN RESULT IN PERSONAL INJURY.
THE USE OF EXTRA PERSONNEL IS RECOMMENDED TO LIFT AN OBJECT WEIGHING MORE THAN 35 POUNDS AND LESS THAN 50 POUNDS. THE USE OF MECHANICAL LIFT ASSISTANCE DEVICES IS REQUIRED FOR OBJECTS WEIGHING MORE THAN 90 POUNDS.

WARNING

PERSONAL INJURY POSSIBILITY!
LIFTING MR PARTS, SUCH AS FRUS, TOOLKITS AND PARTS OF THE SYSTEM WITHOUT ASSISTANCE (MECHANICAL OR ADDITIONAL PERSONNEL), CAN RESULT IN PERSONAL INJURY.
THE USE OF EXTRA PERSONNEL IS RECOMMENDED TO LIFT AN OBJECT WEIGHING MORE THAN 35 POUNDS AND LESS THAN 50 POUNDS. THE USE OF MECHANICAL LIFT ASSISTANCE DEVICES IS REQUIRED FOR OBJECTS WEIGHING MORE THAN 90 POUNDS.

WARNING

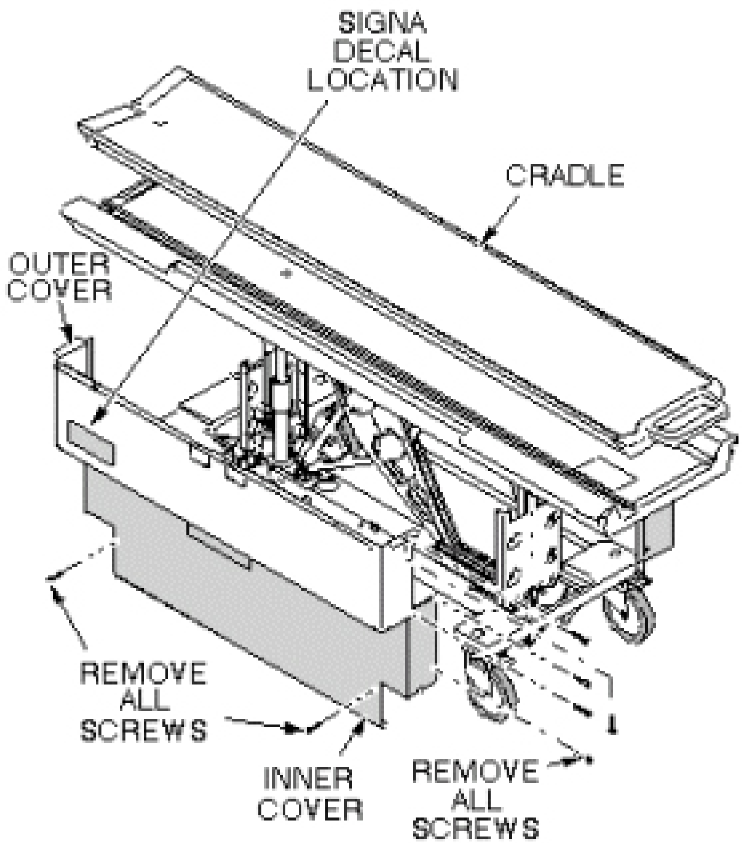
PERSONAL INJURY POSSIBILITY!
WHEN PERFORMING MAINTENANCE ON THE TABLE IN ITS UP POSITION, SEVERE PERSONAL INJURY CAN OCCUR IF TABLE IS ACCIDENTALLY LOWERED DURING MAINTENANCE PERFORMANCE.
ENSURE THAT THE LOCK BAR IS SAFELY IN PLACE.

6.4 Procedure

6.4.1 Remove Hydraulic Cylinder

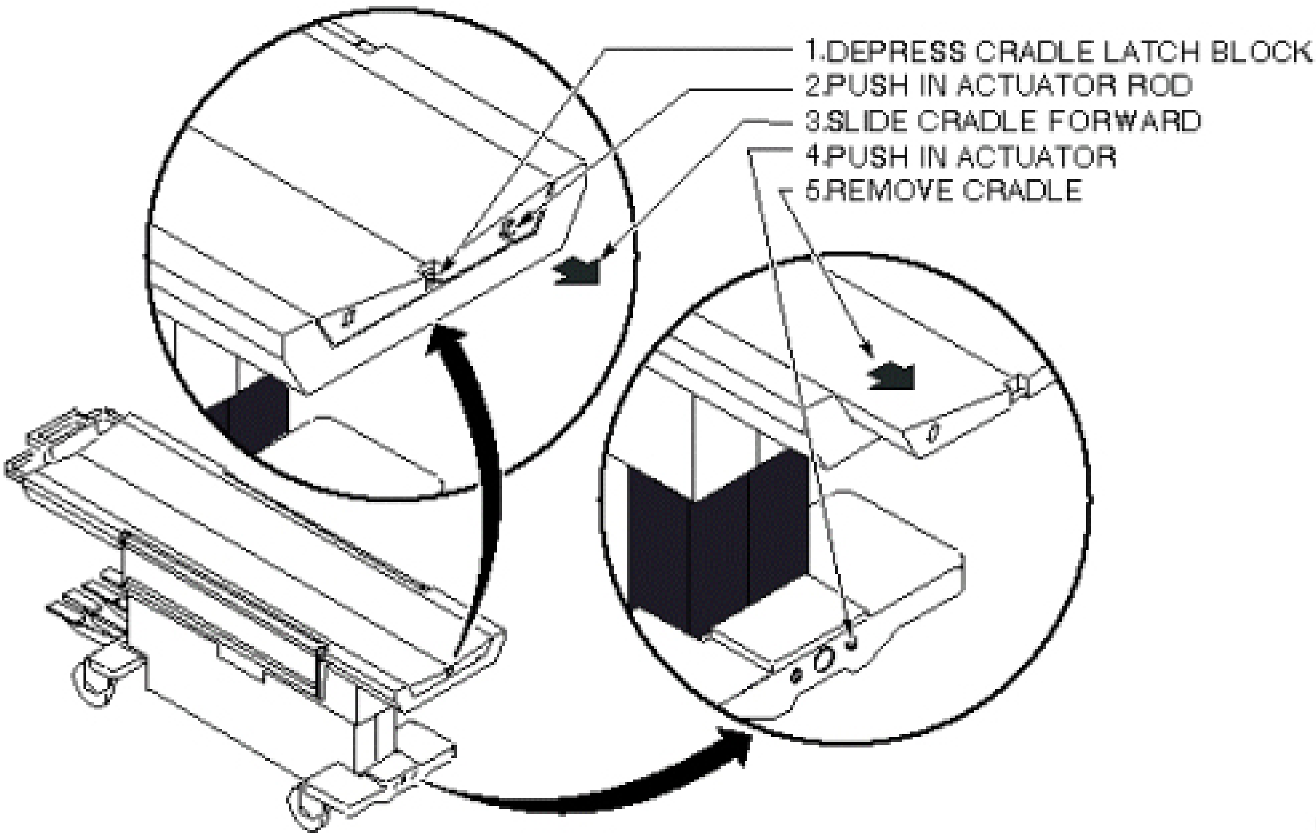
1. Undock patient transport, and move it out of exam room.
2. Remove both outer covers by removing mounting screws (see Illustration 6-26).

Illustration 6-26: REMOVAL OF PATIENT TABLE COVERS



3. Remove both inner covers by removing mounting screws.
4. Remove cradle assembly (see Illustration 6-27 .

Illustration 6-27: PATIENT TRANSPORT CRADLE REMOVAL



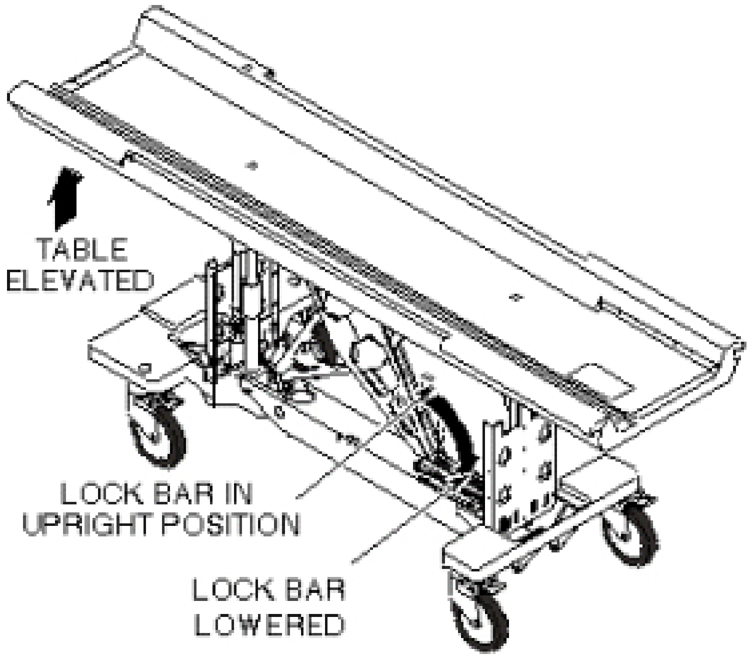
5. Raise table to Up position.

WARNING

PERSONAL INJURY POSSIBILITY!
WHEN PERFORMING MAINTENANCE ON THE TABLE IN ITS UP POSITION, SEVERE PERSONAL INJURY CAN OCCUR IF TABLE IS ACCIDENTALLY LOWERED DURING MAINTENANCE PERFORMANCE.
ENSURE THAT THE LOCK BAR IS SAFELY IN PLACE.

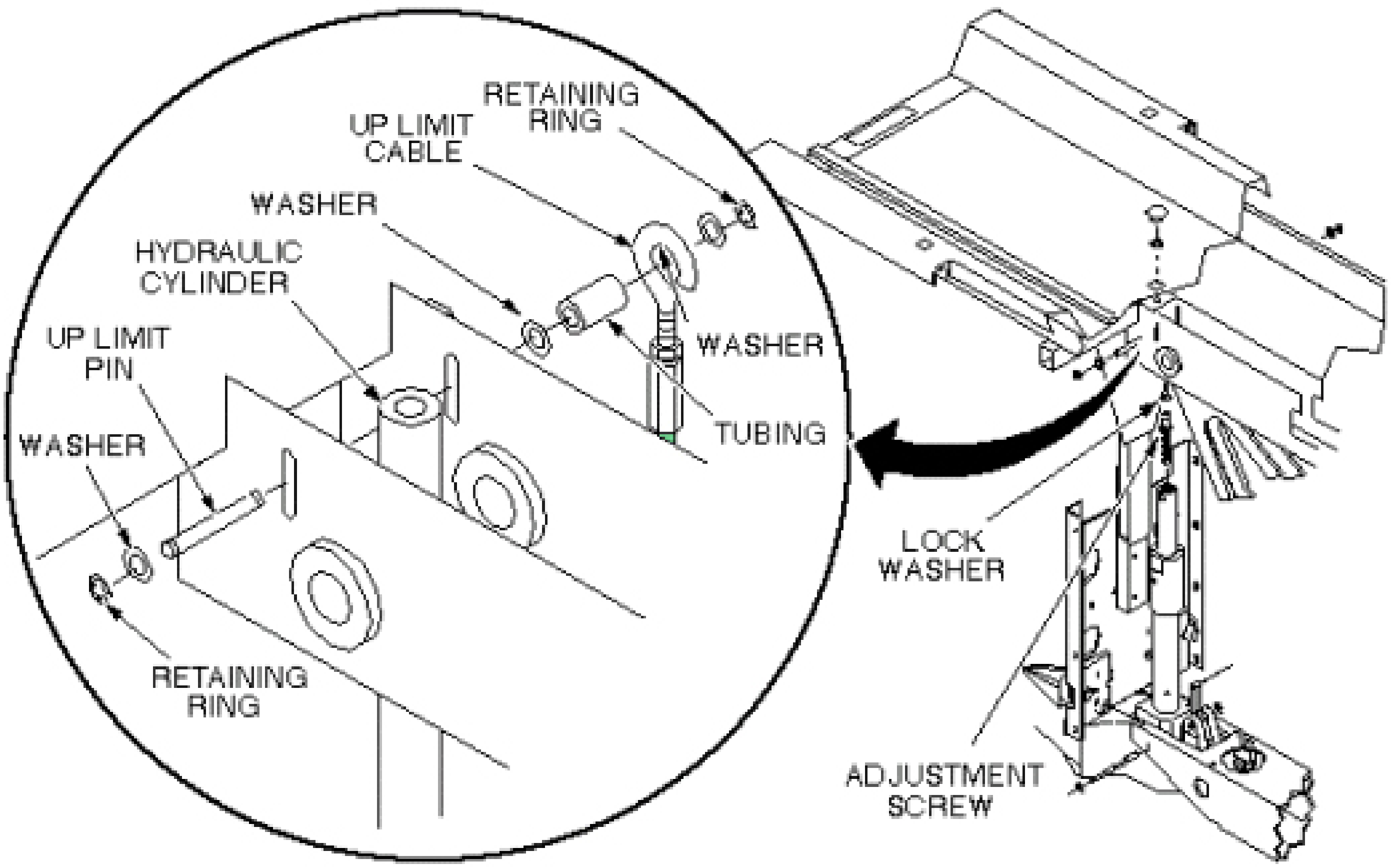
6. Unhook lock bar spring, and lower bar to horizontal position, as shown in Illustration 6-28 , then lower table until lock bar fully supports table top.

Illustration 6-28: PATIENT TABLE LOCK BAR



7. Remove plug from patient transport (see Illustration 6-29).
8. Remove retaining ring from end of adjustment screw.
9. Remove retaining ring, washer, and Up Limit cable attachment pin. Then remove tubing and washer. All are in Illustration 6-29 .

Illustration 6-29: PATIENT TABLE HYDRAULIC CYLINDER REPLACEMENT



10. Remove Up Limit cable attachment pin that extends through hydraulic cylinder.

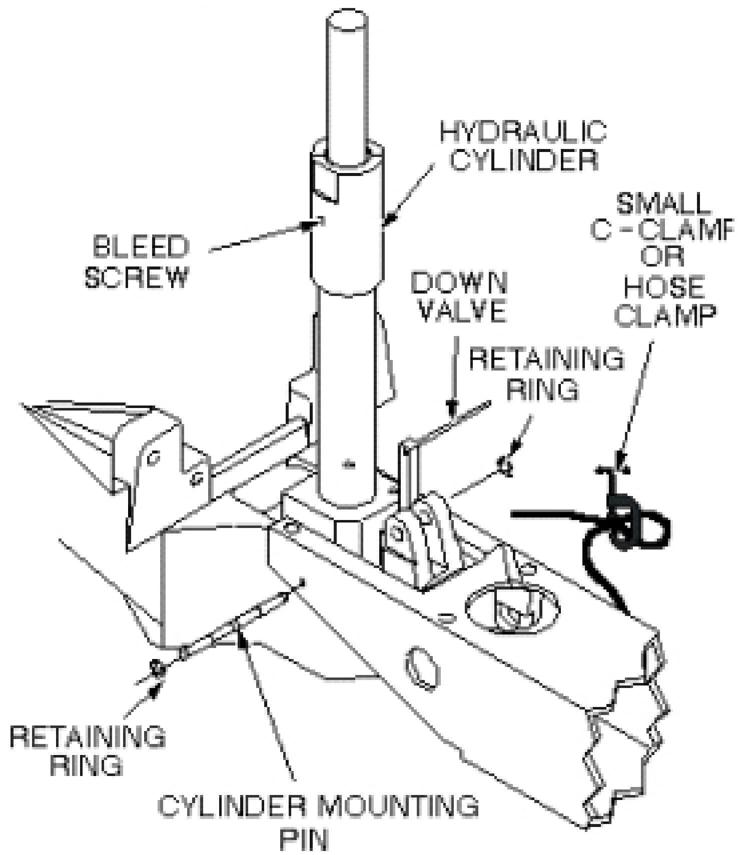
NOTE: The hydraulic cylinder cannot be pressed unless the Down valve is opened. See Illustration 6-32 .

11. Manually press hydraulic cylinder down until it is fully retracted.

12. Disconnect two hydraulic lines from base of hydraulic cylinder, and fasten upward and aside to avoid excess spillage. Immediately fold and clamp the translucent (yellow) hose to prevent oil from draining from the reservoir (see Illustration 6-30).

13. Remove retaining ring from either end of cylinder mounting pin at base of cylinder assembly (see Illustration 6-30).

Illustration 6-30: REMOVAL OF HYDRAULIC CYLINDER



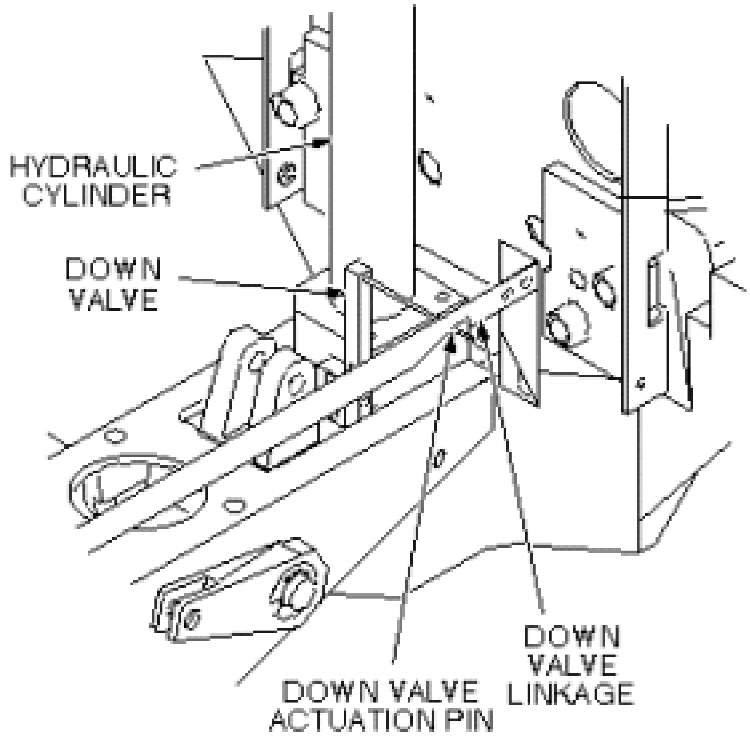
14. Remove cylinder mounting pin from base of hydraulic cylinder.
15. Carefully move linkage to free actuation pin on Down valve (see Illustration 6-31).
16. Remove hydraulic cylinder from base of table assembly.

6.4.2 Install Replacement Hydraulic Cylinder

1. Prepare replacement cylinder for installation by pressing cylinder to its fully retracted position.

NOTE: It may be necessary to move linkage to fit actuation pin of Down valve into groove of linkage, as shown in Illustration 6-31 .

Illustration 6-31: DOWN VALVE ACTUATION PIN



2. Carefully set cylinder in base of table, taking care not to bend actuation pin on Down valve.

3. Insert cylinder mounting pin securing base of hydraulic cylinder to frame of table (see Illustration 6-30).

4. Insert retaining ring.

5. Connect two hydraulic lines to base of hydraulic cylinder.

6. Insert adjustment screw into top of hydraulic cylinder (see Illustration 6-29).

7. Insert lock washer on adjustment screw.

8. Position hydraulic cylinder so that top of adjustment screw is aligned with hole in table (see Illustration 6-29).

9. Pump up cylinder until table goes up slightly.

10. Insert retaining ring at top of adjustment screw to secure cylinder to table (see Illustration 6-29).

11. Insert Up Limit cable attachment pin through hole in hydraulic cylinder (see Illustration 6-29). Do not attach the cable at this time.

12. Pump table up to the top of its travel.

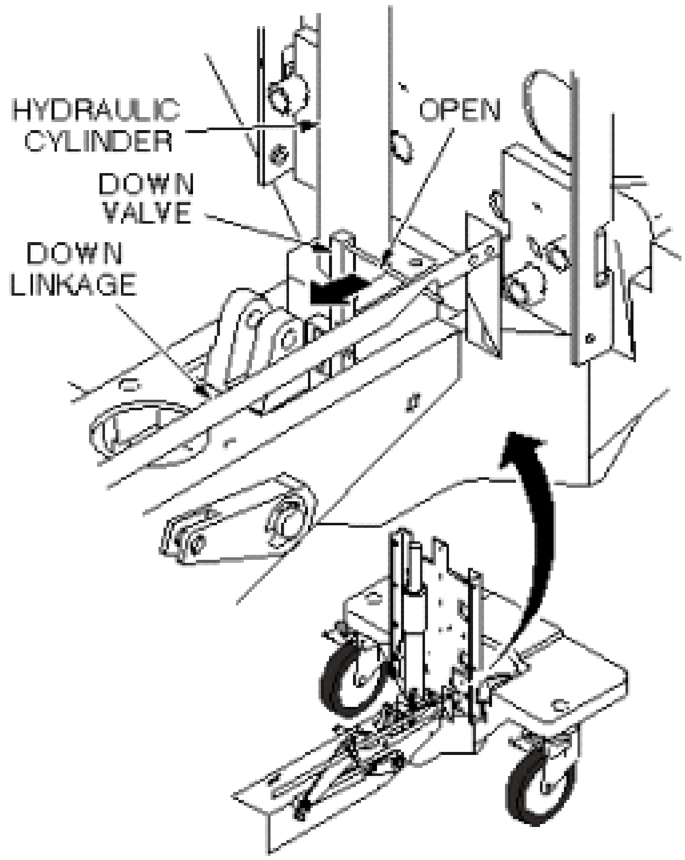
13. Attach washer and tubing to Up Limit cable attachment pin.

14. Attach cable to end of Up Limit pin, and secure with washer and retaining ring. Adjust length if necessary to install, and to remove any slack.

15. Ensure that lock bar is at its vertical (Unlocked) position as shown in Illustration 6-28 , with the lock bar spring secured.

16. Partially open bleed screw on hydraulic cylinder until no air bubbles are visible (see Illustration 6-32).

Illustration 6-32: HYDRAULIC CYLINDER DOWN VALVE



17. Open Down valve and lower table to its lowest position.

18. Hold Down valve open and exercise Up Pump foot pedal. This bleeds the hydraulic system by shunting air to the hydraulic reservoir.

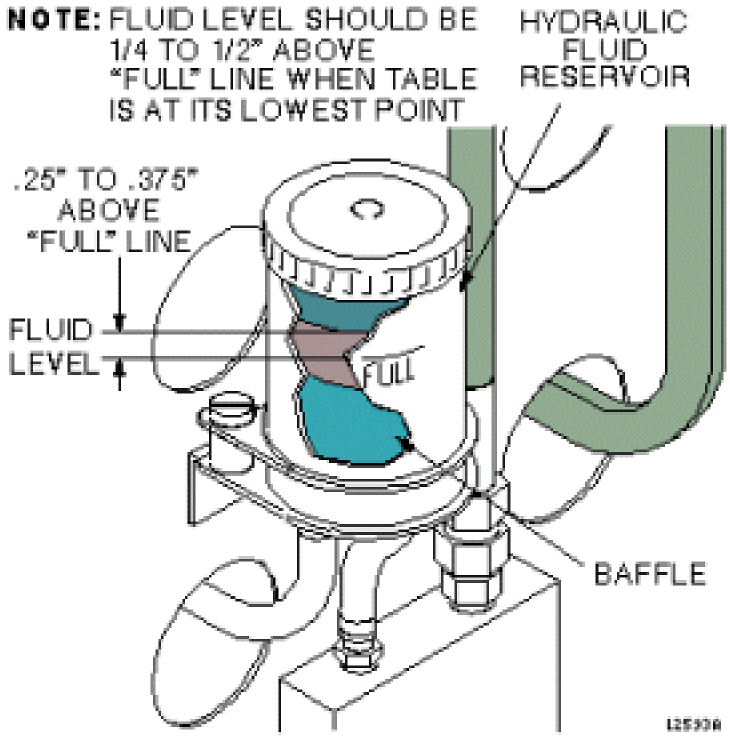
19. Bleed the cylinder again by partially opening bleed screw, when table is fully up, until no air bubbles are visible.

6.4.3 Final Calibration and Checks

1. Refer to procedure for Chapter 2, Section 3, Table Top Height Adjustment; perform those procedures, then perform the following steps.

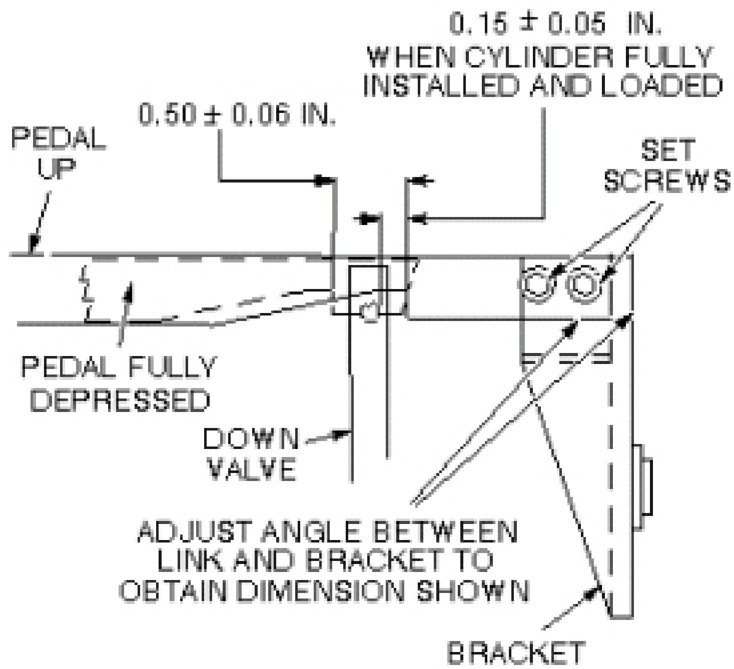
2. Fill table reservoir so that fluid is $^1/_4$” to $^1/_2$” above full line when table is at its lowest position (see Illustration 6-33).

Illustration 6-33: PATIENT TABLE RESERVOIR FILL LEVEL



3. Check Down valve linkage adjustment as shown in Illustration 6-34 . Obtain desired tolerance between Down valve linkage and bracket by adjusting two set screws on bracket.

Illustration 6-34: DOWN VALVE LINKAGE ADJUSTMENT



4. Replace plug in top of table (see Illustration 6-29).
5. Replace inner and outer covers. Note position of Signa logo (if applicable) on outer covers and the semicircle cut out on the inner covers, and then move that end towards the front of table (see Illustration 6-26).
6. Replace cradle assembly (see Illustration 6-27).

7 Patient Transport Side Cover Removal

7.1 Personnel Requirements

Required Persons	Preliminary Reqs	Procedure	Finalization
1	Not Applicable	Not Applicable	Not Applicable

7.2 Overview

This procedure is applicable to the following table products:

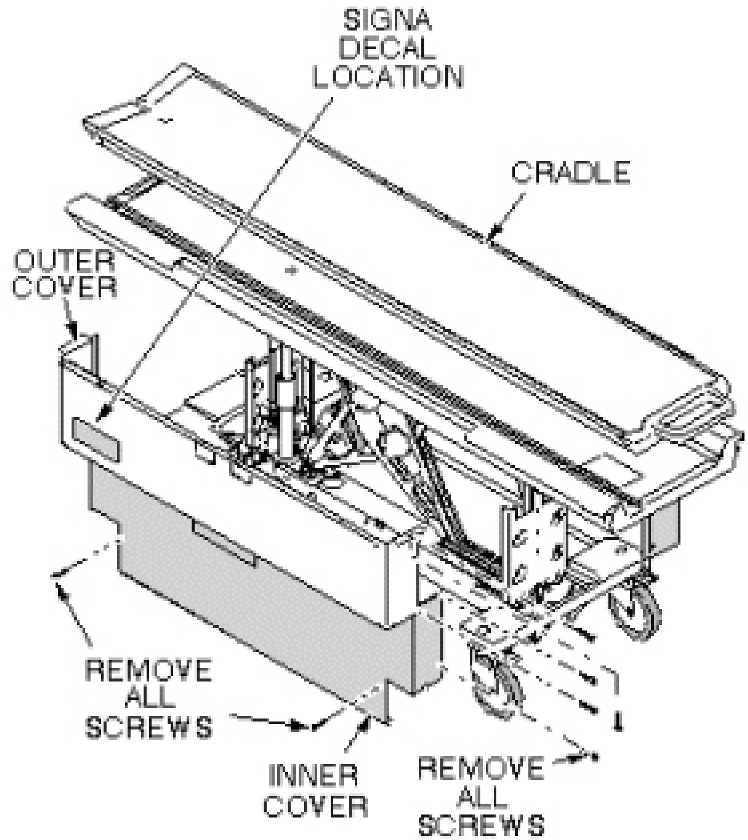
- Signa Lightweight Patient Transport Table
- Signa Oncology Patient Transport Table Option
- Signa OR-Compatible Patient Transport Table Option

See Illustration 6-35 for location of upper and lower side covers on patient transport and attaching hardware.

7.3 Procedure

1. Undock patient transport and move it out of exam room.
2. Remove both outer covers by removing mounting screws (see Illustration 6-35).
3. Remove both inner covers by removing mounting screws.

Illustration 6-35: REMOVAL OF PATIENT TABLE COVERS



8 Removing Cradle Assembly

8.1 Personnel Requirements

Required Persons	Preliminary Reqs	Procedure	Finalization
1 or 2, depending on system	Not Applicable	10 mins	Not Applicable

8.2 Overview

This procedure is applicable to the following table products:

- Signa Lightweight Patient Transport Table
- Signa Oncology Patient Transport Table Option
- Signa OR-Compatible Patient Transport Table Option

The cradle removal procedure requires:

- One person for a Signa Lightweight Patient Transport Table
- Two people for a Signa Oncology or Signa OR-Compatible Patient Transport Table Option

8.3 Preliminary Requirements

8.3.1 Safety

DANGER

STRONG MAGNETIC FIELD!
FERROUS MATERIALS CAN BECOME DANGEROUS PROJECTILES IN THE PRESENCE OF THE MAGNETIC FIELD PRODUCED BY THE SIGNA MAGNET. DO NOT BRING ANY FERROMAGNETIC TOOLS OR EQUIPMENT INTO THE MAGNET ROOM.

WARNING

RISK OF PERSONAL INJURY!
LIFTING MR PARTS, SUCH AS FRUS, TOOLKITS AND PARTS OF THE SYSTEM WITHOUT ASSISTANCE (MECHANICAL OR ADDITIONAL PERSONNEL), CAN RESULT IN PERSONAL INJURY.
THE USE OF EXTRA PERSONNEL IS RECOMMENDED TO LIFT AN OBJECT WEIGHING MORE THAN 35 POUNDS AND LESS THAN 50 POUNDS, BUT REQUIRED FOR OBJECTS WEIGHING MORE THAN 50 POUNDS. THE USE OF MECHANICAL LIFT ASSISTANCE DEVICES IS REQUIRED FOR OBJECTS WEIGHING MORE THAN 90 POUNDS.

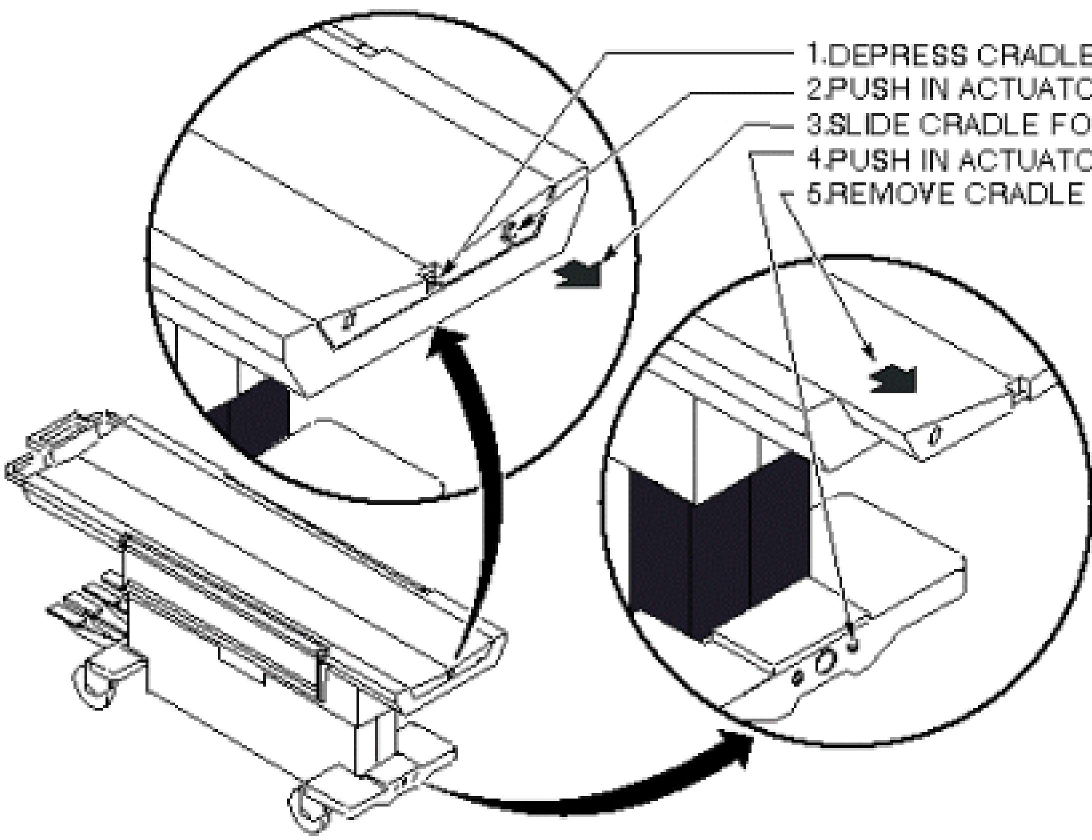
WARNING

RISK OF PERSONAL INJURY!
LIFTING MR PARTS, SUCH AS FRUS, TOOLKITS AND PARTS OF THE SYSTEM WITHOUT ASSISTANCE (MECHANICAL OR ADDITIONAL PERSONNEL), CAN RESULT IN PERSONAL INJURY.
THE USE OF EXTRA PERSONNEL IS RECOMMENDED TO LIFT AN OBJECT WEIGHING MORE THAN 35 POUNDS AND LESS THAN 50 POUNDS, BUT REQUIRED FOR OBJECTS WEIGHING MORE THAN 50 POUNDS. THE USE OF MECHANICAL LIFT ASSISTANCE DEVICES IS REQUIRED FOR OBJECTS WEIGHING MORE THAN 90 POUNDS.

8.4 Procedure

1. Undock patient transport.
2. At the same time, depress cradle latch block at front end of transport and push in actuator rod at front of cradle. See Illustration 6-37 .

Illustration 6-37: PATIENT TRANSPORT CRADLE REMOVAL

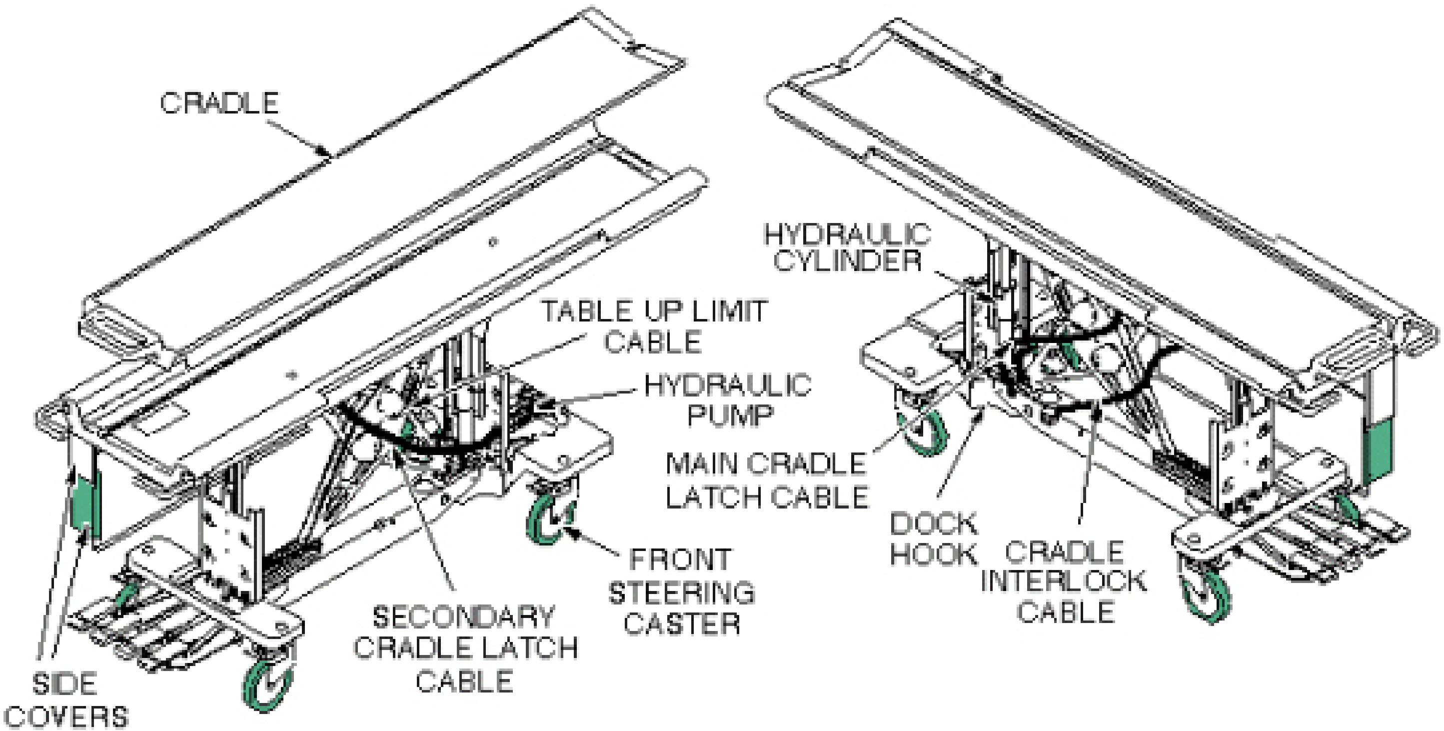


3. While holding these actuators, slide cradle forward on patient transport approximately three inches. The actuators can now be released.
4. Push in right actuator in base at front end of transport and slide cradle the rest of the way off the patient transport. See Illustration 6-37 .

9 Patient Transport Component Location

Most replacement procedures covering components of the patient transport require the removal of the transport side covers. Refer to Section 7, Patient Transport Side Cover Removal. See Illustration 6-37 for location of major components of the patient transport.

Illustration 6-37: LIGHTWEIGHT PATIENT TRANSPORT



10 Secondary Cradle Latch Actuating Cable Replacement

10.1 Personnel Requirements

Required Persons	Preliminary Reqs	Procedure	Finalization
1 or 2, depending on system	Not Applicable	60-90	Not Applicable

10.2 Overview

This procedure is applicable to the following table products:

- Signa Lightweight Patient Transport Table
- Signa Oncology Patient Transport Table Option
- Signa OR-Compatible Patient Transport Table Option

This procedure requires:

- One person for a Signa Lightweight Patient Transport Table
- Two people for a Signa Oncology or a Signa OR-Compatible Patient Transport Table Option

10.3 Preliminary Requirements

10.3.1 Replacement Parts

Item	Qty	Effectivity	Part#	Manufacturer
Cable	1	-	46-271511p1	-

10.3.2 Safety

WARNING

TO REDUCE THE RISK OF EXPOSING LOOSE FERROUS MATERIAL TO THE MAGNETIC FIELD, IT IS MANDATORY THAT THE PATIENT TRANSPORT IS REMOVED FROM MAGNET ROOM DURING THIS PROCEDURE.

WARNING

RISK OF PERSONAL INJURY!
LIFTING MR PARTS, SUCH AS FRUS, TOOLKITS AND PARTS OF THE SYSTEM WITHOUT ASSISTANCE (MECHANICAL OR ADDITIONAL PERSONNEL), CAN RESULT IN PERSONAL INJURY.
THE USE OF EXTRA PERSONNEL IS RECOMMENDED TO LIFT AN OBJECT WEIGHING MORE THAN 35 POUNDS AND LESS THAN 50 POUNDS. THE USE OF MECHANICAL LIFT ASSISTANCE DEVICES IS REQUIRED FOR OBJECTS WEIGHING MORE THAN 90 POUNDS.

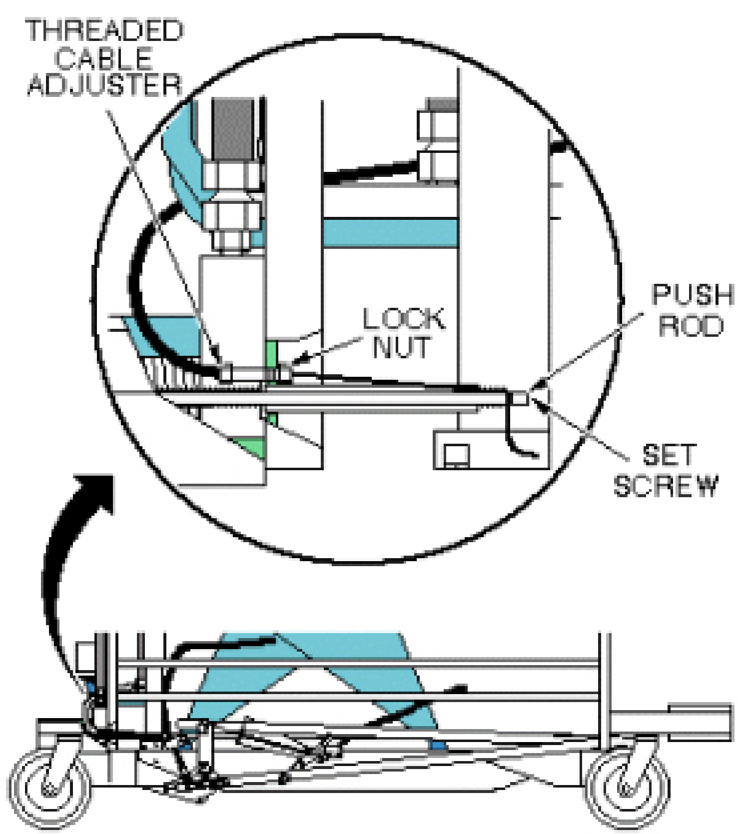
WARNING

RISK OF PERSONAL INJURY!
LIFTING MR PARTS, SUCH AS FRUS, TOOLKITS AND PARTS OF THE SYSTEM WITHOUT ASSISTANCE (MECHANICAL OR ADDITIONAL PERSONNEL), CAN RESULT IN PERSONAL INJURY.
THE USE OF EXTRA PERSONNEL IS RECOMMENDED TO LIFT AN OBJECT WEIGHING MORE THAN 35 POUNDS AND LESS THAN 50 POUNDS. THE USE OF MECHANICAL LIFT ASSISTANCE DEVICES IS REQUIRED FOR OBJECTS WEIGHING MORE THAN 90 POUNDS.

10.4 Procedure

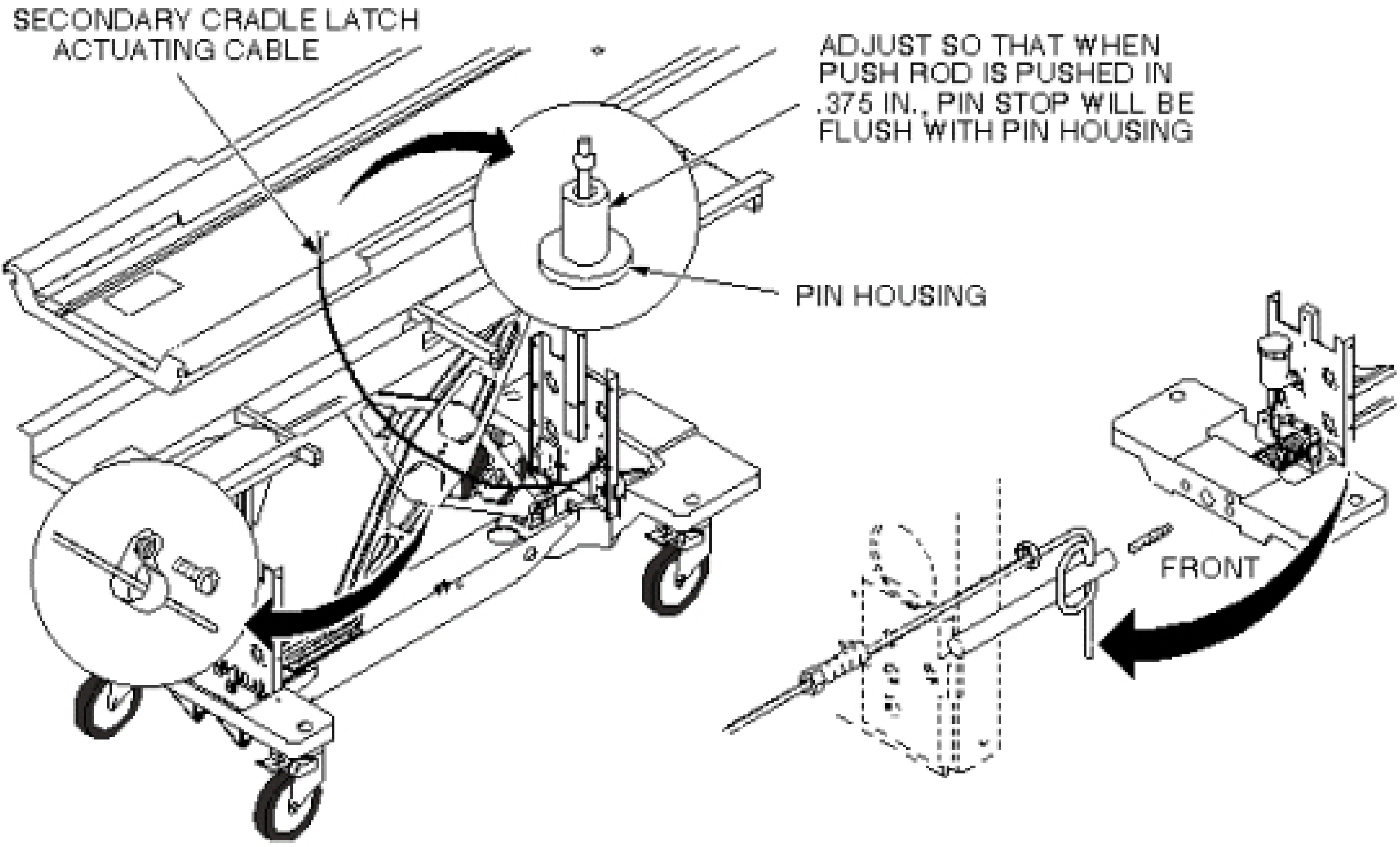
1. Undock patient transport and move from exam room.
2. Remove upper and lower left side covers (see Chapter 2, Section 4, Removing Cradle Assembly).
3. Remove cradle (see).
4. Unfasten cable end from end of push rod (see Illustration 6-38).

Illustration 6-38: SECONDARY CRADLE LATCH ACTUATING CABLE



5. Pull cable out of pin housing (see Illustration 6-39).

Illustration 6-39: SECONDARY CRADLE LATCH CABLE ROUTING



6. Insert new cable through cable casing (see Illustration 6-39).
7. Route cable through cable casing (see Illustration 6-39).
8. Loop cable twice through hole in push rod and secure with set screw. Tighten securely to prevent cable from slipping (see Illustration 6-39).
9. Adjust cable so that when push rod is pushed in 3/8 inch (9.5 mm); pin stop is flush with pin housing. Use threaded cable adjuster for fine adjustment. Tighten lock nut after adjustment is complete, then cut off excess length of cable (see Illustration 6-39).
10. Return patient transport to scan room.
11. Dock patient transport.
12. Verify that release pin is flush to 1/16 inch (1.6 mm) below flush when table is docked.
13. Replace cradle and upper and lower side covers. Note position of Signa logo on outer covers, and position toward front of table.

11 Table Up Limit Cable Replacement

11.1 Personnel Requirements

Required Persons	Preliminary Reqs	Procedure	Finalization
1	Not Applicable	Not Applicable	Not Applicable

11.2 Overview

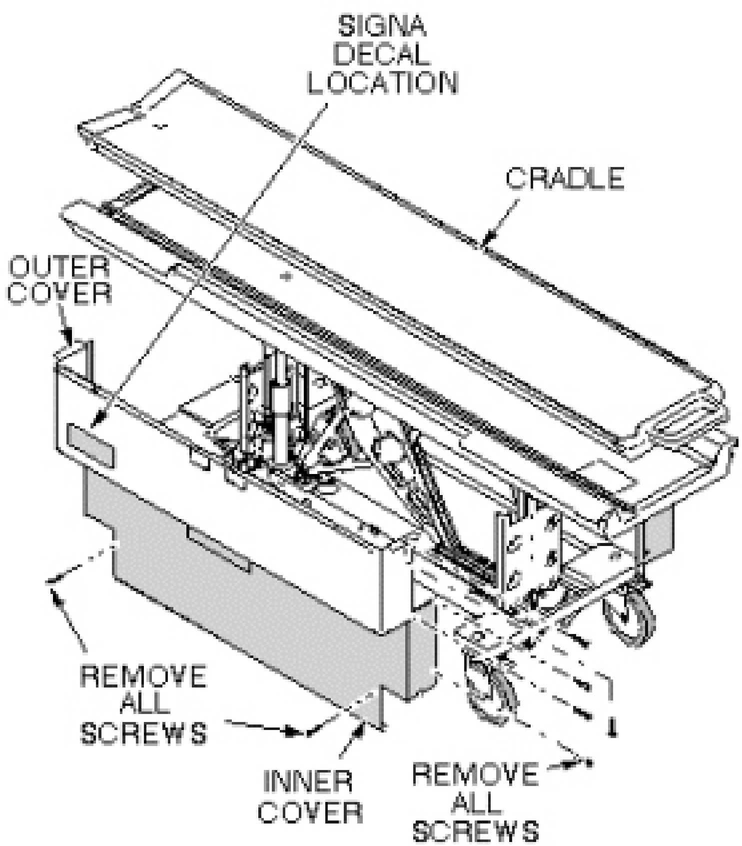
This procedure is applicable to the following table products:

- Signa Lightweight Patient Transport Table
- Signa Oncology Patient Transport Table Option
- Signa OR-Compatible Patient Transport Table Option

11.3 Procedure

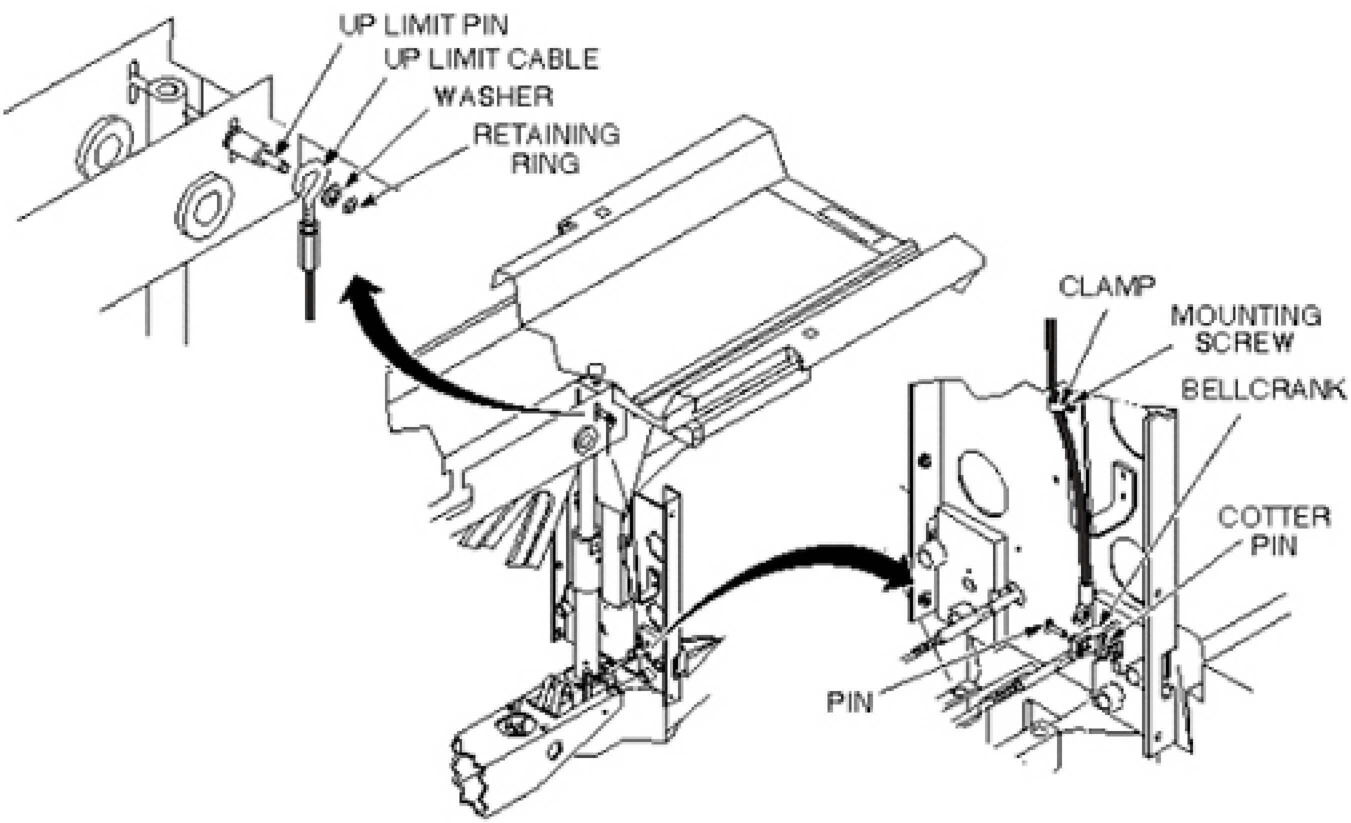
1. Undock patient transport and move from exam room.
2. Remove upper and lower side covers (see Illustration 6-40).

Illustration 6-40: REMOVAL OF PATIENT TABLE COVERS



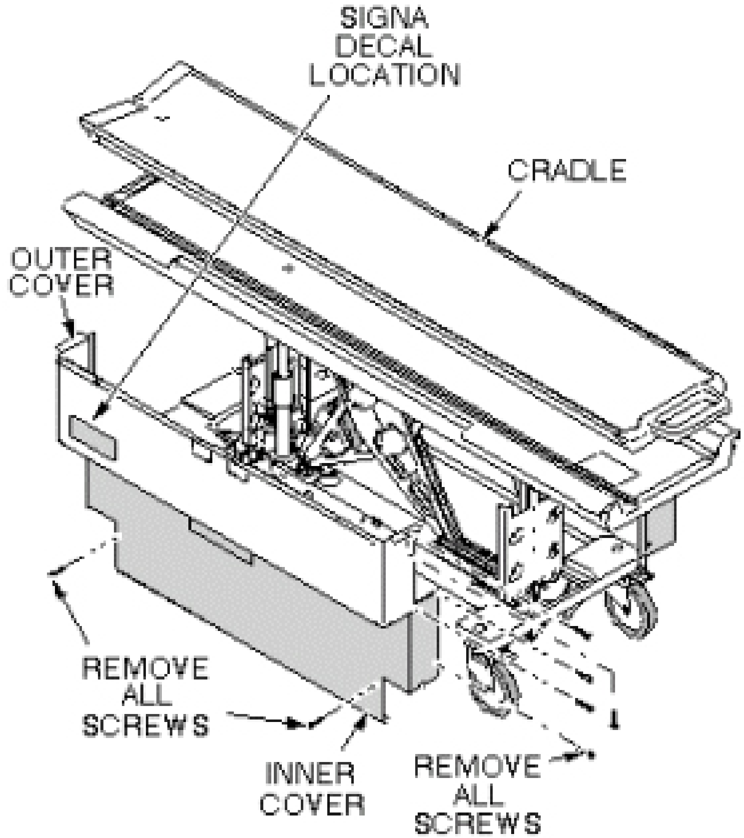
3. Raise table to maximum height.
4. Remove cotter pin and retaining pin securing bottom end of cable to bell crank (see Illustration 6-41).
5. Remove clamp securing cable to bracket (see Illustration 6-41).
6. Remove retaining ring and washer securing end of cable to Up Limit pin (see Illustration 6-41).
7. Install new cable and attach to Up Limit pin per above.
8. Fasten other end of cable to bell crank and secure retaining pin with cotter pin (see Illustration 6-41).
9. Attach clamp securing cable to bracket (see Illustration 6-41).

Illustration 6-41: TABLE UP LIMIT CABLE REPLACEMENT



10. Adjust table Up Limit cable to remove any slack when hydraulic cylinder is fully extended.
11. Check for proper operation of Up Limit switch when table is docked.
12. Replace upper and lower side covers. Note position of Signa logo (if applicable) on outer covers and the semicircle cut out on the inner covers, and then move that end towards the front of table (see Illustration 6-42).

Illustration 6-42: REPLACE PATIENT TABLE COVERS



12 Table Arm Board Bumper Field Repair/Replacement Procedure

12.1 Personnel Requirements

Required Persons	Preliminary Reqs	Procedure	Finalization
1	Not Applicable	60 mins	Not Applicable

12.2 Overview

This procedure is applicable to the following table products:

- Signa Lightweight Patient Transport Table
- Signa Oncology Patient Transport Table Option
- Signa OR-Compatible Patient Transport Table Option

This procedure outlines how to replace or repair the Table Arm Board Bumper which can be damaged or break off after long periods of active use.

12.3 Preliminary Requirements

12.3.1 Tools and Test Equipment

Item	Qty	Effectivity	Part#	Manufacturer
Bumper for Arm Board (if lost)	1	-	46-271017P2	-

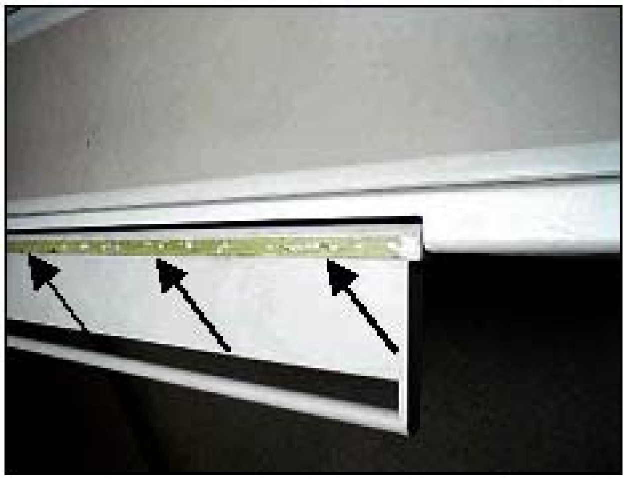
12.3.2 Consumables

Item	Qty	Effectivity	Part#	Manufacturer
Sharp Knife or Utility Knife	1	-	Local Purchase	-
Narrow (1 inch) Putty Knife	1	-	Local Purchase	-
Towels or Rags	As required	-	Local Purchase	-
Adhesive Remover (such as “GOO GONE” or equivalent)	1	-	Local Purchase	-
Isopropyl Alcohol	1	-	46-183000P164	-
Neoprene Gloves (100 count box)	1	-	46-194427P400 (lg) P401 (xlg)	-
Adhesive (one quart …fixes many bumpers)		-	46-170079P1	-
Brush (for application of adhesive)	1	-	46-194427P218	-

12.4 Procedure

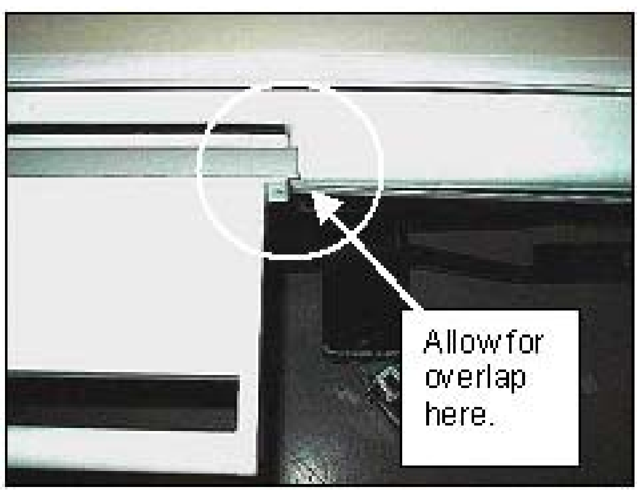
1. Remove table from magnetic field to a ventilated location.
2. Remove partially attached bumper(s) (if present).
3. Clean arm board bumper channel(s) completely with alcohol and/or adhesive remover, putty knife, and towel/rag. See Illustration 6-43 .

Illustration 6-43: ARM BOARD BUMPER CHANNEL



4. Remove old glue from bumper(s) with alcohol or adhesive remover/clean. Or if installing new bumpers, clean new bumper(s) with alcohol.
5. Apply adhesive to bumper channel and bumper surface per manufacturer directions (use gloves and brush). Install bumper in channel with some over lap at each end. Apply pressure to entire bumper to ensure good adhesion. Repeat on second bumper if necessary. See Illustration 6-44 .

Illustration 6-44: BUMPER INSTALLED WITH OVERLAP



6. Cut bumper ends at ~45 degree angles to match arm board edges. This will prevent catching on sheets, patients, etc. See Illustration 6-45 .

Illustration 6-45: CUT BUMPER ENDS

7. Return table to customer for use.

13 Cradle Wheel and Housing Replacement

13.1 Personnel Requirements

Required Persons	Preliminary Reqs	Procedure	Finalization
1 or 2, depending on system	Not Applicable	30 mins	Not Applicable

13.2 Overview

This procedure is applicable for the following table products:

- Signa Lightweight Patient Transport Table
- Signa Oncology Patient Transport Table Option
- Signa OR-Compatible Patient Transport Table Option

The cradle used on the Signa Oncology Patient Transport Table Option weighs approximately 74 lbs (34 kgs), which may require 2 persons for placing the cradle into a serviceable position.

13.3 Preliminary Requirements

13.3.1 Tools and Test Equipment

Item	Qty	Effectivity	Part#	Manufacturer
Non-Magnetic Hand Tool Kit	1	-	5113258 or 5112581	-
Flat Blade Screwdriver (from kit)	1	-	-	-
Pliers (from kit)	1	-	-	-

13.3.2 Replacement Parts

Item	Qty	Effectivity	Part#	Manufacturer
Wheel and Housing Kit (containing 6 Wheel and Housing assemblies)	As required	-	5179918	-

13.3.3 Safety

CAUTION

POTENTIAL FOR INJURY TO SERVICE PERSONNEL.
THE CRADLES USED ON THE SIGNA ONCOLOGY TABLE AND THE SIGNA OR-COMPATIBLE TABLE WEIGH APPROXIMATELY 74 LBS (34 KGS).
THE CRADLE CAN BE TURNED OVER AND SUPPORTED BY THE TABLE FOR THIS REPLACEMENT PROCEDURE. USE YOUR JUDGMENT AS A SECOND PERSON MAY BE REQUIRED TO ASSIST IN THE CRADLE POSITIONING PROCESS.

NOTICE

Follow ALL required safety and PPE procedures customary for your organization when working on this product.

- **Wear gloves for hand protection.**
- **Wear safety glasses for eye protection.**

13.4 Procedure

1. Undock the Patient Transport Table from the magnet and move it to a suitable location for servicing.
2. Lock the table castors to prevent table movement during servicing.
3. Release the cradle from the Patient Transport Table. See Illustration 6-46 for sequence.

NOTE: For purposes of reference, the Oncology cradle is shown in .

Illustration 6-46: RELEASE THE CRADLE

4. Flip the cradle over to access the Wheel and Housing assemblies underneath. See Illustration 6-47 .

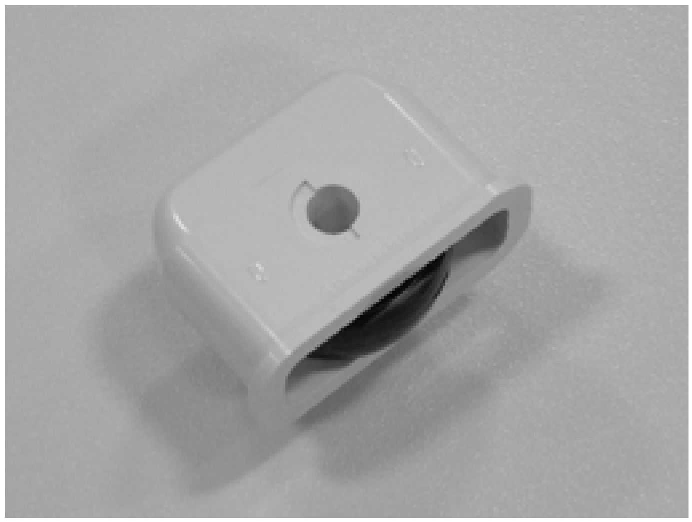
Illustration 6-47: REMOVE THE WHEEL AND HOUSING

NOTE: The Wheel and Housing assembly is press fit into the pocket on the cradle. The choice of tools for removing the Wheel and Housing is dependent on how tight the fit is.

5. Pry and lift the defective Wheel and Housing assembly out from the pocket.

6. Install the new Wheel and Housing assembly.

Illustration 6-48: WHEEL AND HOUSING ASSEMBLY



7. Verify the wheel spins freely within the housing.

8. Replace other defective Wheel and Housing assemblies as required.

14 Front and Rear Spacer Replacement

14.1 Personnel Requirements

Required Persons	Preliminary Reqs	Procedure	Finalization
1 or 2	Not Applicable	1 hour	Not Applicable

14.2 Overview

This procedure is applicable to the Signa Oncology Patient Transport Table Option.

The purpose of the Front Table Spacers is to eliminate a finger pinch hazard between the cradle and the IV Pole mounting hole as well as the tabletop surface. The Rear Table Spacers address the issue of finger pinch hazard between the tabletop surface and the cradle overhang. Repair or replacement is required if the spacers are loose or missing.

14.3 Preliminary Requirements

14.3.1 Tools and Test Equipment

Item	Qty	Effectivity	Part#	Manufacturer
Ruler or tape measure	1	-	-	-
Utility knife	1	-	-	-
Putty knife, 1 inch wide	1	-	-	-

14.3.2 Consumables

Item	Qty	Effectivity	Part#	Manufacturer
Towels or cloths	As required	-	-	-
Isopropyl Alcohol	1	-	46-183000P164	-
Masking tape	As required	-	-	-

14.3.3 Replacement Parts

Item	Qty	Effectivity	Part#	Manufacturer
Table Spacer Kit (See Note in Step 1)	1	-	5308567	-
Double-faced adhesive tape	1	-	46-170106P1	-

14.3.4 Safety

CAUTION

POTENTIAL INJURY TO SERVICE PERSONNEL.
THE CRADLE USED ON THE SIGNA ONCOLOGY TABLE WEIGHS APPROXIMATELY 74 LBS (34 KGS).
USE YOUR JUDGMENT AS A SECOND PERSON MAY BE REQUIRED TO ASSIST IN THE CRADLE POSITIONING PROCESS. IF THE SCANNER IS AVAILABLE THE CRADLE MAY BE MOVED INTO THE SCANNER BORE TO PROVIDE THE NECESSARY CLEARANCE FOR THIS PROCEDURE. TWO PEOPLE ARE REQUIRED TO LIFT AND REMOVE THE CRADLE FROM THE TABLE.

NOTICE

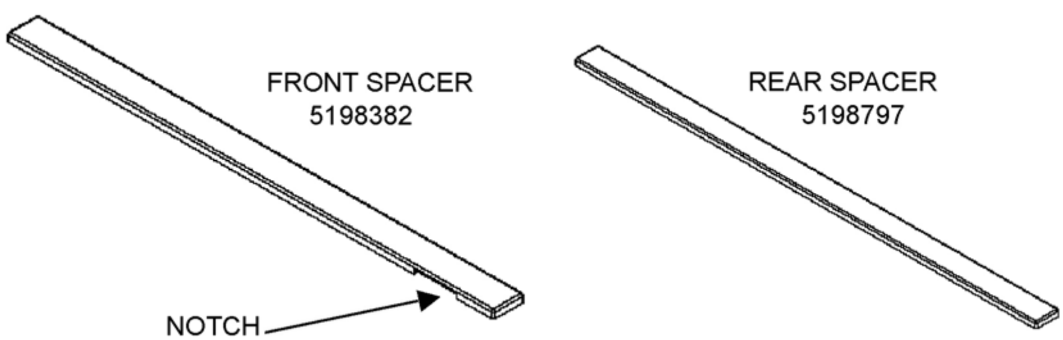
Follow ALL required safety and PPE procedures customary for your organization when working on this product.

- **Wear gloves for hand protection.**
- **Wear safety glasses for eye protection.**

14.4 Procedure

NOTE:

1. This kit contains 2 front spacers and 2 rear spacers.
2. The right and left front spacers are identical parts.
3. The right and left rear spacers are identical parts.

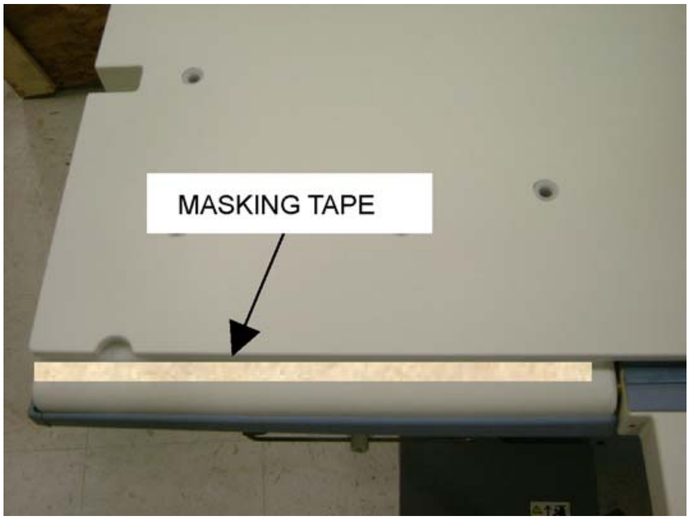


NOTE: This procedure assumes a second person is assisting in the lifting and removal of the cradle.

1. Undock the Signa Oncology Patient Transport Table from the magnet and move it to a suitable location for servicing.

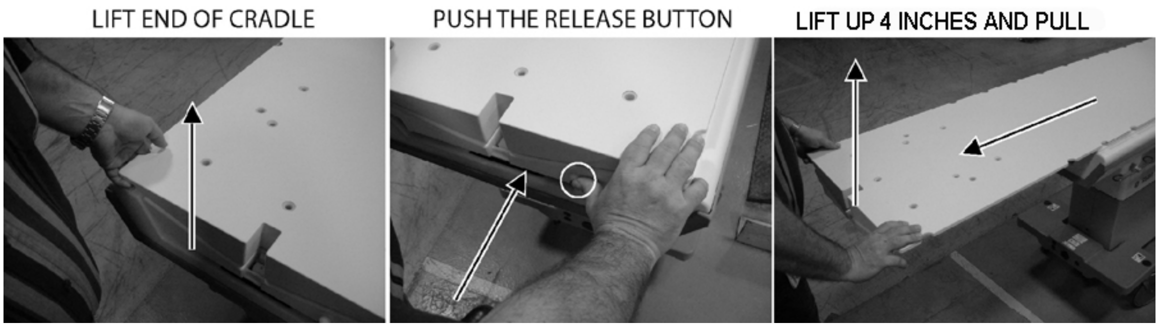
2. With the cradle in its *Home* position, apply a strip of masking tape to the tabletop surface adjacent to the location for the spacer that is being serviced. This will be used as a visual aid to align the spacer so it coincides with the side edge of the cradle. See Illustration 6-49

Illustration 6-49: MASKING TAPE ALIGNMENT AID



3. Release the cradle from the Patient Transport Table. See Illustration 6-50 for sequence.

Illustration 6-50: CRADLE REMOVAL SEQUENCE



4. Remove the cradle from the Patient Transport Table surface and store it in a suitable place.

5. Remove the loose spacer (if applicable) using a putty knife. Clean both the table surface and the bottom surface of the spacer using a clean towel or cloth and Isopropyl Alcohol. Allow both surfaces to dry completely.

6. Apply a strip of double face adhesive tape down the full length of the bottom side of the spacer. Trim off any excess using the utility knife.

7. For Front Spacer Alignment, make sure the following conditions are met before adhering it to the table (see Illustration 6-51):

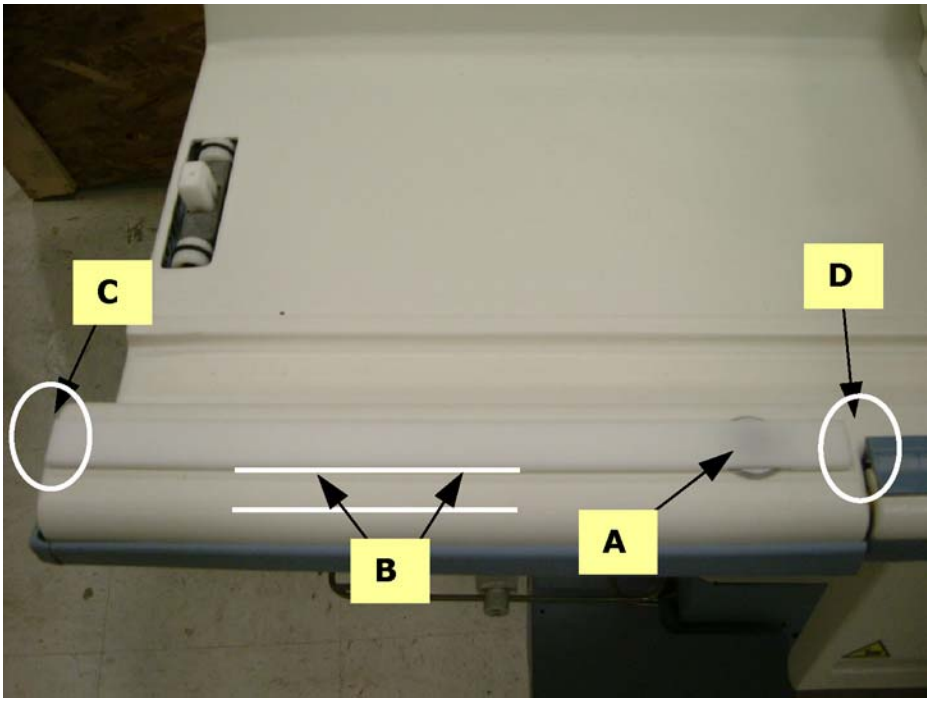
    a. The notch is centered over the IV Pole mounting hole.

    b. The side of the spacer is parallel to the side of the table.

    c. The end closest to the end of the table should not extend beyond the end of the table.

d. There is about a 0.25-inch gap between the end closest to the Arm Board and Arm Board hinge.

Illustration 6-51: FRONT SPACER ALIGNMENT



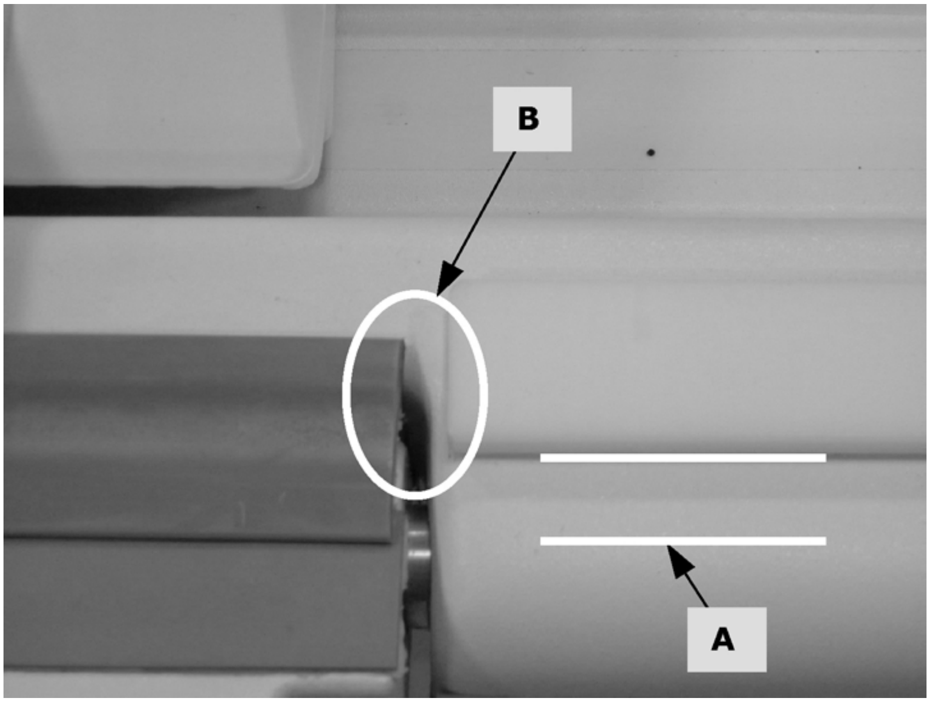
8. When ready, remove the protective strip from the adhesive tape and install the spacer into its final position.

9. For Rear Spacer Alignment, make sure the following conditions are met before adhering it to the table (see Illustration 6-52):

   a. The side of the spacer is parallel to the side of the table.

   b. There is about a 0.25-inch gap between the end closest to the Arm Board and Arm Board hinge.

Illustration 6-52: REAR SPACER ALIGNMENT



10. When ready, remove the protective strip from the adhesive tape and install the spacer into its final position.

11. Raise the Arm Board and make sure the spacer does not interfere with its operation. Correct as necessary.

12. When finished, reinstall the cradle onto the table.

15 Safety Decal Location and Replacement

15.1 Personnel Requirements

Required Persons	Preliminary Reqs	Procedure	Finalization
1	Not Applicable	30 mins	Not Applicable

15.2 Overview

This procedure is applicable to the Signa Oncology Patient Transport Table Option.

NOTE:

1. The sheet of decals contains items that are used on both the Signa Oncology Patient Transport Table and the Signa OR-Compatible Patient Transport Table. The Heavy Object decal, P/N 5306483F is the only one from this sheet used for the Signa Oncology Patient Transport Table.

2. Hand Crush Warning Label, IEC 60878 Safety Label

15.3 Preliminary Requirements

15.3.1 Consumables

Item	Qty	Effectivity	Part#	Manufacturer
Isopropyl Alcohol	As required	-	-	-
Towels or Cloths	As required	-	-	-

15.3.2 Replacement Parts

Item	Qty	Effectivity	Part#	Manufacturer
Sheet of decals (See Note 1 in Procedure Overview.)	1	-	5306483	-
Hazard decal, Hand Crush Warning (See Note 2 in Procedure Overview.)	4	-	5142065	-

15.3.3 Safety

NOTICE

Follow ALL required safety and PPE procedures customary for your organization when working on this product.

- **Wear gloves for hand protection**
- **Wear safety glasses for eye protection**

15.4 Procedure

1. Undock the Signa Oncology Patient Transport Table from the magnet and move it to a location suitable for servicing.

2. Clean the area for the decal using a clean towel or cloth and Isopropyl Alcohol. Allow the surface to dry before applying the new decal.

3. Remove the protective backing from the decal and install the decal. Make sure to smooth out any air bubbles. The Heavy Object decal is located on the cradle surface near the release handle. See Illustration 6-53 . The Hand Crush decal is located on each end of each Arm Board. See Illustration 6-54 .

Illustration 6-53: SIGNA ONCOLOGY TABLE CRADLE

Illustration 6-54: SIGNA ONCOLOGY TABLE ARM BOARDS

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